

# A SEARCH FOR TIME-VARIABLE AND FREQUENCY-DRIFTING NARROWBAND TECHNOSIGNATURES TOWARD ALMA-OBSERVED STARS WITHIN 40 PC OF THE SUN: SURVEY DESIGN AND FIRST RESULTS

GLENN J. WHITE<sup>1,2</sup> AND ROBIN DEY<sup>3</sup>

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## ABSTRACT

We have searched public archival ALMA observations of an archive-defined sample of 168 stars within 40 pc of the Sun – every such star with usable public ALMA coverage, which is the intersection of a volume-limited catalogue with a strongly selected archive rather than a volume-limited stellar sample for narrowband technosignatures. We find no evidence for a technosignature in any of the data searched. The experiment constrains carriers that are time-variable on the timescale of an observing track and/or drift in frequency: the per-channel temporal-median subtraction that rejects stationary ground-based interference also suppresses phase-steady carriers, so a stable beacon lies outside the class this pipeline can constrain, and we make no statement about it. At the epoch of this release, 104 of 208 planned star/band datasets across 85 census stars (79 independent systems) have been searched to completion: 417 spectral windows spanning 1.3–38.8 pc and 92.4 GHz of unique sky frequency between 89.6 and 495.1 GHz. Every window’s stellar position is compared with 512 control positions carried through an identical search, which calibrates the false-alarm rate empirically rather than assuming Gaussian statistics. Four windows pass the joint criterion: three are circumstellar CO (two toward  $\beta$  Pictoris, one toward HD 48370) and one, toward CP–72 2713, is a marginal crossing consistent with the 0.8 flags expected by chance across 417 windows.

Nominal  $5\sigma$  detection thresholds span  $1.6 \times 10^{13}$ – $2.8 \times 10^{17}$  W in EIRP (median  $1.3 \times 10^{15}$  W). Because most windows are channelised at 15.625 MHz these are native-channel power thresholds, not Hz-resolution narrowband sensitivities. A 1200-trial injection campaign into real calibrated visibility data recovers the searched class at  $42^{+16}_{-14}$  per cent at the nominal threshold, so these thresholds must not be read as 95-per-cent completeness limits. The principal result of the paper is that non-detection together with the threshold range: a reproducible archival pipeline has searched this volume to these depths and found nothing of the searched class. We also derive, as a secondary and deliberately conditional calculation, an effective occurrence bound *given* that a transmitter occupies the particular frequencies observed towards its own host and emits throughout the archival epochs:  $< 6.5$  per cent of the 56 injection-calibrated systems at  $\text{EIRP} \geq 10^{16}$  W, rising to 13.0 per cent at duty cycle 0.5 and 65 per cent at 0.1. That bound is pilot-grade – the completeness behind it is measured on six configurations and three stars – and it is conditional in a way that matters: under any transmitter-frequency prior spread across ALMA’s tuning range the searched islands cover too little spectrum to exclude anything at all. It should not be read as a prevalence measurement.

*Subject headings:* extraterrestrial intelligence, radio lines: stars, submillimetre: stars, surveys, astrobiology

## 1. INTRODUCTION

A technological civilisation, signalling deliberately or simply going about its own affairs, may leave a detectable imprint on the electromagnetic spectrum: a *technosignature* (Sheikh 2020; Vidal et al. 2026). Six decades of radio searches now cover many thousands of stars, yet, as Wright et al. (2018) quantified, sample only an infinitesimal fraction of the multi-dimensional space of transmitter frequency, bandwidth, power, and duty cycle. Three gaps persist: the millimetre and submillimetre regime above  $\sim 50$  GHz is essentially unsearched; most surveys target interest-selected classes of star rather than the local stellar population; and few systematically reprocess *existing* interferometric archives.

This paper addresses all three. We take a census of every star within 40 pc of the Sun with public ALMA archival coverage, irrespective of spectral type, disc-hosting status, or known planets, and mine that archive for narrowband technosignature candidates and anomalous millimetre continuum, correcting for stellar proper motion and for Doppler drift rates plausible for a transmitter on an orbiting platform. The processing sets the search class – it suppresses phase-steady carriers – so this paper constrains *time-variable and/or sufficiently frequency-drifting narrowband emission* only (Appendix E measures both sides of that definition). Four archival-specific analyses supplement the classical search: closure-phase vetting, a continuum anomaly screen, a chirp and periodicity re-analysis, and a cross-target frequency-occupancy check.

Per-target tables are deferred to online-only Appendix I. We publish at 50 per cent completion because the citable content is methodological – the census definition, the empirically calibrated false-alarm machinery, the measured

<sup>1</sup> School of Physical Sciences, The Open University, Walton Hall, Milton Keynes, MK7 6AA, UK; glenn.white@open.ac.uk

<sup>2</sup> RAL Space, STFC Rutherford Appleton Laboratory, Harwell Campus, Didcot, OX11 0QX, UK

<sup>3</sup> VBRL Holdings Inc.

completeness – with a first set of per-target limits, every number tied to a frozen pipeline and a named data snapshot that later work will supersede row by row without invalidating this one.

## 2. BACKGROUND

Six decades of searches divide into wide-field commensal surveys and targeted programmes on preselected stars, latterly dominated by Breakthrough Listen, whose GBT, Parkes, and MeerKAT programmes published multi-thousand-star limits from 1.1 to 8 GHz (Isaacson et al. 2017; Enriquez et al. 2017; Price et al. 2020; Czech et al. 2026), with parallel efforts on FAST, LOFAR, NenuFAR, the Sardinia Radio Telescope, and commensal VLA, MeerKAT, and MWA back-ends (Zhang et al. 2020; Johnson et al. 2023; Valente et al. 2025; Tremblay et al. 2024); machine-learning rejection of RFI-dominated hit lists is now central (Ma et al. 2023; Choza et al. 2024; Jacobson-Bell et al. 2025; Poznanski et al. 2025; Garrett et al. 2026) – a capability our classical threshold search lacks (§6). As Margot et al. (2023) note, a non-detection bounds the transmitter fraction only if the sample represents the underlying population, which interest-selected samples do not. Essentially all this activity lies below  $\sim 10$  GHz; above it the mm/submm regime has been searched once, also with ALMA (Mason et al. 2024). Sixteen programmes are tabulated in Appendix D.

Mason et al. (2024) searched archival Band 3 windows toward 28 “stellar bycatch” stars, placing  $\text{EIRP}_{\min} > 7 \times 10^{17}$  W for the nearest and identifying high-frequency work’s challenges: drift rates scale with frequency, smearing erodes sensitivity without a drift search, and the molecular line forest confuses. Our pilot (White 2026; under review, citation provisional pending acceptance) extended that methodology to TRAPPIST-1 across Bands 3, 6, and 7 ( $\text{EIRP}_{\min} \sim 5 - 10 \times 10^{13}$  W, no detection) and built the ranked 100-star list from which this survey grew. New here are the census selection (§3), the 512-position control-ring false-alarm calibration, closure-phase vetting, the spectral-type-normalised continuum screen, the chirp and periodicity extension, and the cross-target frequency-occupancy check – each, to our knowledge, new *in an ALMA archival technosignature survey*, though we claim no wider priority: several have centimetre-wave counterparts, and we have not made the exhaustive search a general claim would require. No execution block is analysed in both papers: the pilot’s 100-star search products are superseded, not re-used, here, and every window analysed in this paper was re-reduced and re-searched from the raw archive data with the pipeline of §4.

Figure 1 places this release’s nominal EIRP thresholds among published searches in *total power*. Earlier versions gave each threshold divided by its channel width, in  $\text{W Hz}^{-1}$ , as the primary comparison; that was wrong for the transmitter class considered here, and we withdraw it.  $\text{EIRP}_{5\sigma}/\Delta\nu_{\text{ch}}$  is a channel-averaged equivalent spectral luminosity, correct only for emission that *fills* the channel, not the spectral power density of an unresolved carrier: a 1 Hz carrier in our deepest window,  $1.6 \times 10^{13}$  W in a 15.625 MHz channel, still needs all  $1.6 \times 10^{13}$  W to be detected, so its intrinsic spectral power is of order  $10^{13} \text{ W Hz}^{-1}$ , not the  $10^6 \text{ W Hz}^{-1}$  the division returns – which, beside Hz-scale backends, flatters ALMA by some six orders of magnitude against exactly the signal class sought. The meaningful comparison, and what every threshold here is, is the total EIRP required at each instrument’s own native resolution (Figure 1). Our linewidth convention is set out in §4.

A non-detection must become a quantitative exclusion statement. Wright et al. (2018) formalised the “Cosmic Haystack” search volume, Margot et al. (2023) its conversion into an upper bound on the transmitter-hosting fraction, and Sheikh et al. (2025a) asked where Earth’s own technosignatures would be detectable – a calibration point for any survey’s sensitivity.

Hence the case for a census over interest-selected lists. Most surveys select by proximity, spectral type, planets, or Earth Transit Zone membership – reasonable per unit time – while the ALMA-specific form, disc selection, skews the sample young and dynamically active, ALMA-detected discs being orders of magnitude brighter than the Solar System’s zodiacal cloud at interstellar distances. We instead take every star within 40 pc in the usable field of at least one public ALMA observation (§3): a bycatch census of the ALMA-observed subset of the local stellar population in a fixed volume, not of that population itself. This removes technosignature-specific but not archival astrophysical selection; our 168 stars are  $\sim 1$  per cent of the catalogued stellar population within 40 pc, and that residual coverage selection is characterised as a completeness function (§3, Table 3).

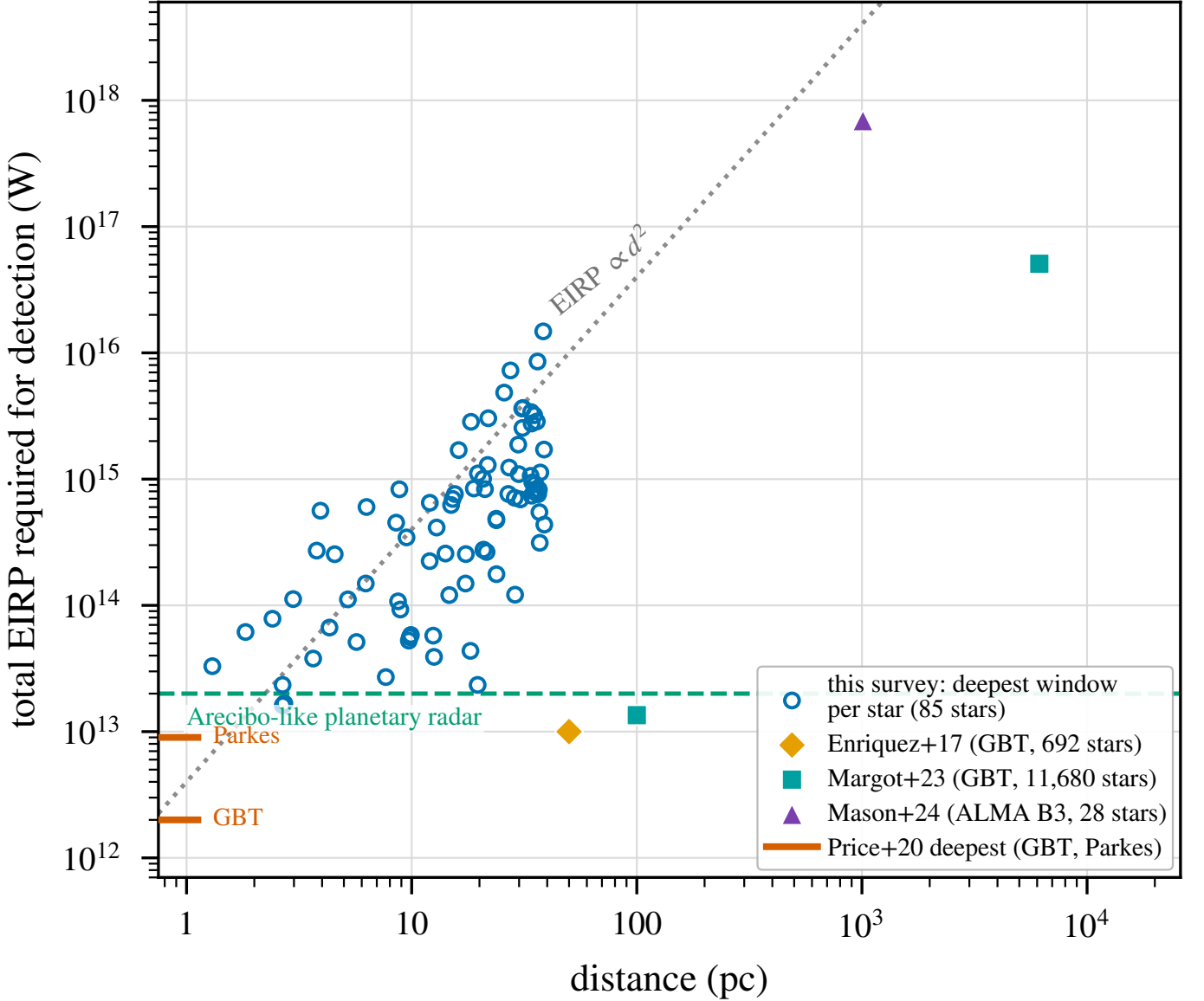
The SKA (Siemion et al. 2015; Tremblay et al. 2026) and ngVLA (Ng et al. 2022) will widen the coming decade’s search with wide bandwidth and commensal architectures below  $\sim 50$  GHz, but neither matches ALMA’s resolution and collecting area above it near-term, so a systematic mm/submm census stays ALMA-specific for some years.

## 3. SAMPLE SELECTION

Our sample comprises every star within 40 pc of the Sun (Gaia DR3 parallaxes, distances direct inversions  $d = 1/\pi$ ; every entry has  $\pi \geq 25$  mas with  $\sigma_\pi/\pi \leq 0.8$  per cent) whose proper-motion-corrected position at the observation epoch falls in the field of view of at least one public ALMA execution block, not necessarily at the pointing centre: mosaics and wide pointings serendipitously cover more than one catalogued star. Disc, planet and spectral type are descriptive metadata here, not selection criteria. This gives 168 unique target stars, processed nearest-first – a sensitivity-ranked order, not an unbiased draw, on which the prevalence statements of §6 are conditional.

Composition follows from the archival selection: 76 of the 168 stars are M dwarfs (63 per cent), then 12 K, 11 G, and 8 F dwarfs, one A-type star ( $\beta$  Pic), and six white dwarfs; six faint companions lack tabulated types. Eighteen of the 85 processed stars are confirmed exoplanet hosts (§6), and the debris-disc hosts ( $\beta$  Pic, AU Mic,  $\epsilon$  Eri,  $\tau$  Cet) contribute dusty continua, kept from masquerading as anomalies by the continuum lane’s spectral-type-normalised screen (§4).

The 168 stars are every star within 40 pc with public ALMA coverage: complete with respect to, and conditional only on, the intersection of the 40-parsec reference catalogue with the public-archive snapshot (2026 September 8)



Total power for an unresolved carrier at each survey's native resolution; not a spectral-power density.

FIG. 1.— Literature context of this release's thresholds, in total power – the meaningful quantity for an unresolved narrowband carrier, not the channel-averaged equivalent spectral luminosity that dividing by channel width returns (see text; the earlier  $\text{W Hz}^{-1}$  panel is withdrawn). Blue: this survey, deepest window per star (85 stars, 1.30–38.83 pc),  $1.6 \times 10^{13}$ – $1.5 \times 10^{16}$  W, median  $5.6 \times 10^{14}$  W. Orange diamond: Enriquez et al. (2017),  $\sim 10^{13}$  W within 50 pc. Aqua squares: Margot et al. (2023),  $1.35 \times 10^{13}$  W within 100 pc and  $5.08 \times 10^{16}$  W within 6135 pc. Violet triangle: Mason et al. (2024), archival ALMA Band 3,  $6.91 \times 10^{17}$  W for their closest star at 1.01 kpc. Left-edge ticks: deepest per-target limits of Price et al. (2020),  $2 \times 10^{12}$  W (GBT) and  $9 \times 10^{12}$  W (Parkes), no single sample distance. Grey dotted:  $\text{EIRP} \propto d^2$  at constant received flux, normalised to  $4 \times 10^{12}$  W at 1 pc, a guide not a fit. Green dashed: an Arecibo-like planetary radar ( $2 \times 10^{13}$  W), purely a technological power scale; only our G 272-61, in its two resolved components, falls below it. Thresholds, drift coverage and confidence conventions differ among these surveys: context, not a ranking.

defining this release. Against the reference Gaia DR3 catalogue of the solar neighbourhood (Figure 13, online-only appendix; 17 566 stars within 40 pc passing the parallax-quality cut of Table 15) the 168 stars are  $\sim 1\%$ ; that list is itself incomplete at both ends, so the denominator is a convention and the fraction says only that coverage is small. Table 15 (online-only appendix) quantifies how the covered stars differ, leaving the selection function inspectable: strong enrichment of the nearest systems falling towards parity with distance, mild M-dwarf depletion with F/G enrichment, and planet-hosting in 20 of the 168 census entries. Age and disc-hosting flags are not uniformly available and are documented qualitatively; the release includes a machine-readable selection-function table.

ALMA coverage is itself astrophysically selected by the proposers' interests, so prevalence statements here concern the ALMA-observed subset of the solar neighbourhood, not the neighbourhood itself. Stellar counts quoted throughout are catalogue-entry counts: the 120 entries comprise 113 distinct physical systems (seven designation-linked component pairs),<sup>4</sup> and since the 44 narrowband-covered entries include the unresolved UV Ceti and AT Mic pairs, the 104

<sup>4</sup> Designations follow SIMBAD; in the per-target file (Appendix I) each entry has its own row and distance, and spatially unresolved

star/band datasets, 417 windows and 57 execution blocks derive from 79 independent systems – the paired rows are bookkeeping, not independent trials.

“Usable public ALMA coverage” is a five-criterion decision tree, frozen for this release absent an erratum: (i) at least one public execution block (EB) whose QA2-calibrated visibilities cover the star’s epoch-propagated position in the primary beam, with no minimum integration, sensitivity, antenna count, mosaic-tile count or array configuration beyond QA2 itself; (ii) propagated position within one primary-beam FWHM of the field centre (largest accepted offset  $\sim 0.7$  FWHMs, Sirius B); (iii) each spectral window admitted individually, on the pipeline’s quality vetting alone; (iv) no programme-category restriction, science, calibration and observatory projects entering on identical terms; and (v) a fixed archive snapshot date (2026 September 8), re-evaluated only at release boundaries. No star or window was excluded by an unlisted criterion, none for short integration (the shortest retained are 1.0 and 1.5 min).

ALMA project metadata quantify pointed versus serendipitous coverage: almost every narrowband-covered star is the pointed science target in at least one execution block, the exceptions being a few blocks on another programme’s field or calibrator-only, leaving Wolf 219 the sole genuinely serendipitous star. An archival census’s residual selection function is the union of ALMA target lists – a contrast with interest-selected technosignature samples that is one of degree, but of *quantified* degree: a bycatch fraction of 1 star in 44, 1 EB in 57.

#### 4. DATA AND METHODOLOGY

Survey vocabulary – target/band dataset, spectral window, execution block, control ring, candidate, molecular-line exclusion mask – is defined in Table 11, Appendix B.

Figure 3 summarises the design: one shared archive-ingestion stage feeds two independent search lanes. For each target we mine, within a bounded budget, all calibrated (or locally recalibrated) archival execution blocks covering the star’s Gaia-propagated position, and search their visibilities in (i) the *primary* lane, a narrowband, Doppler-drift-corrected candidate search following Mason et al. (2024) and Enriquez et al. (2017) that converts each non-detection into an equivalent isotropic radiated power limit  $\text{EIRP}_{5\sigma} = 4\pi d^2 S_{\min} \Delta\nu$  ( $S_{\min}$  five times the searched window’s per-channel rms, primary-beam corrected;  $\Delta\nu$  its native channel width;  $d$  the target distance), and (ii) an *ancillary* line-excluded continuum lane (Appendix L). “ $5\sigma$ ” throughout is the local instrumental threshold of Table 2, separately and empirically calibrated (§5.4). The chirp/periodicity extension and molecular-line census are in Appendices G and N; every limit, disposition, false-alarm calibration and completeness statement derives from the narrowband lane.

Both lanes exclude catalogued molecular transitions at tolerances suited to their statistics. The continuum lane masks the eight common species (CO,  $^{13}\text{CO}$ ,  $\text{C}^{18}\text{O}$ , HCN,  $\text{HCO}^+$ , CS, CN, SiO) at native resolution before integrating; the narrowband lane applies a  $\pm 50 \text{ km s}^{-1}$  stellar-frame velocity exclusion mask (§§4.2, 5.4) to any threshold crossing, and also sets the effective searched bandwidth ( $\approx 87$  of the 92.4 GHz unique frequency union). Masking reduces the frequency space a candidate can occupy, not the per-channel noise, so no threshold in the per-target file changes.

All flux densities and EIRP thresholds are Stokes  $I$ , so the limits apply to total transmitted power independent of polarisation geometry. What is lost is diagnostic, not sensitivity: polarisation is no discriminator here (no Stokes  $V/Q/U$  extraction or circularity test), and a polarisation-sensitive test is a natural future addition (§6.2).

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**How to read every limit in this paper.** *Every  $\text{EIRP}_{5\sigma}$  value here is a nominal, local instrumental detection threshold: five times the per-channel rms of the searched window, at that window’s native channel width, for a transmitter at the star’s position matching the searched temporal and spectral morphology – time-variable on track timescales, since phase-steady carriers dwelling in one channel for most of the track are suppressed by the statistic’s own continuum subtraction (Appendix E). It is not a calibrated completeness limit: the injection campaign measures a recovery of roughly one half at the threshold itself ( $\text{EIRP}_{50} = 1.10 \text{ EIRP}_{5\sigma}$  for the searched class), rising to 83 per cent at twice the threshold power ( $\text{EIRP}_{90} > 2 \text{ EIRP}_{5\sigma}$ ; Appendix E). It is a total-power threshold set by ALMA’s coarse native channelisation, not a Hz-resolution narrowband sensitivity. The same reading applies to every subsequent “ $5\sigma$ ” and to the per-target file (Appendix I); it is stated once here and not repeated at each occurrence.*

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##### 4.1. The search statistic, and the words used for its output

The primary statistic is formed on extracted spectra, not images or visibilities. For a sky position  $\mathbf{x}$  the per-channel time series is median-subtracted along the time axis, the residual dynamic spectrum stacked along each trial linear drift rate  $\dot{\nu}$ , and the statistic is the largest resulting single-channel amplitude in units of the window’s noise,

$$T(\mathbf{x}) = \max_{\dot{\nu}, \nu} \frac{S(\mathbf{x}, \nu, \dot{\nu})}{\sigma(\mathbf{x})}, \quad (1)$$

where  $S$  is the drift-stacked flux density in one native channel and  $\sigma(\mathbf{x})$  a single robust scale per position and window, estimated as  $1.4826\times$  the median absolute deviation of that position’s median-subtracted spectrum over all channels: local in position, global in frequency.  $T_\star \equiv T(\mathbf{x}_\star)$  at the proper-motion-propagated stellar position; the same operation,

pairs (e.g. the UV Ceti entries at 2.7 pc, with blended fluxes) are marked in the notes column.

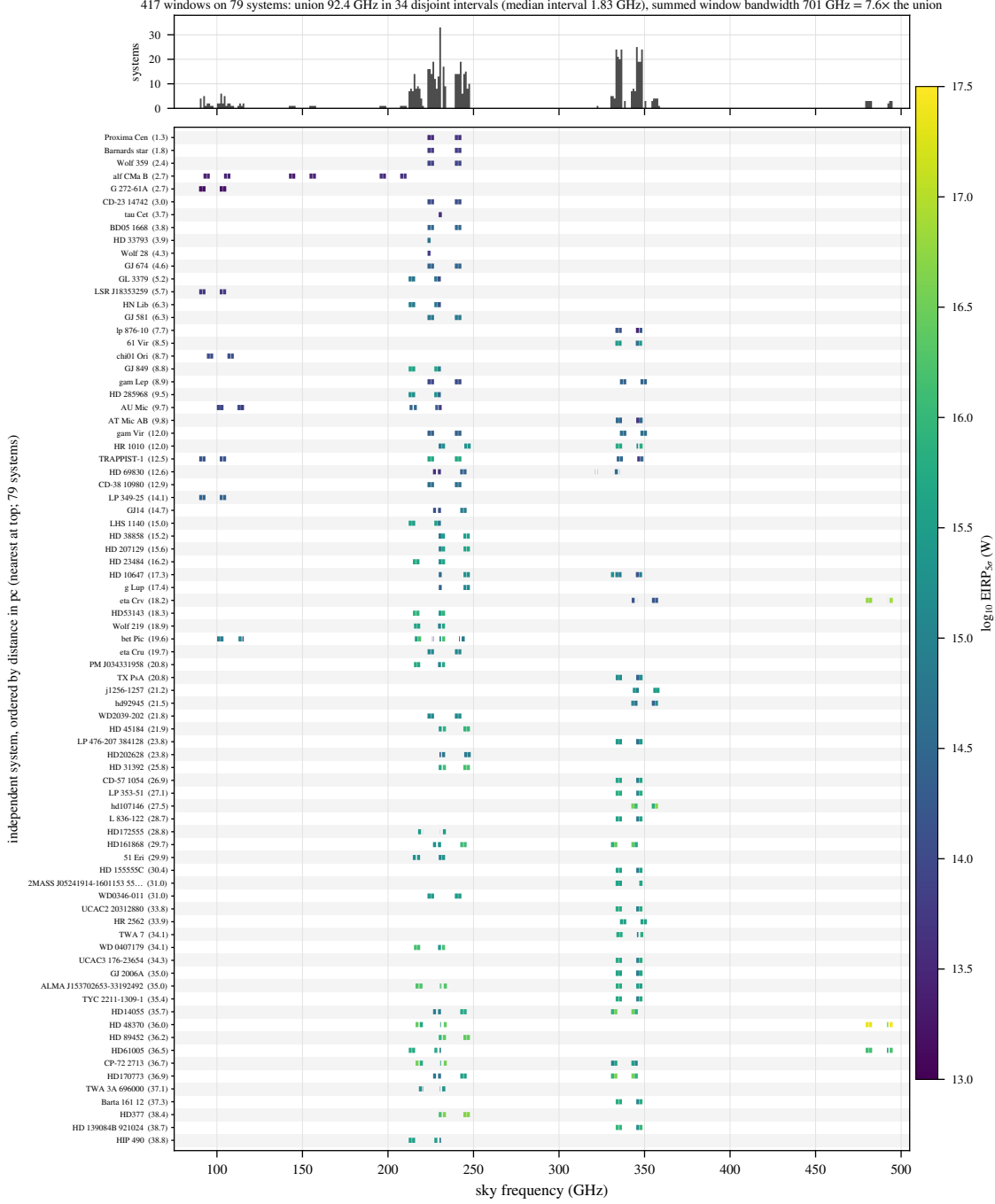


FIG. 2.— What this survey sampled in frequency. Each searched window is a horizontal segment on its host system’s row, rows ordered by distance, coloured by nominal  $\text{EIRP}_{50}$ ; the upper panel counts systems per gigahertz of sky frequency. The 92.4 GHz frequency union comprises 34 disjoint islands of median width 1.8 GHz, concentrated near 230 and 345 GHz, and the summed window bandwidth exceeds the union by a factor 7.6 – coverage narrow, clustered and repeatedly revisited, not broad. The occurrence limits of §6 are conditioned on these intervals. Thresholds are nominal in the sense of §4.

drift grid, channelisation and noise estimator give  $\{T_i\}$  at each of the 512 control positions. Table 1 sets out the data path in order, so the statistic can be reimplemented without the appendices.

The output vocabulary nests strictly (Table 2): *hit*, *spatially significant hit*, *candidate*. No object of this release is a candidate.

The searched drift-rate range is the larger of (a) a generic frequency-scaled default ( $12\text{--}13 \text{ Hz s}^{-1} \text{ GHz}^{-1}$ ), the fiducial of Mason et al. (2024) with a threefold safety margin – a design boundary, not a claim that faster drift is implausible – and (b) for confirmed exoplanet hosts, the maximum line-of-sight orbital acceleration of the innermost known planet, so a transmitter co-located with a short-period planet cannot drift out of the window. Drift rate maps onto line-of-sight

solid border: validated component

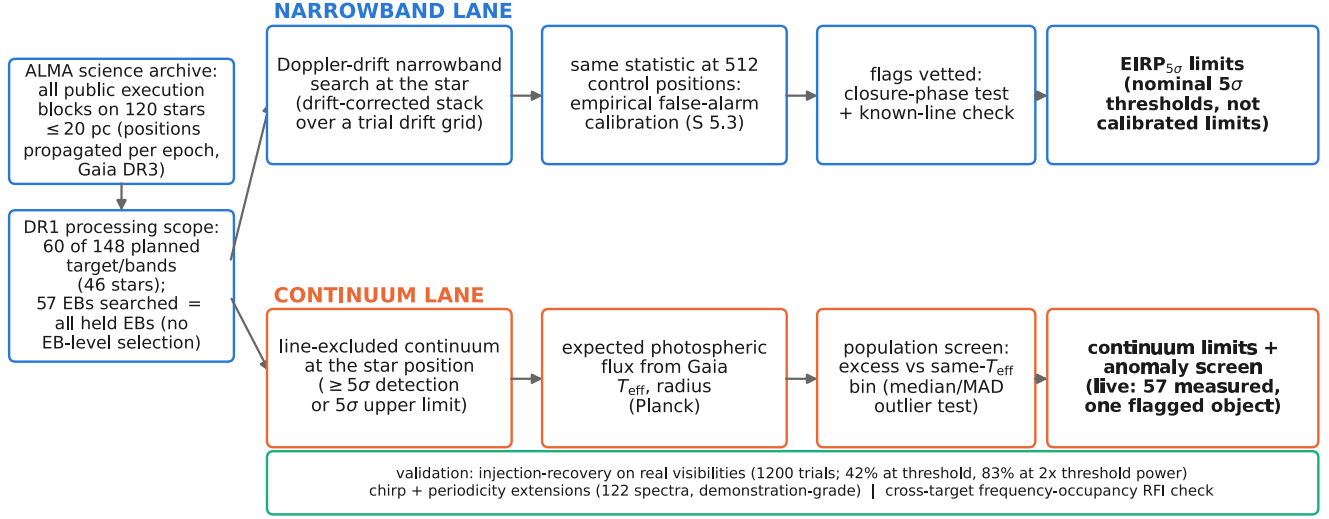


FIG. 3.— Design of the survey. All public execution blocks on the 168 targets are mined from the ALMA archive, positions propagated to each epoch from Gaia DR3, and searched along the two independent lanes described in the text. Solid borders mark validated components; the validation rail summarises the injection-recovery (1200 trials, Appendix E), chirp/periodicity and frequency-occupancy analyses.

TABLE 1

THE STATISTICAL DATA PATH, IN ORDER. STEPS 1–8 RUN IDENTICALLY AT THE STELLAR POSITION AND ALL 512 CONTROLS; ONLY STEP 9 ONWARD DISTINGUISHES THEM.

|    |                                                                 |
|----|-----------------------------------------------------------------|
| 1  | calibrated measurement set from the ALMA archive                |
| 2  | stellar position propagated to the observation epoch (Gaia DR3) |
| 3  | spectral extraction at that position, native channelisation     |
| 4  | per-channel subtraction of the time median                      |
| 5  | stack along each trial linear drift rate                        |
| 6  | robust per-position noise scale $\sigma(\mathbf{x})$ (MAD)      |
| 7  | $T(\mathbf{x})$ from Eq. 1                                      |
| 8  | repeat 3–7 at 512 control positions on a ring                   |
| 9  | threshold: $T_* \geq 5$ (a hit)                                 |
| 10 | rank test: $T_* > \max_i T_i$ (spatially significant hit)       |
| 11 | molecular-line mask, $\pm 50 \text{ km s}^{-1}$ stellar frame   |
| 12 | vetting: recurrence, closure phase, cross-target occupancy      |

TABLE 2

NOMENCLATURE FOR SEARCH OUTPUT AND THE TWO SENSES IN WHICH A PROBABILITY IS QUOTED.

| Term                          | Meaning                                                                                                          |
|-------------------------------|------------------------------------------------------------------------------------------------------------------|
| hit                           | one channel $\times$ drift cell at the stellar position with $T \geq 5$                                          |
| spatially significant hit     | a window whose $T_*$ exceeds all 512 control statistics                                                          |
| candidate                     | a spatially significant hit surviving every veto; none in this release                                           |
| nominal threshold             | $5\sigma$ local instrumental threshold, <i>not</i> a completeness limit                                          |
| completeness                  | measured recovery of injected signals by the unmodified pipeline                                                 |
| $p_{\text{rank}, \text{min}}$ | $1/(N_{\text{ctrl}} + 1) = 1/513$ , the rank <i>resolution</i> of the control ensemble                           |
| $P(\text{false flag})$        | the survey false-positive probability after thresholding, correlation and non-Gaussianity; not the same quantity |

acceleration through the first-order Doppler relation

$$a_{\text{los}} = c \frac{\dot{\nu}}{\nu}, \quad (2)$$

so the ceiling is most compactly quoted in cross-band-comparable form  $\dot{\nu}/\nu$ :  $12\text{--}13 \text{ Hz s}^{-1} \text{ GHz}^{-1}$  is  $\dot{\nu}/\nu = (1.2\text{--}1.3) \times 10^{-8} \text{ s}^{-1}$ , i.e.  $|a_{\text{los}}| = 3.6\text{--}3.9 \text{ m s}^{-2}$ , independent of observing frequency – the acceleration at  $\sim 0.04 \text{ au}$  around a solar-type star. Earth-motion contributions ( $\lesssim 0.13 \text{ Hz s}^{-1} \text{ GHz}^{-1}$ ) lie two orders of magnitude below, subsumed by the grid. These are *apparent* topocentric drift rates, including deliberate transmitted chirp as well as acceleration-

induced drift, which the survey does not distinguish; our constraints concern linear apparent drift within the searched range only.

Across the 417 windows maximum searched drift rates span  $1.1\text{--}5.9\text{ kHz s}^{-1}$ , native channel widths  $15.3\text{ kHz--}31.25\text{ MHz}$  (median  $15.625\text{ MHz}$ ; Appendix I), trial drift rates per window  $2\text{--}1944$  (median 4). The threefold margin matters: at  $1\times$  the release would have been blind to two of its eight threshold-crossing windows, including the AU Mic candidate; at  $5\times$  no disposition changes: transmitters drifting faster than  $\sim 13\text{ Hz s}^{-1}\text{ GHz}^{-1}$  remain unconstrained. Intra-integration smearing is negligible almost everywhere: at most 1.10 channels, median 0.001; only three windows have  $\eta_{\text{smear}} < 0.99$  (marked \* in Appendix I; Table 12, Appendix C), the worst the  $\beta$  Pic Band 6 fine window (effective threshold  $1.74\times$  nominal). Channel widths are effective resolutions, verified against the measurement sets’ spectral window tables (Appendix C).

#### 4.2. Frequency and velocity reference frames

Every frequency axis here – searched spectra, tabulated ranges, candidate frequencies – is *topocentric*, as delivered by the ALMA correlator, no velocity-frame shift being applied to any data product at any stage. Velocities enter only in the vetting stage’s stellar-frame line-exclusion mask, whose conversion chain is in Appendix C. The stacking is coherent in frequency and in the drift-rate trial but preserves no electric-field phase coherence between integrations.

The native channel width is also the linewidth convention for every EIRP threshold here – the commonest source of confusion when comparing technosignature limits across surveys. The quoted  $\text{EIRP}_{5\sigma}$  applies to *any* transmitter narrower than the native channel; dividing by it gives the minimum detectable spectral power for wider ones. Channelisation is the dominant price the archive exacts, but we deliberately do not project an Hz-resolution sensitivity: a radiometer-equation scaling  $P_{\text{min}} \propto \sqrt{\Delta\nu}$  across six orders of magnitude in spectral resolution is a thought experiment, not an achievable ALMA sensitivity, and quoting it invites the misreading this paper avoids. The worked example, sub-channel qualifications, smearing table and channel-width verification are in Appendix C.

The narrowband search covers every spectral window of an observation, capped at 8 per target to bound compute cost; the cap bound exactly one target/band (61 Vir Band 7: four of eight windows searched; Appendix I), whose frequency-space selection function is therefore tabulated. Of the 104 star/band datasets, 53 have every configured window searched; the remaining 3 ( $\tau$  Cet, Kapteyn’s Star, Van Maanen’s Star) are represented by their deepest single window, marked in Appendix I, full multi-window coverage awaiting the next release. Each searched window yields its own  $\text{EIRP}_{5\sigma}$  row (Appendix I); Figure 8 plots only the deepest per target/band.

Velocity coincidence, not drift, is the operative discriminant when a threshold crossing is checked against the eight masked species (§4): circumstellar emission sits at rest offset by the gas’s kinematics, whereas a technosignature may appear at any frequency and drift rate. The searches as executed used a narrower one-channel-width implementation, agreeing with the velocity-space form on every disposition here; its justification and the bandwidth removed are in §5.4.

Both narrowband thresholds and continuum fluxes are corrected for ALMA’s primary-beam response at the star’s offset from the phase centre, negligibly for most targets (median factor 1.0001, 90th percentile 1.012, none above 2%). Two sit at large offset.  $\epsilon$  Eri Band 6 lies at  $\sim 0.7$  primary-beam FWHM, where the correction reaches  $2.9\text{--}3.4\times$  and the Gaussian beam form fails; rather than publish provisional values we *withhold* those four windows, which enter no number in this paper but are released, flagged, in the per-target file and return once the holography beam model is applied. Sirius B is retained and quoted as such: corrections  $3.6\text{--}16$  per cent across bands, a bounded systematic rather than an invalid model. Excluding Sirius B too would change neither endpoint of the quoted EIRP range nor the median ( $1.3 \rightarrow 1.4 \times 10^{15}\text{ W}$ ) nor any conclusion.

#### 4.3. Reference transmitter benchmarks

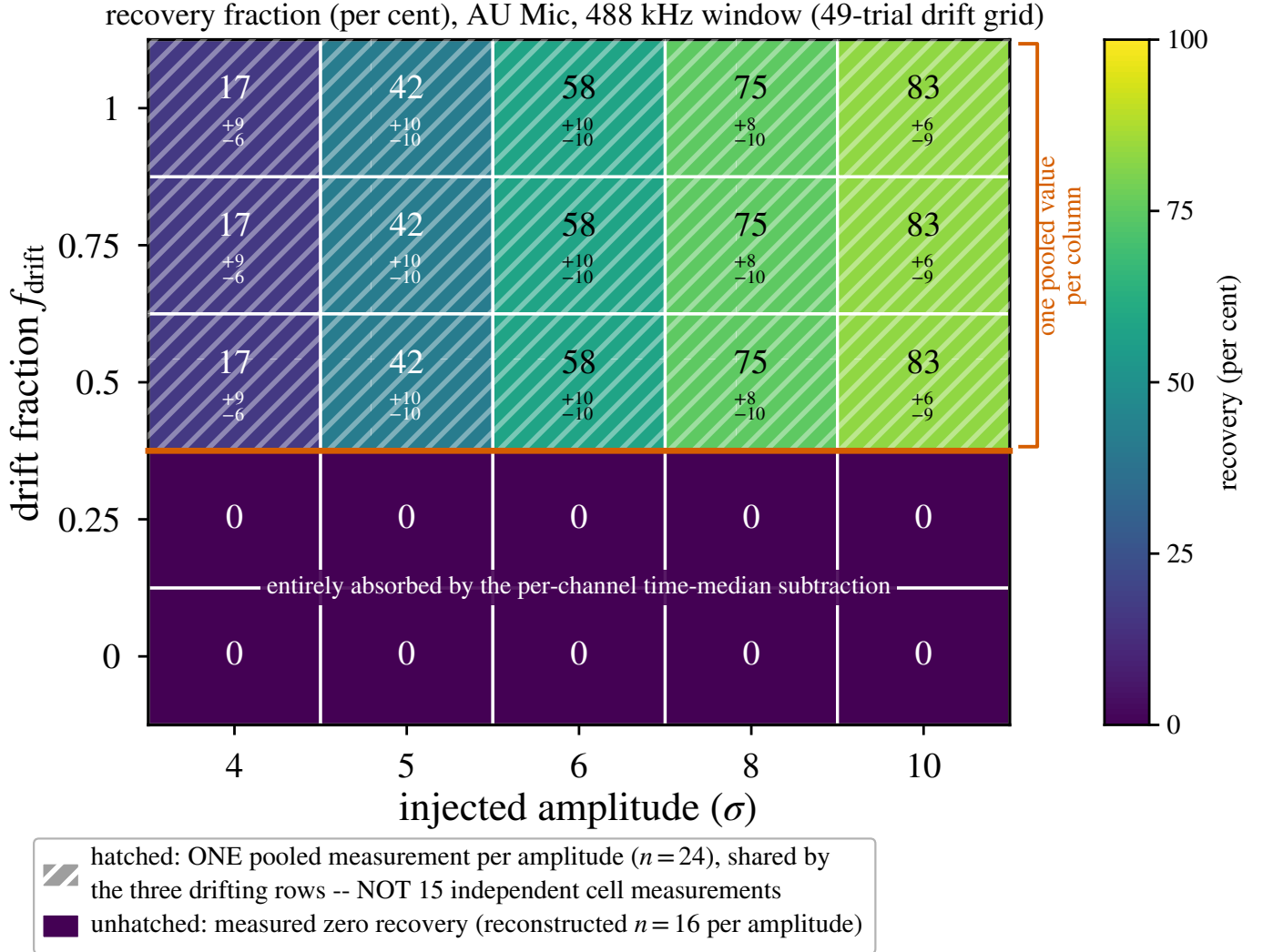
The EIRP thresholds of §5 are compared throughout to two human-transmitter benchmarks, both from Sheikh et al. (2025a): an Arecibo-like planetary radar (S-band  $\text{EIRP } 2 \times 10^{13}\text{ W}$ , the most powerful directed transmitter an Earth-equivalent civilisation could point at us, purely a technological-power benchmark), and unintentional Earth mobile-network leakage with no signalling intent (EIRP of order  $4 \times 10^9\text{ W}$ ).

#### 4.4. Ancillary and validation analyses

Four analyses beyond the narrowband search are executed here, each specified in full online (Table 14) and summarised only as far as the main text depends on it.

*Injection-recovery validation* (Appendix E). Synthetic tones of known amplitude were injected into real visibilities and recovered with the unmodified pipeline (1200 trials across six window configurations, three bands and both resolution classes). Per-channel time-median subtraction absorbs phase-steady carriers, totally so on the coarse ( $15.625\text{ MHz}$ ) windows (0 of 500 coarse injections recovered from  $4\sigma$  to  $10\sigma$ ), so coarse-window limits bound transmitters time-variable on track timescales, not phase-steady carriers. For the drifting class on the fine window recovery is 17, 42, 58, 75 and 83 per cent at 4, 5, 6, 8 and  $10\sigma$ , giving  $\text{EIRP}_{50} = 1.10\text{ EIRP}_{5\sigma}$  and, as a bound,  $\text{EIRP}_{90} > 2.0\text{ EIRP}_{5\sigma}$  (Fig. 12, Table 17). Tables here stay in nominal  $\text{EIRP}_{5\sigma}$ ; the occurrence limit of §6 is the one place measured completeness is folded in, and Band 8 completeness is transferred by shared construction, not measured.

*Drift-ceiling caveat.* The searched ceiling is a fixed  $12\text{ Hz s}^{-1}\text{ GHz}^{-1}$ , i.e. line-of-sight  $|a| = 3.6\text{ m s}^{-2}$ , uniform across every window: two orders of magnitude clear of Earth-rotation and Earth-orbit drifts, but 11 per cent *below* the  $4.0\text{ m s}^{-2}$  of a close-in planet on a TRAPPIST-1 b-like orbit. Carriers on the most rapidly accelerating bodies are therefore not fully covered, and the limits of §6 are conditional on that ceiling (Figure 14).



Trial-level records were not retained; errors are binomial Wilson 68 per cent intervals on the pooled class.

FIG. 4.— What the pipeline recovers, for the best-calibrated fine-channel configuration, against injected amplitude and drift fraction  $f_{\text{drift}}$  (injected drift as a fraction of the searched ceiling). Carriers with  $f_{\text{drift}} \leq 0.25$  are absorbed entirely by the time-median subtraction at every amplitude; those with  $f_{\text{drift}} \geq 0.5$  recover progressively, from 42–83 per cent between  $5\sigma$  and  $10\sigma$ . Trial-level records were not retained, so surviving fractions pool grid phase, sub-channel placement and the three drifting rows: the figure is drawn as discrete cells, not an interpolated surface, and each column’s hatched rows share one pooled measurement ( $n = 24$  per amplitude) rather than fifteen independent ones. Intervals are binomial Wilson 68 per cent; absorbed rows are measured zeros.

*Closure-phase vetting* (Appendix F). Threshold crossings face an automated point-source test unavailable to single-dish and beamformed surveys, needing multiple independent baselines; it is one-sided – evidence against celestial point-source behaviour, never an authentication. It is implemented and simulation-validated, but at this survey’s flux densities has *no discriminating power*: per-baseline S/N never exceeds 5.1 in the positive-control fields, so no flagged window could have failed it whatever its true nature. We report it as a proof-of-principle diagnostic for deeper releases, not an operative filter here, and describe no window as having “survived” it: the flagged windows are formally consistent with a point source, which here carries no evidential weight.

*Extended and cross-target screens* (Appendices G and H). A chirped (quadratic-drift) extension and a per-channel periodicity search ran on the 122 retained spectra as a demonstration, not a survey-grade screen; null behaviour is as expected (star-exceeds-ensemble rates 10.7 and 13.1 per cent against an 11-per-cent chance rate), no instance is a candidate, and no limit derives from them. Separately, crossings towards different targets were cross-matched for recurrence at the same sky frequency – the archival substitute for the ON/OFF cadence this single-pointing survey lacks. Of the 366 star pairs with overlapping searched ranges, none falls within the matching kernel, subject to one bookkeeping caveat (64 crossings of one  $\beta$  Pic window untested) in the appendix.

## 5. RESULTS

Strict volume ordering (nearest star first; §3) weights these results to the very nearest stars, spanning 1.3–19.6 pc. Per-target values, including each band’s searched frequency range, are online (Appendix I).



TABLE 3

THE SURVEY AS FROZEN FOR THIS RELEASE. EVERY COUNT QUOTED IN THIS PAPER RECONCILES WITH THIS TABLE; SUBSET NUMBERS ARE IDENTIFIED AS SUCH AT THE POINT OF USE. THRESHOLDS ARE NOMINAL (BOXED RULE, §4).

| Quantity                                        | Value                                                                             |
|-------------------------------------------------|-----------------------------------------------------------------------------------|
| Census stars within 40 pc                       | 168                                                                               |
| Stars searched; independent systems             | 85; 79                                                                            |
| Star/band datasets                              | 104 of 208 planned (50%)                                                          |
| Spectral windows searched                       | 417                                                                               |
| extracted; repeats, defective, withheld         | 448; 21, 6, 4                                                                     |
| fine (<5 MHz) / coarse channels                 | 111 / 306                                                                         |
| Execution blocks                                | 100                                                                               |
| Distance range                                  | 1.30–38.83 pc                                                                     |
| Unique frequency union                          | 92.4 GHz (34 intervals, 89.6–495.1 GHz)                                           |
| Effective after line-exclusion mask             | ≈87 GHz                                                                           |
| Native channel widths                           | 15.3 kHz–31.25 MHz (median 15.625 MHz)                                            |
| Drift-rate ceilings                             | 1.1–5.9 kHz s <sup>−1</sup> (12.0–13.3 Hz s <sup>−1</sup> GHz <sup>−1</sup> )     |
| Per-window on-source time                       | 21–5274 s (median 2177 s)                                                         |
| Exposure×bandwidth                              | 1.4 × 10 <sup>6</sup> s GHz                                                       |
| Nominal $S_{\min}$                              | 0.48 mJy–1.30 Jy (median 6.7 mJy)                                                 |
| Nominal EIRP <sub>5σ</sub>                      | 1.6 × 10 <sup>13</sup> –2.8 × 10 <sup>17</sup> W (median 1.3 × 10 <sup>15</sup> ) |
| Measured recovery at threshold (searched class) | 42 <sup>+16</sup> <sub>−14</sub> %                                                |
| Hits (≥ 5σ at the stellar position)             | 18 windows                                                                        |
| Spatially significant hits                      | 4 windows                                                                         |
| circumstellar or foreground CO                  | 3                                                                                 |
| consistent with the control ensemble            | 1                                                                                 |
| Credible technosignatures                       | 0                                                                                 |

#### 5.1. Per-star summary

Table 23 (online-only) gives the one-line-per-star view complementing the per-window file (Appendix I); two stars of the release are continuum-only. All values are read from that file: EIRP values are nominal native-channel thresholds (boxed rule, §4), continuum fluxes primary-beam corrected.

#### 5.2. Barnard’s Star and Wolf 359

Barnard’s Star and Wolf 359 – the second- and fourth-nearest stellar systems – have not, to our knowledge, previously been searched for technosignatures at ALMA frequencies. Both yield non-detections from public Band 6 archival data: EIRP<sub>5σ</sub> limits of 6.15–6.89 × 10<sup>13</sup> W (Barnard’s Star, 4 windows) and 7.84–9.10 × 10<sup>13</sup> W (Wolf 359), with continuum upper limits of 0.575 and 0.453 mJy.

#### 5.3. Data-quality exclusions and open items

Of the 208 planned target/bands, 104 are searched to completion and reported here, the rest queued for later releases bar a few failures excluded from all counts. Four classes of entry within the 104 are absent from the released per-target file (Appendix I) and itemised with forensics in Appendix R: (i) four process-level search failures, each still carrying a continuum non-detection; (ii) one pair excluded for data quality; (iii) windows partially withheld for a noise-handling defect; and (iv) one relabelled entry.

A reproducible defect produces implausibly small noise in a minority of coarse-channel windows; naming the two affected did not generalise as the sample grew, so item (iii) is now caught by a physical test. In a correctly weighted measurement  $\sigma\sqrt{t_{\text{on}}\Delta\nu_{\text{ch}}}$  is set by system temperature and collecting area and varies by factors of a few across a heterogeneous archive; we exclude windows more than 100× below the sample median. Six of 417 windows fail, across five stars, a factor 1.5 × 10<sup>3</sup> below the lowest retained across an empty gap of 3.2 decades (Figure 5); the threshold is not fine-tuned, any value between 1.5 × 10<sup>1</sup> and 2.2 × 10<sup>4</sup> selecting the same six. Their nominal thresholds, 2.0–7.8 × 10<sup>12</sup> W, would otherwise have set this paper’s headline depth; excluding them, the best is 1.6 × 10<sup>13</sup> W. 21 of the 448 extracted rows repeat a (star, execution block, frequency range, channel width) key, and we keep one per key. Most repeats are identical, but two are not – two reductions of the same  $\tau$  Ceti Band 6 window differ by 44 per cent in threshold, and the two HR 1010 Band 6 rows straddle the 5σ threshold (5.10 and 4.95) – so the hit count is rule-dependent at the level of one window. The four spatially significant windows are not: no discarded row is both a hit and star-beats-ring, and every row, kept or discarded, ships with its status. The audit is otherwise clean apart from four windows of elevated system temperature, and every census identifier re-resolves independently in SIMBAD and *Gaia* (Appendix R).

#### 5.4. Technosignature search

All 104 star/band datasets of the searched sample yielded at least one narrowband result (process-level failures, §5.3, uncounted), spanning 417 spectral-window measurements after 21 duplicate rows were removed in the catalogue audit (Appendix I). EIRP<sub>5σ</sub> limits span 1.6 × 10<sup>13</sup>–2.8 × 10<sup>17</sup> W, median 1.3 × 10<sup>15</sup> W (Figures 8, 2 and 1); as expected for a volume-ordered survey, the deepest are the nearest systems’ (1.6 × 10<sup>13</sup> W for the UV Ceti pair at 2.7 pc), the

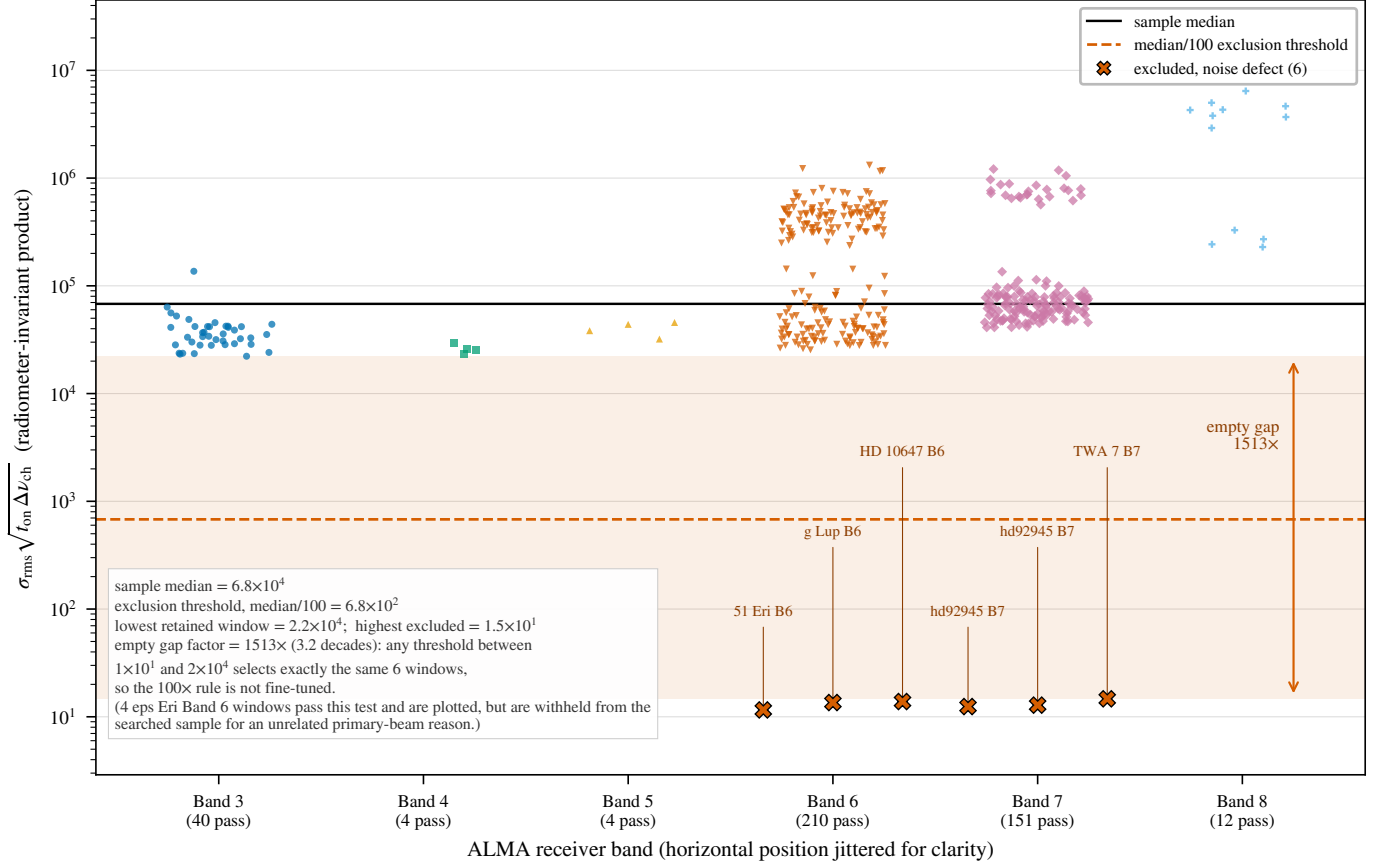


FIG. 5.— Quality control on the noise estimate (§5.3). The radiometer-invariant product  $\sigma \sqrt{t_{\text{on}} \Delta \nu_{\text{ch}}}$  varies by factors of a few across a heterogeneous archive; the six excluded windows lie a factor  $1.5 \times 10^3$  below the lowest retained, across an empty gap.

TABLE 4  
THE NARROWBAND DECISION CASCADE OF THIS RELEASE, FOR THE 417 WINDOWS SEARCHED AFTER QUALITY CONTROL (§5.3); VOCABULARY AS IN TABLE 2.

| Stage                                                                     | Count              |
|---------------------------------------------------------------------------|--------------------|
| Channel $\times$ drift trial cells (all windows)                          | $3.97 \times 10^7$ |
| Control positions compared per window                                     | 512                |
| Windows searched after quality control                                    | 417                |
| Windows containing $\geq 1$ hit ( $\geq 5\sigma$ at the stellar position) | 18                 |
| Windows containing a spatially significant hit                            | 4                  |
| attributed to circumstellar or foreground CO                              | 3                  |
| indistinguishable from the control ensemble                               | 1                  |
| Candidates (surviving every veto)                                         | 0                  |

shallowest  $\eta$  Corvi, the first Band 8 target searched, where distance, coarse channels and receiver performance set the ceiling. Against the benchmarks of §4.3, only one bound system – the two resolved components of G 272-61 at 2.7 pc – reaches below the Arecibo-like planetary-radar EIRP, and none approaches the  $\sim 4 \times 10^9$  W of unintentional Earth leakage.

Four windows contain a *spatially significant hit*; none survives vetting, so this release reports *zero candidates*. Table 4 follows the cascade, Table 5 each flagged window and the evidence dispositioning it.

Do four spatially significant hits exceed this survey’s own calibration (a counting question), and is any distinguishable from the empirical background or known astrophysical emission (the dispositions below)? Both answers are no.

#### SURVEY-LEVEL FALSE-ALARM CALIBRATION

Each window-level test compares the peak signal-to-noise ratio at the star’s position with the same statistic at 512 controls; the drift-rate grid and thousands of channels apply identically to both, so it absorbs the per-window trials factor empirically. The statistic is now *symmetric*: the on-source quantity is the single propagated stellar position, as each control is. Earlier versions maximised over a small region around it while controls stayed single, an asymmetry that was not benign: under a pure-noise null with exchangeable positions the maximum over  $n_{\text{src}}$  source-region positions beats that over  $n_{\text{ctrl}}$  single controls with probability  $n_{\text{src}}/(n_{\text{src}} + n_{\text{ctrl}})$ , 21 per cent for a typical window’s  $n_{\text{src}} = 135$  region and 512 controls – the order of the measured region-max star-beats-ring rate, 64 of 417 windows (15.3 per cent;

TABLE 5

THE FOUR WINDOWS CONTAINING A SPATIALLY SIGNIFICANT HIT AND THE EVIDENCE DISPOSITIONING EACH.  $T_*$  IS THE SEARCH STATISTIC AT THE STELLAR POSITION (Eq. 1), ‘RING MAX’ THE LARGEST OF THAT WINDOW’S 512 CONTROL STATISTICS, AND  $p$  THE ADD-ONE RANK PROBABILITY, FLOOR  $1/513 = 0.0019$ . VELOCITY OFFSETS ARE TOPOCENTRIC, FROM THE NEAREST OF THE EIGHT MASKED SPECIES’ CATALOGUED TRANSITIONS; THE  $\pm 50 \text{ km s}^{-1}$  MASK APPLIES IN THE STELLAR FRAME AFTER THE CONVERSION CHAIN OF APPENDIX C. NO DISPOSITION RESTS ON A SINGLE TEST.

| Window         | $\nu$ (GHz) | $T_*$ | ring max | $\Delta v$ (km s $^{-1}$ ) | Disposition                 |
|----------------|-------------|-------|----------|----------------------------|-----------------------------|
| $\beta$ Pic B3 | 115.2605    | 14.65 | 7.27     | −26.9 (CO 1→0)             | circumstellar CO            |
| $\beta$ Pic B6 | 230.5164    | 11.68 | 9.12     | −28.4 (CO 2→1)             | circumstellar CO            |
| HD 48370 B6    | 230.7305    | 27.10 | 26.83    | −34.2 (CO 2→1)             | CO cloud emission           |
| CP−72 2713 B7  | 345.1152    | 5.81  | 5.68     | −1326 (CO 3→2)             | control-ensemble background |

Wilson 95 per cent interval 12.2–19.1). Symmetrically, star-beats-ring occurs in 5 of 417 windows (1.2 per cent; Wilson interval 0.5–2.8), consistent with the exchangeable expectation  $1/513 = 0.19$  per cent per window inflated by the real maps’ heavy tails, and stable on the  $\leq 20$  pc subsample alone. That rate is thus operative, the survey-level false-alarm expectation following from it directly rather than through the bookkeeping ladder the asymmetry required.

The joint criterion gating a candidate is dominated by its threshold leg: only 18 of 417 windows contain any  $\geq 5\sigma$  crossing at the stellar position. Each contributes at most the exchangeable rate  $1/(1+n_{\text{ctrl}}) = 1/513$ , so the pure-noise null predicts  $417/513 = 0.81$  flags. Four flagged windows exceed that budget ( $P(\geq 4|0.81) = 0.9$  per cent), but with no tension: three are identified astrophysical emission, never chance events, so the budget need explain only the residual – one unattributed crossing against 0.78 expected across the 402 non- $\beta$  Pictoris windows, the order of the expectation. This operative estimate absorbs execution-block systematics into the fixed block margins, assuming only that crossing and star-beats-ring status are exchangeable within a block. Secondary bookkeeping models span 1.0–9.2 per cent (Appendix P); the dominant residual uncertainty is the star-beats-ring rate (Wilson 95 per cent interval 12.2–19.1 per cent), giving a 4–19 per cent bracket on  $P(F \geq 3)$ .

One ensemble limit follows: with 512 controls the smallest resolvable per-window  $p$ -value is  $1/513$ , so a Bonferroni threshold over 417 windows ( $1.1 \times 10^{-4}$ ) lies below what the controls can measure. They can show a crossing unremarkable but cannot certify one as significant survey-wide – a burden falling on the localisation, recurrence and astrophysical attribution that decide each flag below. Stated in advance for later releases: each contributes  $\sim N/513$  chance flags for  $N$  windows, so a programme reaching the full  $\sim 2000$ -window census expects  $\sim 4$ , and one unattributed marginal crossing per release is the *expected* outcome, not an accumulating anomaly. We therefore pre-commit that no crossing in this programme will be advanced as a candidate on rank statistics alone: promotion requires evidence the ensemble cannot supply – recurrence at an independent epoch, or a ring enlarged until the rank probability resolves survey-wide significance ( $\gtrsim 10^4$  positions for a Bonferroni threshold over the completed census; §6.2). Until then every such crossing is reported, dispositioned as consistent with chance, and released with its products. No model in the bracket promotes a flagged window: the three CO dispositions rest on kinematic attribution to emission resolved, mapped or independently published, and the CP−72 2713 standing is a rank statement invariant to the model.

*Is the ring exchangeable with the star?* The  $1/513$  expectation holds only if the star and its 512 controls are exchangeable under the null, which spatially varying noise, sidelobes, correlated pixels or phase-centre-localised calibration residuals could spoil; we measure it by stratum (Fig. 6). The median per-window control maximum is flat across bands – 4.46, 4.59, 4.42, 4.52, 4.63 and 4.69 in Bands 3 to 8 – and a Kolmogorov–Smirnov test of the star’s rank among its controls against  $U(0,1)$  is consistent in every band ( $p = 0.14$ – $0.83$ ) and the coarse-channel stratum ( $p = 0.71$ ). Fine-channel windows sit higher (control maximum 5.76 against 4.42), as expected from their far larger channel $\times$ drift trial count; not itself a violation, the test being conditional on the window.

Those tests are conditional on the stellar rank; for the stronger question – exchangeability of the ensemble itself – we ranked each of a window’s 512 controls against the other 512-1 by the statistic and add-one rule applied to the star, pooled over the release. The 213 504 pseudo-star ranks are uniform to the resolution the ensemble can express (mean 0.5010 against 0.5; the Kolmogorov–Smirnov  $D = 1/512$  is the discreteness floor of a 512-point rank statistic,  $p = 0.39$ ), in every band and at both channelisations, and a two-sample comparison of the 417 real stellar ranks with that pooled distribution gives  $p = 0.31$ : the stellar position behaves like one more position in its own ring – the null calibration the control-ring argument needs, measured not assumed.

One stratum does depart, reported rather than pooled away: the fine-channel windows fail the same  $U(0,1)$  test at  $p = 0.018$  ( $n = 111$ ), containing four windows whose on-star peak exceeds their own control maximum against 0.22 expected – precisely the four flagged windows of Table 5. That is astrophysical signal concentrated where channels are finest and narrow lines detectable, not a misbehaving ring; but the stratum-level departure from exchangeability is genuine, so fine-window rank probabilities are slightly optimistic. Two stratifications the retained products cannot support, storing no per-control radial position or primary-beam gain: with the ring radius small compared with the beam (targets lie a median 0.1 arcsec from the pointing centre) all controls should share the target’s response, but we have not measured that and say so. Remaining robustness checks – trials-count reconciliation under channel adjacency, control-ring geometry – are in Appendix P.

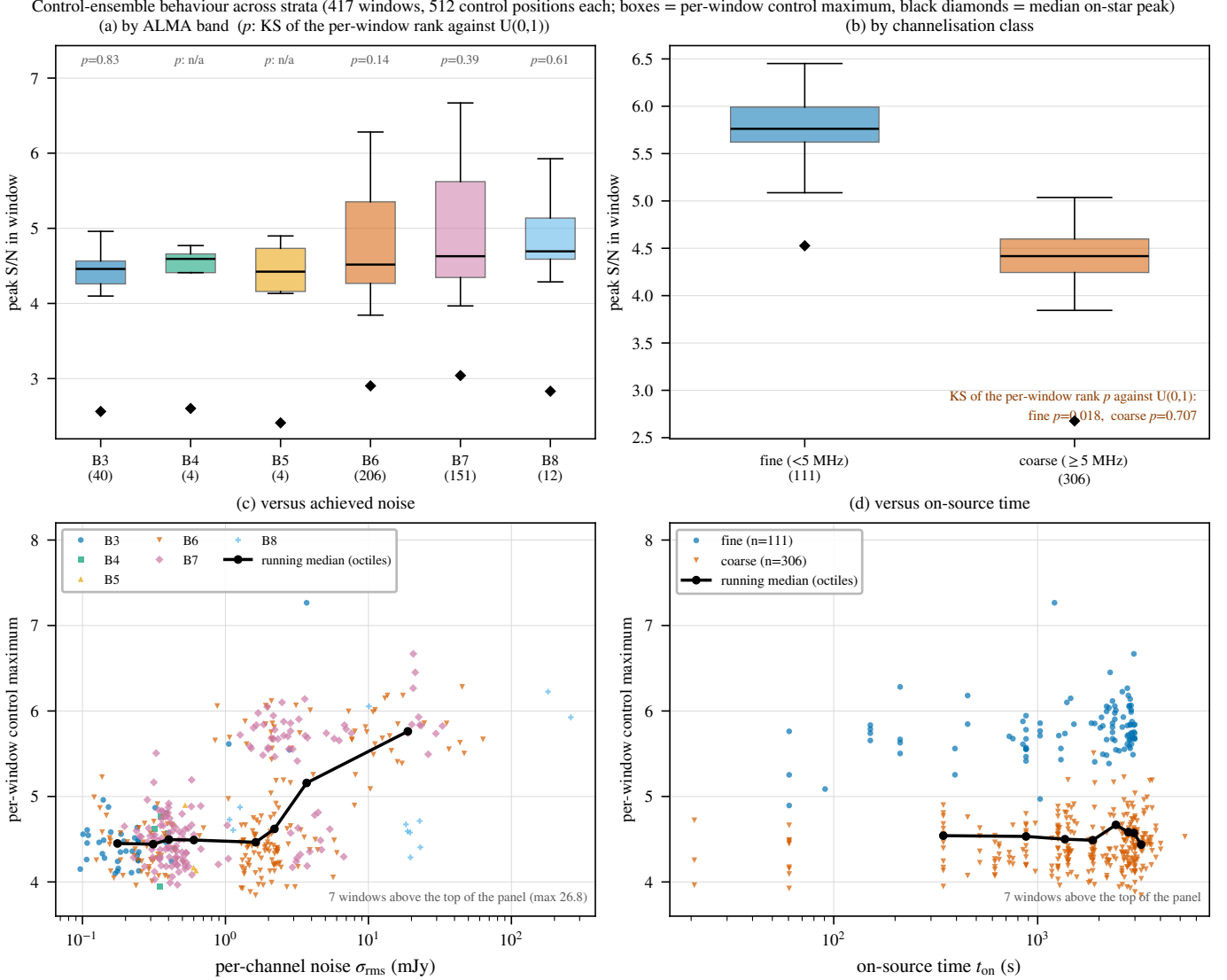


FIG. 6.— Exchangeability of the 512-position control ring by stratum. Boxes give the per-window control maximum, diamonds the median on-star peak,  $p$  the Kolmogorov-Smirnov test in the text. The ensemble tracks the on-star statistic in every band and at both channelisations; the one departure is the fine-channel stratum ( $p = 0.018$ ), driven by the four flagged windows of Table 5.

The designed principal test applies identical source-region, drift-rate and frequency maximisation at the star's position and at each of the 512 controls under one search rule. The frozen release products cannot meet that, so its domain is the seven locally reprocessed windows – six validation, plus the AU Mic candidate's first-epoch block – all giving  $T_{\star} = 3.7$ – $4.9$  against control maxima of  $6.7$ – $7.7$ ,  $p = 0.81$ – $1.00$  (Figure 11, online-only appendix): the star is exchangeable with its ring wherever the test runs at full symmetry. Extending it to all 417 windows needs per-window raw-visibility reprocessing the retained products do not support, so the empirical star-versus-ring calibration above remains the operative release-wide estimate, the symmetric statistic a validation subset. The machinery ships with the release; the full statement, with the family-wise trials factors of Table 16, is in Appendix P.

#### DISPOSITIONS OF THE FOUR FLAGGED WINDOWS

Two of the four are towards  $\beta$  Pictoris, a debris disc with reported CO emission (Matrà et al. 2017): one in Band 3 at 115.2606 GHz, 10.56 MHz below the CO( $J=1 \rightarrow 0$ ) rest frequency, the other in Band 6 at 230.5161 GHz, 21.87 MHz below CO( $J=2 \rightarrow 1$ ). Neither coincidence is exact, so the automated check missed them. Through the frame chain (Table 6; Fig. 7) the offsets are one kinematic component –  $-2.7$  and  $-3.5$  km s $^{-1}$  from stellar systemic,  $0.74$  km s $^{-1}$  apart – in the two lowest CO rotational transitions of a system with detected circumstellar CO; the coarse Band 6 window without a candidate, from a third epoch six years earlier, independently shows CO at  $-3.6$  km s $^{-1}$  in the same frame. All three epochs and both transitions land within  $0.9$  km s $^{-1}$  of each other and  $3.6$  km s $^{-1}$  of systemic, deep inside the  $\pm 50$  km s $^{-1}$  mask. The frame chain matters: raw topocentric offsets are  $-27.5$  and  $-28.4$  km s $^{-1}$ . We attribute both to this kinematically offset CO.

Their empirical significance under the release background is nonetheless extreme, so the attribution is no dismissal: the Band 3 region peak of 20.3 exceeds all 417 per-window control maxima (add-one  $p < 0.0044$ ), the Band 6 peak of

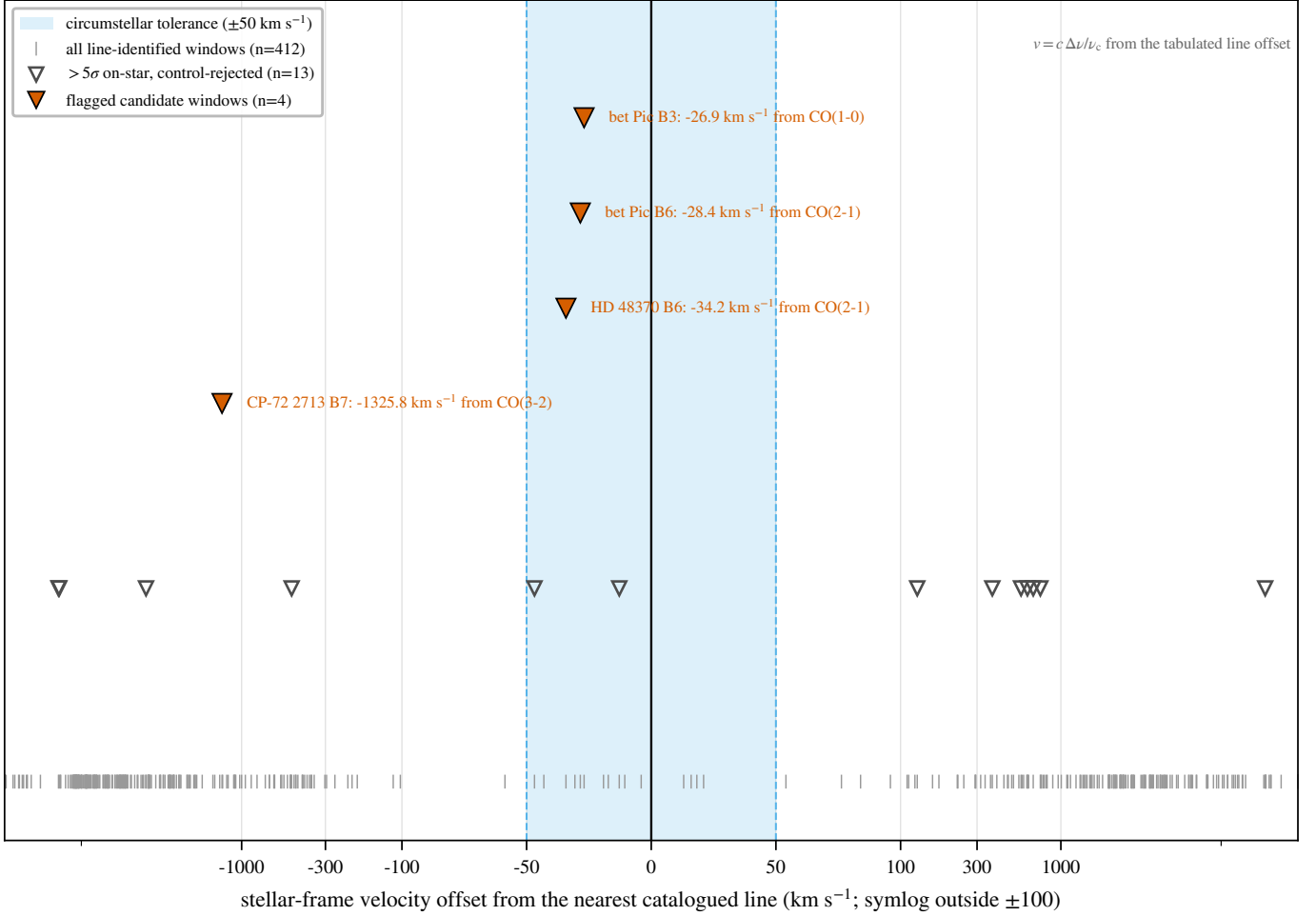


FIG. 7.— Stellar-frame velocity offsets of the candidate windows, through the frame chain (topocentric  $\rightarrow$  barycentric  $\rightarrow$  stellar; Table 6). Shaded band: the  $\pm 50 \text{ km s}^{-1}$  molecular-line exclusion mask. *Top*: the two  $\beta$  Pictoris candidates fall well inside the mask as one kinematic component, resolved in the inset. *Bottom*: the flagged window towards CP-72 2713 and the AU Mic crossing the symmetric statistic does not flag (§5.5) lie far outside any circumstellar tolerance, dispositioned by their spatial behaviour, not the mask.

12.8 all but one – that window’s own CO-filled control ring, reaching 15.2, by far the release maximum ( $p \approx 0.0088$ ). Standing out this sharply is why they demanded astrophysical attribution rather than rejection as noise, obtained by inspecting the offsets against known CO kinematics; the alternative needs a transmitter coincidentally placed at matched offsets from two CO transitions in the same star.

The third, towards HD 48370 in Band 6, is the release’s largest excess in absolute and smallest in relative terms:  $T_{\star} = 27.10$  against a control-ring maximum of 26.83, the star exceeding its ring by one per cent of the statistic. Star and ring rising together is what emission filling the primary beam looks like, and the window sits  $-34.2 \text{ km s}^{-1}$  topocentric from CO( $J=2 \rightarrow 1$ ). The attribution is independently published: Cataldi et al. (2023) report prominent CO( $2 \rightarrow 1$ ) and  $^{13}\text{CO}(2 \rightarrow 1)$  emission towards HD 48370 at  $v_{\text{bary}} \approx 41 \text{ km s}^{-1}$ , well separated from its  $23.6 \text{ km s}^{-1}$  systemic velocity and attributed by them to foreground cloud contamination; our window recovers it. It is also a stated limitation of the spatial test: this window is identified by the velocity coincidence and near-equality of star and ring, not the rank statistic.

The fourth, towards CP-72 2713 in Band 7, is the only flagged window with no astrophysical attribution, and is marginal:  $T_{\star} = 5.81$  against a ring maximum of 5.68, a margin of 0.13 attaining only the rank-probability floor  $p = 1/513$ , the smallest value 512 controls can express and not a measurement of significance. That margin is smaller than the scatter across this star’s own windows, whose other fine-channel control maxima are 5.16, 5.41, 5.67 and 5.85: its rings reach 5.81 routinely where the star does not cross threshold. The nearest transition of the eight masked species lies  $1326 \text{ km s}^{-1}$  away; since that mask is an automated contamination cut, not a complete line catalogue, and a chemically rich or circumstellar environment can present many further ALMA-band transitions, we re-queried a wider catalogue, whose closest entry is  $\text{H}^{13}\text{CN}(J=4 \rightarrow 3)$  at  $-195 \text{ km s}^{-1}$ , still far outside any circumstellar tolerance. The defensible reading is “no plausible known transition identified”, a statement about the catalogues consulted and astrophysical context, not a proof that none exists. With a single execution block at this frequency no recurrence test is possible. We report a marginal crossing consistent with the 0.81 flags expected by chance across the release, attach no technosignature interpretation, and note that a recurrence observation would settle it.

TABLE 6

LINE-ATTRIBUTION AUDIT OF THE THREE CANDIDATE WINDOWS (FIRST THREE ROWS) AND THE COARSE  $\beta$  PICTORIS BAND 6 WINDOW WITHOUT A CANDIDATE SHOWING THE SAME CO COMPONENT, THROUGH THE FRAME CHAIN OF §4.2. TOPOCENTRIC OFFSETS ARE OBSERVED MINUS CATALOGUED REST FREQUENCY (CO  $J=1\rightarrow0$  115.271202 GHz, CO  $J=2\rightarrow1$  230.538000 GHz, PICKETT ET AL. 1998); THE BARYCENTRIC CORRECTION IS THE EARTH-EPHEMERIS TERM AT THE EXECUTION-BLOCK MID-TIME (BLOCK UIDS, TABLE 10); SYSTEMIC RADIAL VELOCITIES ARE CATALOGUED (SIMBAD; *Gaia* DR3 FOR  $\beta$  PICTORIS, GAIA COLLABORATION ET AL. 2022; DR2 FOR AU MIC, FOUQUÉ ET AL. 2018).

| Flag                        | UTC start        | $f_{\text{obs}}$ (GHz) | Transition            | Topo. offset<br>(MHz; $\text{km s}^{-1}$ ) | Bary. corr.<br>( $\text{km s}^{-1}$ ) | $v_{\text{sys}}$<br>( $\text{km s}^{-1}$ ) | Stellar-frame<br>offset ( $\text{km s}^{-1}$ ) |
|-----------------------------|------------------|------------------------|-----------------------|--------------------------------------------|---------------------------------------|--------------------------------------------|------------------------------------------------|
| $\beta$ Pic B3              | 2022-03-02 23:14 | 115.260644             | CO(1 $\rightarrow$ 0) | −10.56; −27.46                             | −7.90                                 | +16.84                                     | −2.72                                          |
| $\beta$ Pic B6              | 2019-03-12 00:14 | 230.516128             | CO(2 $\rightarrow$ 1) | −21.87; −28.44                             | −8.15                                 | +16.84                                     | −3.45                                          |
| AU Mic B6                   | 2014-08-18 04:01 | 230.825230             | CO(2 $\rightarrow$ 1) | +287.23; +373.51                           | −10.02                                | −4.71                                      | +378.82                                        |
| $\beta$ Pic B6 <sup>†</sup> | 2013-10-06 06:23 | 230.528134             | CO(2 $\rightarrow$ 1) | −9.87; −12.83                              | +7.63                                 | +16.84                                     | −3.62                                          |

<sup>†</sup>Coarse Band 6 window without a candidate (Appendix I), shown because it independently shows the same CO component as that band’s candidate fine window.

### LINE EXCLUSION: DEFINITION, COST, AND LIMITS

The searches used an automated line check matching crossings to catalogued rest frequencies within one spectral channel at zero drift. The  $\beta$  Pictoris candidates showed that inadequate: both fell outside its one-channel tolerance (by 10.6 and 21.9 MHz) while lying, after the frame chain, within  $3.6 \text{ km s}^{-1}$  of stellar systemic of a resolved, mapped disc – exactly the regime a channel-width tolerance is blind to. Every crossing was therefore reclassified before publication with a velocity-space mask ( $\pm 50 \text{ km s}^{-1}$  in the stellar frame around each catalogued transition), the pipeline’s line-exclusion definition from this release on. Its kinematic justification, empirical conservatism, that reclassification, and the “Schelling point” counter-argument that a communicator might deliberately choose these frequencies are in Appendix J; retrospectively the mask rejects exactly the two  $\beta$  Pictoris candidates, changing no other disposition.

The masking cost uses one definition: every JPL catalogued transition of the eight species (276), displaced by each system’s radial velocity,  $\pm 50 \text{ km s}^{-1}$  stellar-frame half-width, intersected with each window and merged. The release effectively searches 87.0 GHz of unique sky frequency – the disjoint union of searched windows minus the mask inside it (Figure 2) – down from 92.4 GHz, a 5.7 per cent reduction; over the 417 windows it removes 41.4 GHz of 700.8 GHz gross bandwidth (5.9 per cent; per-species, per-band breakdown in Table 22, online-only appendix), the per-window usable fraction having median 0.960 (10th percentile 0.790). Union bandwidth is exposure-blind, so we also report the summed per-star union, 371.9 GHz, and the summed product of on-source time and searched bandwidth,  $1.4 \times 10^6 \text{ s GHz}$  (on-source time 12.1–5274 s per window, median 2570 s). The mask governs dispositions, not the noise-derived thresholds, so the quoted EIRP<sub>5 $\sigma$</sub>  values are unchanged; no signal was found outside the excluded velocity regions in any window.

The mask is a design selection effect: a signal coincident with a molecular line is not thereby “false” but *excluded from the technosignature search domain*, the astrophysical prior there being overwhelming and this pipeline unable to separate the two. No constraint is placed on transmitters within the excluded regions – 5.9 per cent of the searched bandwidth – and a search for deliberately line-adjacent carriers would need a complementary strategy. Nor is the exposure confined to the candidate windows: 131 of 417 windows (58 per cent) contain an in-band catalogued transition of the eight species, as do all four flagged windows, yet in none does a crossing fall within one native channel of a rest frequency – precisely the regime the velocity-space criterion covers.

#### 5.5. Changing the detection statistic, and the AU Mic crossing

The initial region-max statistic maximised over an  $n_{\text{src}} = 135$ -position region centred on the star against 512 *single-position* controls; under it a crossing towards AU Mic in Band 6 at 230.83 GHz was carried as a candidate. A criterion revised after inspecting the object it must judge invites distrust, so we set out the sequence and evidence explicitly.

The defect is a property of the statistic, demonstrable without reference to any target: the region-max asymmetry raises the star-beats-ring probability under a pure-noise null to 20.9 per cent (§5.4) against  $1/513 = 0.19$  per cent for a symmetric test, and the measured release-wide region-max rate, 15.3 per cent, is of that order – neither quantity involves a candidate. We replaced it with the symmetric single-position form of Eq. 1, applied identically to star and controls, and re-ran the release.

The change is a bias correction, not a loss of sensitivity, and is checkable. Across the 417 searched windows the symmetric criterion flags a strict *subset* of the region-max set: seven were flagged before, four now, and the three that dropped out (towards ALMA J1537–3319, AU Mic and HD 14055) all have the *stellar position itself never exceeding its own control ring* – star statistics of 5.09, 5.22 and 5.22 against ring maxima of 5.63, 5.85 and 5.74. No window in which the star dominated its controls was removed – the only configuration a localised transmitter can produce, since a point source at the stellar position raises  $T_*$ , the quantity the new test uses. Consistently, the injection campaign recovers signals through the unmodified pipeline on the star position (§4.4), so its completeness is a property of the symmetric statistic as reported.

Under the symmetric criterion the AU Mic crossing is not flagged: 10 of its 512 controls reach or exceed the stellar statistic, an empirical  $p = 0.021$  far from the  $1/513$  floor. It also fails two direct, pre-named tests: it is not localised at the stellar position and does not recur at a second epoch (Table 7).

TABLE 7

THE AU MIC CANDIDATE FREQUENCY (230.8252 GHz) UNDER THE THREE TREATMENTS THAT DECIDE IT: THE RELEASE’S FROZEN PRODUCTS (REGION-MAX STATISTIC), THE FULLY SYMMETRIC REPROCESSING OF THE SAME FIRST-EPOCH BLOCK (2014 AUGUST 18, OBSERVATORY CALIBRATION RECIPE VERBATIM) AND THE SECOND-EPOCH BLOCK OF THE SAME PROGRAMME (2014 JUNE 5, IDENTICAL SEARCH TREATMENT).

| Statistic                         | Release<br>(region-max) | Symmetric<br>reprocess. | Second epoch<br>(recurrence) |
|-----------------------------------|-------------------------|-------------------------|------------------------------|
| Star statistic                    | 5.97                    | 4.93                    | 4.77                         |
| Control maximum (512)             | 5.85                    | 7.70                    | 6.78                         |
| Star rank $p$ (add-one)           | —                       | 0.81                    | 0.97                         |
| Single-position star statistic    | 5.22                    | —                       | —                            |
| Per-channel S/N (zero drift)      | —                       | 2.46                    | 2.22                         |
| Peak drift ( $\text{Hz s}^{-1}$ ) | 1418.5                  | 1593                    | 1281                         |

Localisation, recurrence and statistical scale are in Appendix K, with the numbers of Table 7. Neither the reprocessed first epoch nor the one further public Band 6 block covering 230.83 GHz shows a localised excess at the stellar position, and that second epoch is a clean non-recurrence – rejecting a continuous or repeating source at this frequency and these epochs, though not intermittent transmission.

The archival epochs also bound what this release can say about duty cycle: each star was observed in one to three execution blocks, searched independently with no stacking, so the non-detection concerns the epochs observed, not permanent activity. The surplus of available over searched blocks (§6) is the recurrence pool the second-epoch test drew on, and §6 turns the epoch structure into an explicit duty-cycle constraint.

### 5.6. Ancillary results

Three ancillary results are reported in full online.

*Integration-time audit* (Appendix Q). On-source integrations were reconstructed from pipeline bookkeeping, not retained natively, so all 227 window-level records were audited against the archive’s exposure accounting. Two of the 58 star–execution-block groups over-count – AT Mic by a factor 1.43, an internally diagnostic dump-bookkeeping artefact already corrected to the archive’s 877 s in Table 21, and CD–38 10980 by 7 per cent; the rest are conservative. No EIRP threshold depends on it, thresholds deriving from the measured per-channel rms rather than integration time; the audit is conservative for every window of both candidate hosts, so no disposition changes; and the exposure-weighted coverage of §5.4 is at worst optimistic by under 1 per cent. The class was found by chance in one target, hence the direct per-execution-block export committed for the next release.

*Continuum search* (Appendix M). The continuum lane has run over the 57 target/bands processed far enough for a usable measurement – a subset of the narrowband sample, and the one count here referring to fewer than the 417 searched windows. Seventeen show a  $\geq 5\sigma$  detection, forty are non-detections reaching 0.036–3.6 mJy (median 0.52 mJy); every detection is astrophysically unsurprising, the auroral emitter LSR J1835+3259 meriting deeper follow-up. The lane is exploratory, its statistical power unquantified at this depth, and no technosignature disposition rests on it.

*Serendipitous molecular-line catalogue* (Appendix N). The line-exclusion step identifies every  $\geq 5\sigma$  outlier channel as a byproduct; across the subset scanned, 18 groups were found, six plausibly real. All are reported openly, including the two we think probably spurious.

## 6. DISCUSSION

This first stand-alone processed release is preliminary by construction: 104 of 208 planned target/bands (50 per cent) are complete, and that subset is sensitivity-ranked – nearest, most sensitively-constrained systems first – so it is not an unbiased draw, and the full sample will reach larger distances and shallower faint-end limits. No credible technosignature has been found; the EIRP $_{5\sigma}$  thresholds of Appendix I bound any transmitter above them. Two candidates towards  $\beta$  Pic are attributable to CO disc emission, underlining, as Mason et al. (2024) highlight, the need to check mm-band crossings against known molecular transitions; that check is automated for every crossing (§4) and now uses a velocity-space match applied retroactively to every window (§5.4). The third, marginal candidate (AU Mic, §5.5) is not statistically distinguishable from the empirical background, and is recorded for re-examination rather than dismissed.

The searched space is stated on one definition throughout (§5.4, Figure 2). Of the 37 planets known around the 15 planet-hosting targets only TRAPPIST-1 b reaches the drift ceiling, in its edge-on worst case at  $\sim 13 \text{ Hz s}^{-1} \text{ GHz}^{-1}$ ; GJ 581 e, 61 Vir b, and TRAPPIST-1 c reach roughly half, the remaining 33 well inside the searched grid.<sup>5</sup>

Table 3 (§5) consolidates the axes along which a transmitter is or is not detectable; Table 20 (online-only appendix) decomposes the release by ALMA band.

Coverage eligibility and search completeness differ. The census of public usable ALMA data (§3) sets eligibility; every held execution block in scope was searched, so no EB-level selection lies behind the 57. Availability is measured directly, not inherited from release bookkeeping: a proper-motion-matched ALMA archive census (2026 September 9;

<sup>5</sup> Star–planet relative accelerations  $GM_{\star}/r^2$  from NASA Exoplanet Archive planetary-companion parameters (Mission Star/Planet table, accessed 2026 September), maximised over orbital phase and inclination.

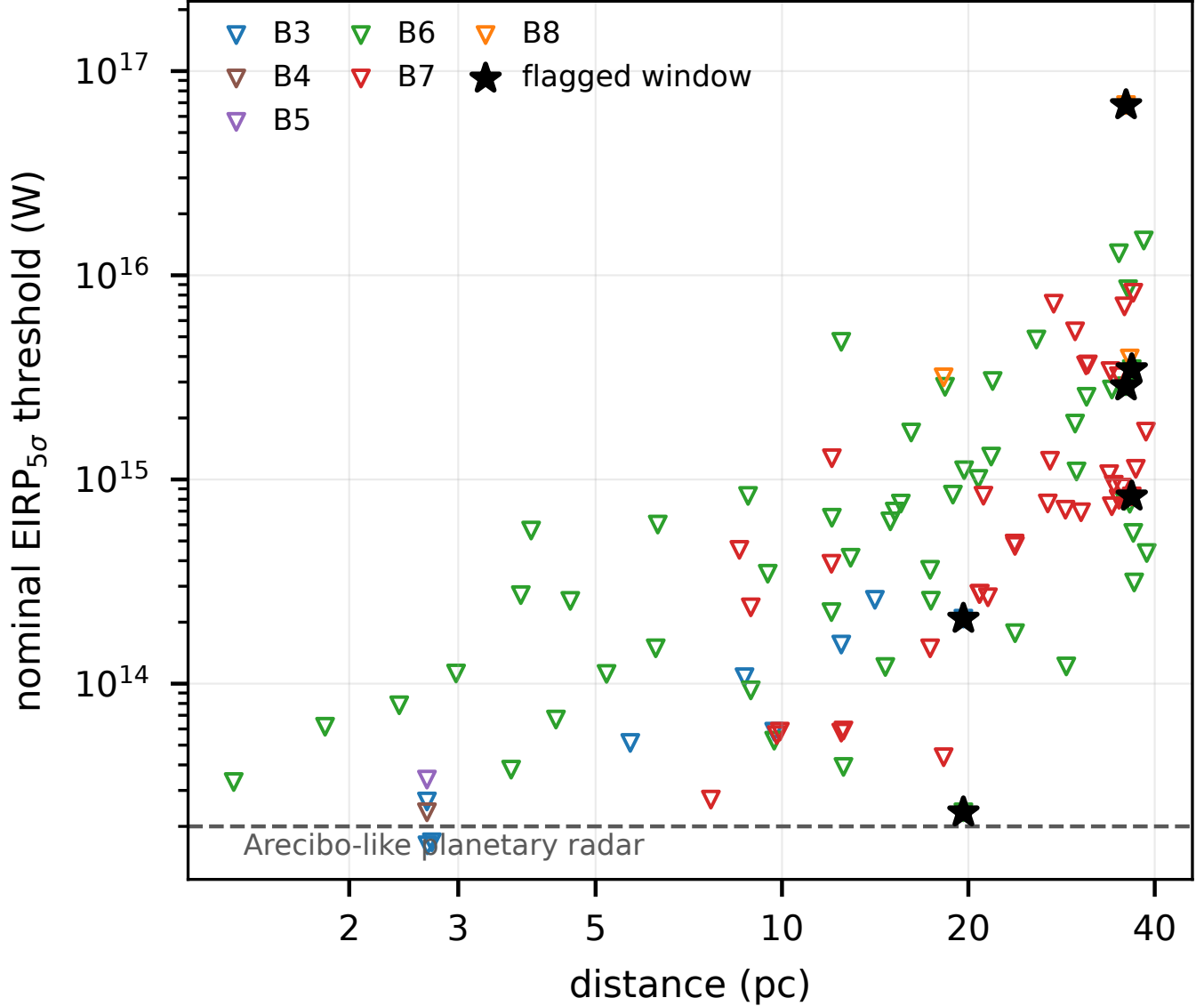


FIG. 8.— Nominal  $\text{EIRP}_{5\sigma}$  thresholds (boxed rule, §4) against distance, one deepest window per star and band (104 datasets, 85 stars); every point is an upper limit. Stars mark the four windows with a spatially significant hit (Table 5). The dashed line is an Arecibo-like planetary radar,  $2 \times 10^{13}$  W, a scale for human transmitter power and not a claim that the survey reaches Earth-technology detectability (§4.3).

public rows, all bands, pointing within 30 arcsec of the propagated stellar position at the observation epoch) counts 107 unique public on-star EBs over the subset run so far, re-finding all but one of its searched EBs – the exception, TRAPPIST-1’s Band 7 execution A002/Xcccc19/X5564, carried by *obscore* only through its calibrator-field row, an archive-representation artefact biasing the census low by one for that star. The selection funnel is Figure 10 (online-only appendix); the ledger `v323_star_coverage.json` gives per-star distance, availability, scope and searched frequency intervals.

Figure 14 (online-only appendix) shows the drift ceiling as an explicit selection function over planet period and host mass: the search fully covers the drift of a transmitter on any known planet in the sample bar the innermost TRAPPIST-1 planets, while one drifting faster falls outside the searched grid, sacrificed by design (Appendix G recovers some non-linear morphologies). The host-mass dependence is weak ( $a \propto M_{\star}^{1/3}$  at fixed period), so the ceiling is effectively a function of orbital period alone; and it constrains only transmitters co-located with *known* planets – one on an undiscovered planet, or on a platform accelerating as it chooses, is covered only in so far as its drift falls in the searched grid.

The release sits against Wright et al. (2018)’s published haystack axis by axis. On frequency (10 MHz–115 GHz) only our Band 3 falls inside (20.6 GHz of 115 GHz, 0.18 of the axis, and one their Hz-resolution surveys reach far more finely); the remaining 66.6 GHz (Bands 4–8, 125–495 GHz) lies wholly outside it, the genuinely new territory. On transmitted bandwidth (0–20 MHz), a single-channel native-channel search probes a fraction set by the channel width (window-mean 0.63, median 0.78). On sensitivity, our median threshold (7.3 mJy across a 15.625-MHz channel,



$\phi \approx 1.1 \times 10^{-21} \text{ W m}^{-2}$ ) detects a  $10^{13} \text{ W}$  transmitter only within  $\sim 0.9 \text{ pc}$ . The sole prior ALMA-archival search, Mason et al. (2024), reaches 28 bycatch targets at  $\geq 1.01 \text{ kpc}$  with a deepest quoted  $\text{EIRP}_{\min}$  of  $6.91 \times 10^{17} \text{ W}$ ; this release’s 42 systems all lie inside  $20 \text{ pc}$  at median per-target thresholds  $\sim 2500\times$  deeper – disjoint corners of one haystack, near versus deep. We do not multiply these fractions into a combined coverage: the axes are strongly coupled (distance and sensitivity above all), so a product of marginals would misstate the overlap by orders of magnitude.

### 6.1. Occurrence limits

What a zero-detection result bounds is fully conditioned. Writing the per-system completeness as

$$C_i(P) = F_i R_i(P) D_i C_{i,\text{drift}}, \quad (3)$$

where  $F_i$  is the probability that the transmitter’s sky frequency falls inside the intervals searched for system  $i$ ,  $R_i(P)$  that a transmitter of  $\text{EIRP } P$  would have crossed system  $i$ ’s threshold had it emitted there,  $D_i$  that it was emitting during the analysed archival interval, and  $C_{i,\text{drift}}$  the recovery probability for a carrier of the searched drift and temporal morphology (Appendix E), the probability of zero credible detections when a fraction  $f$  of systems hosts one is

$$P(N_{\text{det}} = 0 | f) = \prod_{i=1}^{N_{\text{sys}}} (1 - f C_i), \quad (4)$$

and the one-sided 95% upper limit  $f_{95}$  solves  $P(N_{\text{det}} = 0 | f_{95}) = 0.05$ .

*Reading rule.* Among the ALMA-observed systems, fewer than  $f_{95}(P)$  per cent host a transmitter of  $\text{EIRP} \geq P$  that was emitting during the archival observations, at a sky frequency within the particular spectral intervals searched for that system, with drift and temporal morphology lying within the searched class.

*What is and is not being inferred.* Setting  $F_i = D_i = 1$  is not a claim that these factors are near unity, but that the inference is *conditional* on them, in the sense of the reading rule above; Fig. 2 shows which spectral intervals the frequency condition refers to. The proposition differs from star to star – for system  $i$  it is “hosts a transmitter within frequency set  $i$ ” – so we call  $f$  not a prevalence but an *effective conditional occurrence bound given occupancy of each target’s own observed frequencies*, defined on the 79 systems processed here, not the 168-star ALMA-covered census and still less the 40 pc stellar population.

That conditioning can be measured: under the simplest transparent transmitter-frequency prior, a uniform  $p(\nu)$  over ALMA’s tuning range,  $F_i = \int_{\Delta\nu_i} p(\nu) d\nu$  rather than one. The per-system fine-channel frequency union has median  $1.73 \text{ GHz}$ , so  $F_i$  averages  $4.5 \times 10^{-3}$  over Bands 3–8 and  $2.2 \times 10^{-3}$  over Bands 3–10, and the product likelihood excludes nothing whatever:  $\sum_i C_i = 0.27$  and  $0.13$  respectively, so  $f \approx 3.8$  or  $7.8$  transmitters per system would be needed before one detection was expected. Under any frequency prior spread across ALMA’s tuning range, then, this release constrains nothing, and the conditioning supplies the entire content of the numbers below – a conditional bound quoted without it is easy to mistake for a population measurement.

We evaluate Eq. 4 on the 79 independently observed systems (85 narrowband-searched stars; six bound pairs share observations and count once each) in two forms (Table 8). The *thresholded* form is the step function the retained products support:  $R_i(P) = 1$  when system  $i$ ’s deepest threshold lies below  $P$ , else 0. The *measured* form integrates the injection-recovery curve of Appendix E over each system’s searched bandwidth,  $R_i(P) = \sum_w \Delta\nu_w \text{rec}(5P/\text{EIRP}_{5\sigma,w}) / \sum_w \Delta\nu_w$ , folding in the  $17 \rightarrow 83$  per cent recovery of the searched class measured between four and ten times the noise; defined on the 56 systems with a fine-channel window, coarse-channel-only systems lacking measured recovery for the drifting class: a 56-system bound.

*Why this is a pilot number.* The retained products preserve the campaign’s pooled recovery fractions but not its trial-level records, so their denominators come from the design, not a log – weaker provenance than a headline number should have: completeness is measured on six configurations and three stars, the drifting-class curve rests effectively on one fine-channel configuration, and the coarse-window searched class has no calibrated completeness at all. The occurrence bound is therefore a *pilot, configuration-limited* estimate, not a survey-grade measurement, and is secondary to the non-detection and the threshold range. Trial-level logging – identifier, dataset, amplitude, frequency phase, drift, morphology and recovered flag per injected signal – is committed for the next release, with configurations spanning band, channel width, integration time and primary-beam position.

*Which number is the result.* The measured form is this paper’s observational result, the only one whose recovery fraction is measured rather than assumed: at  $\text{EIRP} \geq 10^{16} \text{ W}$  it gives  $f_{95} = 6.5$  per cent. The thresholded form (3.8 per cent) assumes unit recovery, a uniform  $C = 0.42$  gives 9.0 per cent; both are bracketing calculations, not to be quoted as the survey’s constraint. Nor is the measured form simply the weakest of the three: where fine-channel thresholds lie well below  $P$  the measured recovery rises above 0.42 towards its 0.83 plateau, so it falls between the two idealisations. A Bayesian treatment with a uniform prior on  $f$ ,  $p(f | \text{data}) \propto \prod_i (1 - f C_i)$ , gives a 95 per cent credible upper bound of 6.4 per cent for the same model – agreeing to better than the last quoted digit, so nothing turns on the frequentist construction.

Three caveats bound these numbers. Even the measured form fixes  $F_i = D_i = 1$  and the recovery at configuration level – a frequency-integrated completeness for the searched class, not a per-system calibration; the population is heterogeneous in threshold, so each point applies to transmitters above that  $P$ , not one well-defined class; and systems are treated as independent trials in a randomly chosen archival interval, the duty-cycle dependence being quantified

TABLE 8

THE 95 PER CENT UPPER LIMIT  $f_{95}(P)$  ON THE OCCURRENCE OF TRANSMITTERS ABOVE EIRP  $P$ , CONDITIONAL ON EMISSION INSIDE THE SEARCHED WINDOWS AND THROUGHOUT THE ARCHIVAL EPOCHS.  $n$  IS THE NUMBER OF SYSTEMS WITH NON-ZERO COMPLETENESS AT THAT  $P$ ; THE *measured* COLUMN IS THIS PAPER’S RESULT. ‘VACUOUS’ MEANS NO SYSTEM HAS NON-ZERO COMPLETENESS, SO NO BOUND EXISTS.

| $P$ (W)            | thresholded (unit $C$ ) |          | measured $C$ ( <i>result</i> ) |             |
|--------------------|-------------------------|----------|--------------------------------|-------------|
|                    | $n$                     | $f_{95}$ | $n$                            | $f_{95}$    |
| $3 \times 10^{13}$ | 4                       | 52.7%    | 2                              | vacuous     |
| $10^{14}$          | 16                      | 17.1%    | 11                             | 47.1%       |
| $10^{15}$          | 57                      | 5.1%     | 45                             | 10.1%       |
| $10^{16}$          | 78                      | 3.8%     | 56                             | <b>6.5%</b> |
| $10^{17}$          | 79                      | 3.7%     | 57                             | 6.2%        |

TABLE 9

DUTY-CYCLE DEPENDENCE OF THE 95 PER CENT UPPER LIMIT AT  $P \geq 10^{16}$  W, FROM EQ. 4 WITH  $D_i = D$ : THRESHOLDED FORM (79 SYSTEMS, UNIT RECOVERY) AND THE MEASURED-COMPLETENESS FORM (56 FINE-CALIBRATED SYSTEMS), THIS PAPER’S RESULT. ‘VACUOUS’ MEANS NO SYSTEM RETAINS NON-ZERO COMPLETENESS, SO NO BOUND EXISTS.

| Duty cycle $D$ | $f_{95}$ thresholded | $f_{95}$ measured |
|----------------|----------------------|-------------------|
| 1 (continuous) | 3.8%                 | 6.5%              |
| 0.5            | 7.5%                 | 13.0%             |
| 0.1            | 37.7%                | 65.0%             |
| 0.01           | vacuous              | vacuous           |

below. These are, to our knowledge, the first ALMA-side numbers of their kind, and are not competitive prevalence constraints; removing the remaining idealisations needs the per-system  $C_i(P)$  the retained injection campaign supports.

The archival epoch structure gives a duty-cycle constraint directly. Write  $D_i$  for the probability that a transmitter of duty cycle  $D$  emitted during system  $i$ ’s analysed intervals ( $D_i = D$  to the accuracy the epochs support; Table 21 gives per-star epochs and integrations). The same product form gives the  $f_{95}(D)$  ladder of Table 9 and Figure 9: continuous emitters give Table 8’s limits; at  $D = 0.5$  the measured-form limit degrades from 6.5 to 13.0 per cent; by  $D = 0.1$  it is 65 per cent and carries essentially no information; below that no bound of this form exists, transmitters active in a small fraction of random intervals being unconstrained by single-epoch coverage and reachable only by future releases’ recurrence-pool reprocessing. An intermittent transmitter being at least as plausible *a priori* as a continuous one, the  $D = 1$  numbers should never be quoted without this dependence; the non-recurrence logic of §5.5 points the same way.

A cumulative look-elsewhere effect accompanies multi-release use: as releases accumulate, the probability that some unremarkable crossing is reported somewhere grows toward unity even if each release is individually clean. The guard is the per-release procedural pre-commitment of §5.4, not aggregated significance.

The data-quality exclusions of §5.3 carry two open engineering items: the continuum-imaging step still lacks the pointing-offset guard the narrowband step has, and the window-level noise-handling defect affecting HD 10647 and HD 139664 is excluded from results but not yet fixed at source. Both are corrected, and the affected datasets reprocessed, in the single corrected public release (Data Availability); general lessons for archival mining are in online-only Appendix O.

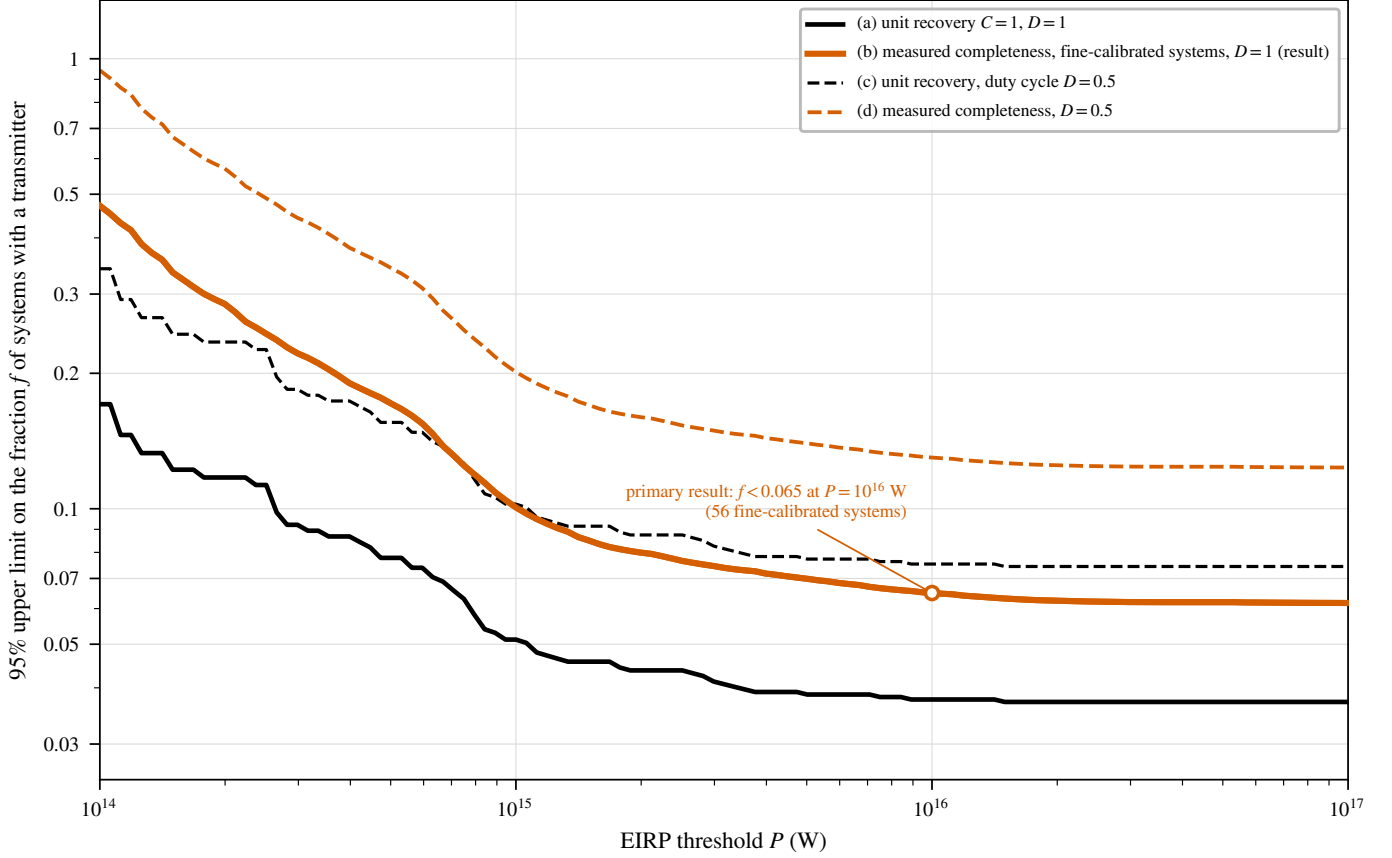
The four supplementary analyses of Table 14 are infrastructure investments, not sources of results at this stage. The Type-I error budget is carried by the primary lane alone (§5.4): the screens gate nothing in it, are correlated with it by construction, and disposition none of the 4 spatially significant windows, so their non-detections must not be multiplied into a stronger constraint than the primary lane’s.

Because selection is on ALMA archival coverage alone, useful subsamples fall out without compromising the census framing, subject to the unresolved-pair caveat of §3. First, 15 of the 46 processed stars (33 per cent; 36 planets) are confirmed exoplanet hosts (Proxima Centauri,  $\tau$  Ceti,  $\epsilon$  Eridani,  $\beta$  Pictoris and the seven-planet TRAPPIST-1 system among them), extractable as a sub-survey from Appendix I; HN Lib and LHS 1140, each with a habitable-zone planet, now carry an ALMA technosignature non-detection beside their planet-search literature. Second, four of the 46 are white dwarfs (Sirius B, Van Maanen’s Star, Wolf 219, CD–38 10980), all continuum non-detections with EIRP<sub>5 $\sigma$</sub>  limits of  $2.3 \times 10^{13}$ – $8.4 \times 10^{14}$  W – among the few mm/submm SETI constraints for white dwarfs (Huang et al. 2026), a channel motivated by post-main-sequence survival and the compact debris discs of metal-polluted ones.

## 6.2. Priorities for the next release

Four changes would enlarge what this experiment can say, ranked by how much signal space they buy.

*First, and by a wide margin: a branch preserving phase-steady carriers.* The per-channel temporal-median subtraction that suppresses stationary ground-based interference also removes a stable narrowband beacon – a canonical technosignature morphology – at every amplitude injected: not a completeness correction, but a whole signal class the pipeline cannot see. An interferometer need not subtract the temporal median to reject terrestrial interference: sky localisation, baseline-dependent phase, delay and fringe-rate signatures and multi-block consistency each separate a celestial source from a ground-fixed emitter without touching time dependence. A branch built on them is the



Product-likelihood limit:  $f$  solves  $\prod_i (1 - fDC_i) = 0.05$ , with  $C_i$  the per-system recovery at threshold  $P$  and  $D$  the assumed duty cycle.

A duty cycle scales every  $C_i$ , so a low-duty-cycle population is bounded only where  $DC_i$  stays large enough to exclude  $f=1$ ; as  $D \rightarrow 0$  no bound exists at any  $P$ . Curve (b) is the primary result quoted in the text.

FIG. 9.— 95 per cent upper limits on the fraction  $f$  of systems hosting a transmitter above EIRP  $P$ , from  $\prod_i (1 - fDC_i) = 0.05$ . Curve (b) is this paper’s result, the injection-measured recovery integrated over each system’s fine-channel bandwidth; curves (a) and (c) assume unit recovery and are bracketing only. Shaded: where the surviving systems cannot exclude  $f=1$  and no bound exists; it grows to cover every  $P$  as  $D \rightarrow 0$ .

highest-value change available.

*Second, use visibilities rather than extracted spectra.* A genuine source at the star should show on every baseline the geometric phase behaviour  $V_{ij}(\nu, t) = A(\nu, t) \exp[-2\pi i(u_{ij}l + v_{ij}m)]$  expected at that sky position. A likelihood ratio between a celestial point source at the stellar position, a common-mode terrestrial interferer and image-plane noise would be strictly more powerful than ranking a peak against a control ring – the distinctive opportunity of ALMA SETI, unavailable to single-dish or beamformed searches.

*Third, enlarge the control ensemble for interesting windows.* The 1/513 rank floor (§5.4) cannot certify significance across a census of thousands of windows. A two-stage scheme – 512 controls routinely, then  $10^4$ – $10^5$  local null evaluations for any window that passes – would settle candidates statistically rather than hand each to astrophysical vetting.

*Fourth, extend the injection campaign and the polarisation axis.* The completeness curve should be measured per band, channel width, integration time, antenna count and primary-beam position rather than transferred by construction, and pushed past  $10\sigma$  until it asymptotes, EIRP<sub>90</sub> being unlocated. The analysis is also Stokes  $I$  only; a polarised carrier discriminates on physical grounds against weakly polarised astrophysical backgrounds here, and polarisation-resolved reprocessing costs no sensitivity.

*A boundary on drift linearity.* The search assumes drift linear over a track. Curvature matters when  $\frac{1}{2}\ddot{\nu}T^2$  exceeds half a native channel; for a circular orbit of period  $P$  this becomes  $P \lesssim 2\pi(\nu/c) a_{\max} T^2 / \Delta\nu_{\text{ch}}$ , i.e.  $P \lesssim 1.6$  d for a typical fine-channel window (230 GHz, 488 kHz channels,  $T \approx 2000$  s) and about an hour for a coarse-channel one. Only the shortest-period orbits reach it: TRAPPIST-1 b, at 1.51 d, sits just inside the boundary and is also the case exceeding the drift ceiling. Elsewhere the constant-drift model is adequate, and the chirp-search extension (Appendix G) covers the remainder.

**What this survey does not constrain.** *Nothing in this paper bounds the prevalence of:* (i) *phase-steady carriers* – a stable beacon dwelling in one channel for more than roughly 40 per cent of a track is removed by the per-channel

temporal-median subtraction, by construction and at every amplitude tested (§4.4); (ii) transmitters emitting *outside the searched spectral windows*, which are 34 narrow islands totalling 92.4 GHz (Fig. 2), not ALMA’s frequency range; (iii) transmitters active *outside the archival epochs* observed – the limits assume continuous emission and are already weak at duty cycle 0.5 and vacuous below  $\sim 0.1$  (Table 9); (iv) drift rates *above* the searched ceiling  $|a| = 3.6 \text{ m s}^{-2}$ , which excludes transmitters on the most close-in orbits, including a TRAPPIST-1 b-like one (Fig. 14); (v) carriers deliberately placed *inside* the  $\pm 50 \text{ km s}^{-1}$  molecular-line mask, which is excluded search space rather than searched-and-empty space; (vi) emission substantially *broader* than one native channel, for which the quoted thresholds must be rescaled; (vii) anything below the nominal threshold, at which measured recovery is  $42^{+16}_{-14}$  per cent, not 95 per cent.

## 7. CONCLUSIONS

We have described the design, methodology and processed release of a technosignature search of the 168 stars within 40 pc with public ALMA archival coverage: a complete census of those stars within 40 pc, with no astrophysical selection of its own, every entry independently audited for position and identity against SIMBAD and *Gaia* (§5.3), but covering only  $\sim 7\%$  of the catalogued stars within 40 pc and inheriting the archive’s non-uniform targeting, so the contrast with interest-selected samples (§3) is one of degree, not kind. This release processes 104 of the 208 planned target/bands – 85 of the 168 census stars, 79 independent systems – sensitivity-ranked, not an unbiased draw.

The release yields EIRP<sub>5 $\sigma$</sub>  limits of  $1.6 \times 10^{13}$ – $2.8 \times 10^{17}$  W (unchanged if the primary-beam-affected Sirius B datasets are also excluded, as the withheld  $\epsilon$  Eri Band 6 windows already are) and forty non-detections in the ancillary continuum screen of unquantified statistical power (median 5 $\sigma$  limit 0.52 mJy). No credible technosignature was identified: nothing matching the searched frequency, drift-rate, temporal and spectral morphology rose above the nominal instrumental threshold during the analysed intervals. Of the 4 windows with a spatially significant hit, 3 are attributed to CO – two towards  $\beta$  Pictoris, one towards HD 48370, the last independently reported by Cataldi et al. (2023) – and 1 towards CP–72 2713 is a marginal crossing consistent with the survey’s own quantified false-flag rate, against an expectation of 0.78 over the non- $\beta$  Pictoris windows ( $P(\geq 1) = 54$  per cent). None survives vetting, so this release reports 0 candidates. The operative criterion, the symmetric single-position statistic, is applied uniformly to all 417 windows; the richer variant maximising over a source region at each control exists only for seven locally reprocessed windows, consistent with exchangeability (Figure 11).

The result is a constraint, not a bare non-detection: the searched windows cover 87.0 GHz of unique effective frequency space after molecular-line masking, at nominal thresholds constraining transmitters of the searched spectral and drift morphology during those intervals, read under the boxed rules of §4 and §6; nothing was found outside the excluded velocity regions. These limits are not narrowband-transmitter parity with Hz-resolution centimetre-wave surveys: the sensitivity comes from proximity overcoming ALMA’s coarse channelisation, not spectral resolution (Figure 2); the contribution is bounding total transmitter power for the nearest stars in a virtually unexplored regime at archive-quality resolution. Epoch coverage bounds it likewise: with one to three execution blocks per star (median 46 minutes on source), transmitters active only outside the sampled epochs, or with duty cycles below these baselines, are unconstrained (§6). As a population statement under the stated idealisations (§6, Eq. 4), zero credible detections among the 79 independently observed systems bound transmitters of the searched class emitting then at  $\text{EIRP} \gtrsim 10^{16}$  W to fewer than 6.5 per cent of the systems carrying calibrated completeness – a bound that weakens to 13.0 per cent at duty cycle 0.5 and 65 per cent at 0.1, and vanishes under any transmitter-frequency prior spread across ALMA’s tuning range (95 per cent confidence, measured completeness, 56 injection-calibrated systems), against 3.8 per cent with the injection-measured, frequency-integrated completeness folded in (24 calibrated systems); a uniform threshold-level completeness of 0.42 puts the limit near 16 per cent.

The release includes the first reported ALMA technosignature searches of Barnard’s Star and Wolf 359 (§5.2) and, to our knowledge, the first use in a technosignature survey of interferometric closure-phase vetting, a population-level spectral-type-normalised continuum anomaly check, a chirped-drift and periodicity extension of the narrowband search re-using retained spectra, and a cross-target frequency-occupancy check for sky-frequency-recurrent interference. All four ship with the pipeline at the two grades of Table 14 and gain statistical power as the survey proceeds. A modest, serendipitous molecular-line catalogue and the exoplanet-host and white-dwarf subsamples are census by-products. The survey continues, and we anticipate extending it beyond 40 pc once the 20 pc sample is complete.

## ACKNOWLEDGEMENTS

This paper uses ALMA data from the public ALMA Science Archive. The execution blocks searched (100 identifiers, with project codes and epochs in Appendix A) should be cited from that list in the standard JAO.ALMA#XXXX.X.NNNNN.S form. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), MOST and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia*, processed by the *Gaia* Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database, operated at CDS, Strasbourg, France.

## COMPETING INTERESTS

G.J.W. declares no competing interests. R.D. is an employee of VBRL Holdings Inc.; VBRL has no financial or intellectual stake in ALMA archival technosignature work, provided no funding for this study, and had no role in its design, analysis, or decision to publish. On that basis the authors declare no competing interest.

#### AUTHOR CONTRIBUTIONS (CREDIT)

**Glenn J. White:** Conceptualisation, Methodology, Supervision, Writing – review & editing.

**Robin Dey:** Software, Investigation, Formal analysis, Data curation, Visualisation, Writing – original draft.

#### DATA AVAILABILITY

The reduced data products underlying this paper, with the full target-selection and analysis pipeline, are at <https://github.com/Tilanthi/SETI>; the repository is live at submission. All numerical values here come from the *submitted-analysis snapshot*: the repository commit tagged at submission, which freezes pipeline code and input data as run, is retained unchanged and reproduces every number, table and figure through the regeneration scripts shipped with it. The *public release* is built from that snapshot with the two pipeline defects recorded in §5.3 (the missing pointing-offset guard in the continuum lane and the window-level noise-handling defect) corrected and the affected datasets reprocessed *before* freezing, so it carries one set of products, not a submitted/corrected pair.

The datasets so reprocessed are exactly those excluded from this paper’s results by §5.3 (the Giclas 9–38 pair and the two defective Band 6 windows); the Sirius B and  $\epsilon$  Eri holography-beam recomputation committed in §4 joins the same freeze, its outcome bounded by the systematic quoted there. No value printed here is silently altered: a change-log names every affected product, table and row with the reason for each change. The machine-readable per-window table carries archive-consistent execution times: where the printed integration-time audit differs (the CD–38 10980 row is optimistic by 7%, Appendix Q), it gives the archive’s. Public-release per-window hit lists are uncapped (the interim product capped them at 50 frequencies; Appendix H).

The complete release will also be deposited on Zenodo, with a DOI minted on final acceptance and inserted here; it comprises (i) per-window search products for all 417 windows (spectral extracts, per-drift-trial peak statistics, the machine-readable line-exclusion mask of §5.4); (ii) control-ring calibration products at all 512 positions; (iii) closure-phase and population-anomaly outputs; (iv) the full injection-recovery campaign products (all 1200 trials of Appendix E, including the 24-tone pilot); (v) the frozen pipeline code and input data snapshot; (vi) the audit tables of online-only Appendix Q, machine-readable; (vii) the machine-readable molecular-line catalogue of Appendix N; and (viii) the proper-motion-matched execution-block availability census of §6 with its per-star ledger. Data products carry the Creative Commons Attribution 4.0 International licence (CC BY 4.0), pipeline code the MIT licence. Raw ALMA visibility data are public from the ALMA Science Archive at <https://almascience.org>.

#### SOFTWARE

The pipeline depends on CASA 6.7.6 (casatools/casatasks 6.7.6-14; calibration, imaging and the local reprocessing of §§5.4–5.5), Python 3 with astropy, numpy and scipy (spectral extraction and statistics), and the ALMA Science Archive and SIMBAD TAP interfaces; all are free and open-source, with exact versions recorded in the frozen release.

#### APPENDIX

##### EXECUTION-BLOCK METADATA

Table 10 lists the 100 unique ALMA execution blocks searched here, with proposal codes and epochs. These metadata were recovered from the archive by execution-block identifier after the search products were frozen, and are not part of them. The per-window linkage between execution block and searched window, with per-observation antenna configurations, synthesised beam sizes and polarisation setup, will accompany the machine-readable table of the corrected tagged release (Data Availability). The epochs span 2013 October 6 to 2025 June 11 (MJD 56571–60837) across 36 unique proposal codes; the AU Mic candidate window of §5.4 comes from the 2014 August 18 execution block of project 2012.1.00198.S. One further public execution block of that project (uid://A002/X7d80c7/X15c6, executed 2014 June 5, 1633 s of science exposure, public since 2015 April 8) also covers the candidate 230.83 GHz window but was not processed; it is the second epoch of the recurrence test of §5.4.

#### GLOSSARY OF SURVEY VOCABULARY

Terms as used here; quantitative definitions are in the sections cited.

#### FREQUENCY CONVENTIONS, LINEWIDTH READING, AND SMEARING

##### REFERENCE FRAMES

The line-exclusion mask’s chain – topocentric  $\rightarrow$  barycentric (Earth ephemeris at the execution-block mid-time)  $\rightarrow$  stellar rest frame (catalogued systemic radial velocity; JPL catalogue rest frequencies, Appendix N) – has as residual error the catalogue radial-velocity uncertainty alone (at most a few  $\text{km s}^{-1}$  within 20 pc); second-order relativistic terms at the  $50 \text{ km s}^{-1}$  scale are  $\lesssim 5 \text{ kHz}$  at 350 GHz, below the finest native channel searched (15.3 kHz). The  $\beta$  Pictoris and AU Mic kinematic attributions (§5.4, Fig. 7) use the same chain, so their offsets are comparable to the exclusion tolerance. Enlarging the drift-rate grid needs no further frame machinery: the searched rates are topocentric, read from the same untransformed axis, and the grid depends only on elapsed time and channel width, not sky position.

TABLE 10

EXECUTION BLOCKS SEARCHED IN THIS RELEASE, ORDERED BY EPOCH. EPOCHS (UTC START, TO THE MINUTE) AND PROPOSAL CODES ARE FROM THE ALMA SCIENCE ARCHIVE `IVOA.OBSSCORE` SERVICE AT `ALMA SCIENCE.ESO.ORG`, QUERIED BY EXECUTION-BLOCK IDENTIFIER ON 2026 SEPTEMBER 8; MJD IS THE ARCHIVE `T_MIN`.

| Execution block           | Proposal       | UTC date         | MJD         | Execution block            | Proposal       | UTC date         | MJD         |
|---------------------------|----------------|------------------|-------------|----------------------------|----------------|------------------|-------------|
| uid://A002/X6f1341/X1484  | 2012.1.00142.S | 2013-10-06 06:23 | 56571.26616 | uid://A002/Xd0fb35/X90dc   | 2017.1.00698.S | 2018-08-22 15:32 | 58352.64755 |
| uid://A002/X74deb6/Xff1   | 2012.1.00268.S | 2013-12-02 07:15 | 56628.30252 | uid://A002/Xd16a1e/X8d4    | 2017.1.01644.S | 2018-08-31 22:26 | 58361.93490 |
| uid://A002/X877e41/X14c1  | 2012.1.00993.S | 2014-05-22 11:24 | 56799.47525 | uid://A002/Xd248b5/X5186   | 2017.1.00698.S | 2018-09-22 11:12 | 58383.46702 |
| uid://A002/X899169/X1838  | 2012.1.00198.S | 2014-08-18 04:01 | 56887.16777 | uid://A002/Xd28a9e/Xbcd    | 2017.1.00698.S | 2018-09-28 10:57 | 58389.45629 |
| uid://A002/X99c183/X4f1e  | 2013.1.00645.S | 2015-01-17 00:35 | 57039.02475 | uid://A002/Xd6ebab/X41a3   | 2018.1.01149.S | 2018-12-20 09:49 | 58472.40910 |
| uid://A002/X9f2ff8/X25b0  | 2013.1.01214.S | 2015-04-26 12:19 | 57138.51354 | uid://A002/Xd86f11/X584b   | 2018.1.01833.S | 2019-01-20 19:05 | 58503.79517 |
| uid://A002/Xb0be8b/X785f  | 2015.1.00307.S | 2016-03-22 23:25 | 57469.97599 | uid://A002/Xd95042/X2d99   | 2018.1.01149.S | 2019-03-09 06:13 | 58551.25945 |
| uid://A002/Xb25e1a/Xed74  | 2015.1.00783.S | 2016-05-01 17:17 | 57509.72060 | uid://A002/Xd9668b/X3a90   | 2018.1.00072.S | 2019-03-12 00:14 | 58554.01019 |
| uid://A002/Xb2e94e/X361a  | 2015.1.00307.S | 2016-05-13 04:19 | 57521.18017 | uid://A002/Xd98580/X3bfe   | 2018.1.01833.S | 2019-03-15 20:39 | 58557.86052 |
| uid://A002/Xb39557/X3789  | 2015.1.00307.S | 2016-05-26 11:30 | 57534.47920 | uid://A002/Xd9d7c7/X6fd2   | 2018.1.01833.S | 2019-03-21 15:03 | 58563.62709 |
| uid://A002/Xb9a2bb/X1899  | 2016.1.00817.S | 2016-10-22 02:07 | 57683.08829 | uid://A002/Xda2c82/X765d   | 2018.1.01833.S | 2019-03-28 21:57 | 58570.91480 |
| uid://A002/Xb9cc97/X150a  | 2016.1.00803.S | 2016-10-25 02:14 | 57686.09356 | uid://A002/Xdc621b/X2bb0   | 2018.1.01833.S | 2019-05-20 04:28 | 58623.18665 |
| uid://A002/Xba3ea7/X2ab4  | 2016.1.00817.S | 2016-11-06 20:01 | 57698.83471 | uid://A002/Xe5808e/X80a1   | 2019.1.01681.S | 2019-12-20 17:03 | 58837.71056 |
| uid://A002/Xbe7502/X2047  | 2016.1.00637.S | 2017-03-28 05:22 | 57840.22390 | uid://A002/Xfd476d/X32f1   | 2021.1.00485.S | 2022-03-02 23:14 | 59640.96830 |
| uid://A002/Xc0677f/X822   | 2016.2.00015.S | 2017-05-17 01:51 | 57890.07740 | uid://A002/Xfdb69b/X667a   | 2021.1.01209.S | 2022-08-30 16:22 | 59821.68235 |
| uid://A002/Xc1946f/X19ce  | 2016.2.00015.S | 2017-07-02 08:43 | 57936.36365 | uid://A002/Xfd45f/X478e    | 2021.1.01209.S | 2022-09-05 02:32 | 59827.10618 |
| uid://A002/Xc6141c/X3af0  | 2017.1.00786.S | 2017-10-27 03:04 | 58053.12832 | uid://A002/Xfeb5e6/X13ee   | 2021.1.01209.S | 2022-09-22 01:39 | 59844.06896 |
| uid://A002/Xc66ce1/Xaa9   | 2017.1.00786.S | 2017-11-04 03:17 | 58061.13698 | uid://A002/Xff294f/X2db6   | 2022.1.00338.L | 2022-10-04 04:49 | 59856.20118 |
| uid://A002/Xc74b5b/X7c16  | 2017.1.01644.S | 2017-11-29 07:29 | 58086.31237 | uid://A002/X106a1f/X6a0c   | 2022.1.01163.S | 2023-05-06 03:35 | 60070.14972 |
| uid://A002/Xc8b2b0/X9c35  | 2017.1.01644.S | 2018-01-05 00:25 | 58123.01783 | uid://A002/X107fa96/Xce38  | 2022.1.01163.S | 2023-06-02 02:51 | 60097.11926 |
| uid://A002/Xc9831a/X443   | 2017.1.00986.S | 2018-01-22 20:39 | 58140.86104 | uid://A002/X10a341d/X82d2  | 2022.1.00134.S | 2023-07-17 18:28 | 60142.76964 |
| uid://A002/Xc99ad7/X348   | 2017.1.00698.S | 2018-01-24 23:51 | 58142.99391 | uid://A002/X10bd07d/X323f  | 2022.1.00134.S | 2023-08-06 09:20 | 60162.83930 |
| uid://A002/Xcbb583/Xb78   | 2017.1.00704.S | 2018-04-06 23:22 | 58214.97419 | uid://A002/X11e10bc/X12662 | 2023.1.00390.S | 2024-09-28 16:03 | 60581.66907 |
| uid://A002/Xcc626d/X579   | 2017.1.00698.S | 2018-04-21 21:13 | 58229.88426 | uid://A002/X1207c39/X9417  | 2024.1.00722.S | 2024-11-15 08:17 | 60629.34515 |
| uid://A002/Xccccc19/X5564 | 2017.1.00215.S | 2018-05-03 11:42 | 58241.48813 | uid://A002/X12277ed/X22e53 | 2024.1.00165.S | 2024-12-22 10:09 | 60666.42301 |
| uid://A002/Xcd97a1/X3f1e  | 2017.1.01583.S | 2018-05-19 07:56 | 58257.33086 | uid://A002/X1237e23/Xdb0   | 2024.1.01095.S | 2025-01-06 18:51 | 60681.78602 |
| uid://A002/Xcda49e/Xdeb   | 2017.1.00561.S | 2018-05-21 09:50 | 58259.41000 | uid://A002/X12925a3/Xa4b1  | 2024.1.01095.S | 2025-05-17 08:00 | 60812.33402 |
| uid://A002/Xcf0c6b/X6513  | 2017.1.00167.S | 2018-06-27 10:14 | 58296.42698 | uid://A002/X12a8fc8/X10b8  | 2024.A.00040.S | 2025-06-11 08:49 | 60837.36781 |
| uid://A002/Xcf196f/Xabc5  | 2017.1.00200.S | 2018-06-29 09:32 | 58298.39773 |                            |                |                  |             |

TABLE 11  
SURVEY VOCABULARY.

| Term                              | Meaning in this paper                                                                                                                 |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Census                            | every star within 40 pc with public ALMA archival coverage (168 entries; §3)                                                          |
| Bycatch                           | census stars observed by ALMA for unrelated science goals                                                                             |
| Target/band dataset               | one star in one ALMA band; the unit of the 208-dataset plan (104 complete)                                                            |
| Spectral window                   | one correlator spw of one execution block, after deduplication; the unit of search and bookkeeping (417)                              |
| Execution block (EB)              | one contiguous ALMA observing session (100 searched; Appendix A)                                                                      |
| MOUS                              | member observation unit set: the ALMA data hierarchy level at which pipeline products are delivered (one calibration recipe per MOUS) |
| QA2                               | the ALMA observatory pipeline’s second-stage quality-assurance calibration and scoring, whose products this survey consumes           |
| obscore                           | the IVOA table protocol of the ALMA Science Archive metadata service from which coverage and EB metadata are queried                  |
| Native channel                    | the correlator channel width of the window, 15.3 kHz–31.25 MHz here                                                                   |
| Threshold crossing (“hit”)        | a $\geq 5\sigma$ spectral feature at the star’s position in a searched window                                                         |
| Candidate                         | a spatially significant hit that survives every astrophysical and instrumental veto (0 in this release)                               |
| Control ring                      | 512 off-source test positions per window, used for empirical false-alarm calibration                                                  |
| Star-beats-ring                   | a window whose star statistic exceeds all 512 control statistics (5/417)                                                              |
| Drift trial                       | one trial Doppler drift rate of the stack grid; 2–1944 per window                                                                     |
| Nominal threshold                 | noise-derived $5\sigma$ detection threshold, <i>not</i> a calibrated completeness limit (boxed rule, §4)                              |
| EIRP $_{5\sigma}$                 | equivalent isotropic radiated power at the nominal threshold; $4\pi d^2 S_{\min} \Delta\nu_{\text{ch}}$                               |
| $t_{\text{on}} / T_{\text{span}}$ | summed on-source integration / elapsed block span per star (Table 21)                                                                 |
| Molecular-line exclusion mask     | the $\pm 50 \text{ km s}^{-1}$ stellar-frame region around catalogued transitions where crossings are excluded (§4.2)                 |
| Closure phase                     | interferometric phase triplet test discriminating point-source structure (Appendix F)                                                 |
| Stellar frame                     | topocentric $\rightarrow$ barycentric $\rightarrow$ systemic-RV frame chain (§4.2)                                                    |
| Funnel                            | the candidate-narrowing cascade from channels searched to vetted candidates (Table 4)                                                 |

### CHANNEL WIDTHS AND THEIR VERIFICATION

Channel widths come from the measurement sets’ own spectral window tables (CHAN\_WIDTH). For the local reprocessing set we verified these are the *effective* resolution, not a pre-averaging bookkeeping value: the tables carry no EFFECTIVE\_BANDWIDTH column (no online spectral averaging was recorded) and the full-resolution windows are named FULL\_RES, so the 15.625-MHz coarse and 488.28125-kHz fine channelisations are the noise bandwidths the thresholds mean.

### LINewidth CONVENTION

The detection statistic is the flux density in a single channel after drift correction, so the quoted EIRP $_{5\sigma}$  applies to any transmitter narrower than the native channel, flat in the assumed linewidth. Two sub-channel qualifications: flatness is exact for a tone centred in a channel (a boundary-straddling tone splits its power between two, at worst a

TABLE 12

THE ONLY WINDOWS WITH NON-NEGLIGIBLE INTRA-INTEGRATION SMEARING AT THE MAXIMUM SEARCHED DRIFT RATE ( $\eta_{\text{smear}} < 0.99$ ; THE OTHER 222 HAVE  $\eta_{\text{smear}} \geq 0.99$ , NOMINAL AND EFFECTIVE THRESHOLDS AGREEING TO BETTER THAN 1%).  $\Delta$  IS THE EXCURSION IN CHANNELS ACROSS ONE INTEGRATION,  $t_{\text{int}}$  THE PER-INTEGRATION ON-SOURCE TIME, AND  $\text{EIRP}_{5\sigma, \text{eff}}$  THE SMEARED THRESHOLD  $\text{EIRP}_{5\sigma}/\eta_{\text{smear}}$ .

| Window         | $\Delta\nu_{\text{ch}}$ (kHz) | $t_{\text{int}}$ (s) | $\Delta$ (ch) | $\eta_{\text{smear}}$ | $\text{EIRP}_{5\sigma, \text{eff}}$ ( $10^{13}$ W) |
|----------------|-------------------------------|----------------------|---------------|-----------------------|----------------------------------------------------|
| $\beta$ Pic B6 | 15.3                          | 6.05                 | 1.096         | 0.574                 | 4.08                                               |
| $\eta$ Crv B8  | 61.0                          | 6.05                 | 0.585         | 0.865                 | 365.8                                              |
| $\eta$ Crv B7  | 61.0                          | 6.05                 | 0.411         | 0.932                 | 4.68                                               |

factor of two; measured in Appendix E); and the transition between adjacent channels follows the correlator’s passband shape (Hanning weighting or spectral averaging included), which varies with correlator generation and is not retained in the frozen products, making the flat-in-linewidth statement a channel-centred idealisation accurate at the tens-of-per-cent level. A robustness test with injected tones at randomised sub-channel positions is committed with the validation items of §6. For the deepest window (UV Ceti Band 3) the quoted limit is  $1.6 \times 10^{13}$  W at a 15.625-MHz channel and the minimum detectable spectral power  $1.0 \times 10^6$  W Hz $^{-1}$ : a 1-Hz carrier there needs the full  $1.6 \times 10^{13}$  W. Channelised at 1 Hz from the start, the same observation would give  $P_{\text{min}} \propto \sqrt{\Delta\nu}$ , falling to  $4 \times 10^9$  W; archival data cannot be re-channelised below native resolution, so the ratio  $\sqrt{\Delta\nu_{\text{ch}}/\Delta\nu'}$  ( $\sim 4000$  here) is the unavoidable price of archival channelisation.

**Worked example: from flux to EIRP.** Take UV Ceti (G 272-61B) Band 3. The searched window’s per-channel rms is 0.244 mJy, so  $S_{\text{min}} = 5 \times 0.244 = 1.22$  mJy in one 15.625-MHz native channel. At 2.67 pc that is  $1.22 \times 10^{-29}$  W m $^{-2}$  Hz $^{-1}$ , and  $\text{EIRP}_{5\sigma} = 4\pi d^2 S_{\text{min}} \Delta\nu = 4\pi (2.67 \text{ pc})^2 \times 1.22 \times 10^{-29} \text{ W m}^{-2} \text{ Hz}^{-1} \times 15.625 \text{ MHz} = 1.6 \times 10^{13}$  W. Any carrier narrower than 15.625 MHz must deliver that *total* power to be flagged; a wider transmitter needs  $S_{\text{min}}$  per channel.

#### INTRA-INTEGRATION SMEARING

The frequency excursion at the maximum searched drift rate across one integration,  $\dot{\nu}_{\text{max}} t_{\text{int}}$ , is evaluated for every window: at most 1.10 channels across the 417 windows (median 0.001, 90th percentile 0.06). A linearly drifting signal smeared by  $\Delta$  channels within an integration retains a fraction  $\eta_{\text{smear}} = \text{sinc}(\pi\Delta/2)$  of its peak-channel amplitude, so a worst-drift transmitter’s effective threshold is  $\text{EIRP}_{5\sigma}/\eta_{\text{smear}}$ ; the three affected windows of 417 are those of Table 12, the median window essentially unsmeared. A signal drifting midway between adjacent trial rates accumulates at most 0.03 channels of residual drift across the grid as realised, so no significant sensitivity is lost between trial rates.

#### PRIOR LOW-FREQUENCY TECHNOSIGNATURE SURVEYS

Table 13 summarises the surveys cited in §2, limited to what their abstracts state; a dash marks quantities the cited work does not quote. Band, backend and threshold conventions differ by construction, so the table is context, not a sensitivity ranking (Figure 1).

TABLE 13  
SELECTED PRIOR TECHNOSIGNATURE PROGRAMMES BELOW 10 GHz AND THE SOLE PRIOR ALMA SEARCH, AS CITED IN THE TEXT.

| Programme / study                                       | Facility   | Scale        | Frequency         | Note as stated by the cited work                                                              |
|---------------------------------------------------------|------------|--------------|-------------------|-----------------------------------------------------------------------------------------------|
| Project Ozma (Drake 1960)                               | —          | 2 stars      | 1.42 GHz          | $\tau$ Cet and $\epsilon$ Eri                                                                 |
| Project Phoenix (Backus & Project Phoenix Team 2002)    | multi-site | —            | —                 | pioneered multi-site RFI verification                                                         |
| Allen Telescope Array (Tarter 2001)                     | ATA        | —            | —                 | first purpose-built commensal SETI instrument                                                 |
| ATA TRAPPIST-1 (Tusay et al. 2024)                      | ATA        | —            | —                 | —                                                                                             |
| ATA ISO survey (Sheikh et al. 2025b)                    | ATA        | —            | —                 | —                                                                                             |
| BL first data release (Enriquez et al. 2017)            | GBT        | 692 stars    | 1.1–1.9 GHz       | EIRP $> 10^{13}$ W                                                                            |
| BL 1327-star search (Price et al. 2020)                 | —          | 1327 stars   | 1.10–3.45 GHz     | —                                                                                             |
| BL Earth Transit Zone (Gajjar et al. 2019)              | GBT        | —            | 3.95–8.00 GHz     | —                                                                                             |
| Margot et al. (Margot et al. 2023)                      | GBT        | 11,680 stars | 1.15–1.73 GHz     | —                                                                                             |
| ML vetting (Ma et al. 2023)                             | GBT        | 820 stars    | —                 | $\beta$ -CVAE cuts human-vetting load by $\sim 100\times$                                     |
| turboSETI correction (Choza et al. 2024)                | —          | —            | —                 | S/N mischaracterisation; retrospective corrections                                            |
| FAST commensal ISOs (Zhang et al. 2020; Li et al. 2026) | FAST       | —            | —                 | interstellar-object targets                                                                   |
| Occultation timing (Tusay et al. 2025)                  | —          | —            | —                 | eclipsing/occultation strategy                                                                |
| COSMIC (Tremblay et al. 2024)                           | VLA        | —            | —                 | commensal system                                                                              |
| MeerKAT dragnet (Czech et al. 2021)                     | MeerKAT    | —            | —                 | —                                                                                             |
| MWA back-ends (Morrison et al. 2023)                    | MWA        | —            | —                 | —                                                                                             |
| ALMA stellar bycatch (Mason et al. 2024)                | ALMA B3    | 28 stars     | 90.642/93.151 GHz | first ALMA technosignature search; $\text{EIRP}_{\min} = 6.9 \times 10^{17}$ W (nearest star) |



TABLE 14  
SCOPE OF THE ANALYSES IN THIS RELEASE.

| Analysis                      | Applied to                          | Validation status                                                                                                     | In headline limits? |
|-------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------|
| Primary narrowband search     | 417 windows / 85 stars (79 systems) | Injection-calibrated for the searched class (1200 trials; Appendix E); false-alarm rate calibrated empirically (§5.4) | Yes                 |
| Closure-phase vetting         | 3 candidates                        | Simulation-validated; sensitivity-limited on sky (Appendix F)                                                         | Disposition only    |
| Injection-recovery validation | 6 configurations / 3 stars          | This release; scope limits stated (Appendix E)                                                                        | Completeness only   |
| Chirp/periodicity screen      | 122 retained spectra                | Demonstration-grade (Appendix G)                                                                                      | No                  |
| Frequency-occupancy check     | 366 overlapping star pairs          | Executed (Appendix H)                                                                                                 | Disposition only    |

TABLE 15

SELECTION FUNCTION OF THE SEARCHED SAMPLE AGAINST A VOLUME-LIMITED REFERENCE CENSUS. CENSUS: GAIA DR3 SOURCES WITH  $\varpi > 0$ ,  $\sigma_{\varpi}/\varpi \leq 0.1$  AND  $1/\varpi \leq 40$  pc (17566 OBJECTS; 8594, 49 PER CENT, CARRY A `TEFF_GSPPHOT` ESTIMATE). SAMPLE: THE 85 STARS WITH AT LEAST ONE SEARCHED WINDOW IN THIS RELEASE. SAMPLE DISTANCES ARE THOSE USED BY THE SEARCH ITSELF AND ARE COMPLETE; TEMPERATURES EXIST FOR 51 OF THE 85 STARS (ALL 85 MATCHED TO THE 168-ENTRY ALMA-COVERED CENSUS, 56 BY EXACT NAME; 51 OF THOSE MATCHES CARRY A TEMPERATURE), SO THE TEMPERATURE PANEL IS NORMALISED WITHIN EACH COLUMN OVER CLASSIFIED OBJECTS ONLY AND  $T_{\text{eff}}$  PROXIES SPECTRAL CLASS. “RATIO” IS THE SAMPLE COLUMN FRACTION OVER THE CENSUS COLUMN FRACTION: 1 MEANS THE SAMPLE TRACKS THE REFERENCE POPULATION,  $> 1$  OVER-REPRESENTATION.

| Property                                                       | Census       | Searched   | Ratio |
|----------------------------------------------------------------|--------------|------------|-------|
| Distance (all 85 searched stars; 17566 census objects)         |              |            |       |
| $d \leq 5$ pc                                                  | 48 (0.3%)    | 12 (14.1%) | 51.7  |
| 5–10 pc                                                        | 240 (1.4%)   | 13 (15.3%) | 11.2  |
| 10–20 pc                                                       | 2069 (11.8%) | 18 (21.2%) | 1.8   |
| 20–30 pc                                                       | 5391 (30.7%) | 18 (21.2%) | 0.7   |
| 30–40 pc                                                       | 9818 (55.9%) | 24 (28.2%) | 0.5   |
| $T_{\text{eff}}$ class <sup>a</sup> (51 searched; 8594 census) |              |            |       |
| A ( $\geq 7500$ K)                                             | 24 (0.3%)    | 1 (2.0%)   | 7.0   |
| F (6000–7500 K)                                                | 402 (4.7%)   | 9 (17.6%)  | 3.8   |
| G (5300–6000 K)                                                | 860 (10.0%)  | 15 (29.4%) | 2.9   |
| K (3900–5300 K)                                                | 1400 (16.3%) | 5 (9.8%)   | 0.6   |
| M ( $< 3900$ K)                                                | 5908 (68.7%) | 21 (41.2%) | 0.6   |
| Other axes                                                     |              |            |       |
| Known planet hosts <sup>b</sup>                                | 20 (11.9%)   | 18 (21.2%) | 1.8   |
| Searched stars / systems                                       | –            | 85 / 79    | –     |

<sup>a</sup>Bins are temperature bins, not MK types: A  $\geq 7500$ , F 6000–7500, G 5300–6000, K 3900–5300, M  $< 3900$  K. Gaia `teff_gspphot` is unavailable for about half the reference census, preferentially the coolest and faintest objects, so the census M fraction is a lower limit and the earlier-type ratios correspondingly upper limits. White dwarfs are not separable here and fall wherever their (unreliable) photometric temperature places them.

<sup>b</sup>The Gaia reference census carries no planet information, so the census column here is the 168-entry ALMA-covered census, of which 20 entries are known planet hosts. The 85 searched stars include 18 hosts of 41 known planets.

#### SUPPLEMENTARY MATERIAL (ONLINE ONLY)

The appendices below are online-only supplementary material, not part of the printed article, but carried in the preprint source so every number in the main text is reproducible. Each, and each relocated figure and table, is cross-referenced where the main text uses it.

#### INJECTION-RECOVERY SENSITIVITY VALIDATION

The quoted limits assume a  $5\sigma$  threshold behaves as idealised Gaussian noise predicts. We test that by injecting synthetic narrowband signals of known amplitude into real calibrated visibilities through the extraction code’s own mechanism – as far upstream of the statistic as the retained products allow – and measuring what the unmodified pipeline recovers. An early 24-tone, zero-drift-only Proxima Cen pilot put 50 per cent recovery near  $6\sigma$ ; the campaign below supersedes it, reproducing that as threshold marginality.

Six configurations span three bands and both resolution classes: three coarse Band 7 spws of AT Mic, one coarse Band 3 and one coarse Band 6 spw of AU Mic, and AU Mic’s 488-kHz flagband window (49-trial drift grid). Into each we place a complex tone at the star position at amplitudes 4–10 $\times$  the statistic’s own noise, drift fractions  $f_{\text{drift}} = 0, 0.25, 0.5, 0.75, 1.0$  of the window’s searched ceiling, on- and half-grid rates, and channel-centred and boundary placements: 200 trials per configuration, 1200 in all, run through the unmodified search under a frozen recovery criterion ( $T \geq 5$ , peak channel within 3 of the injection, peak drift within one grid step; Fig. 12).

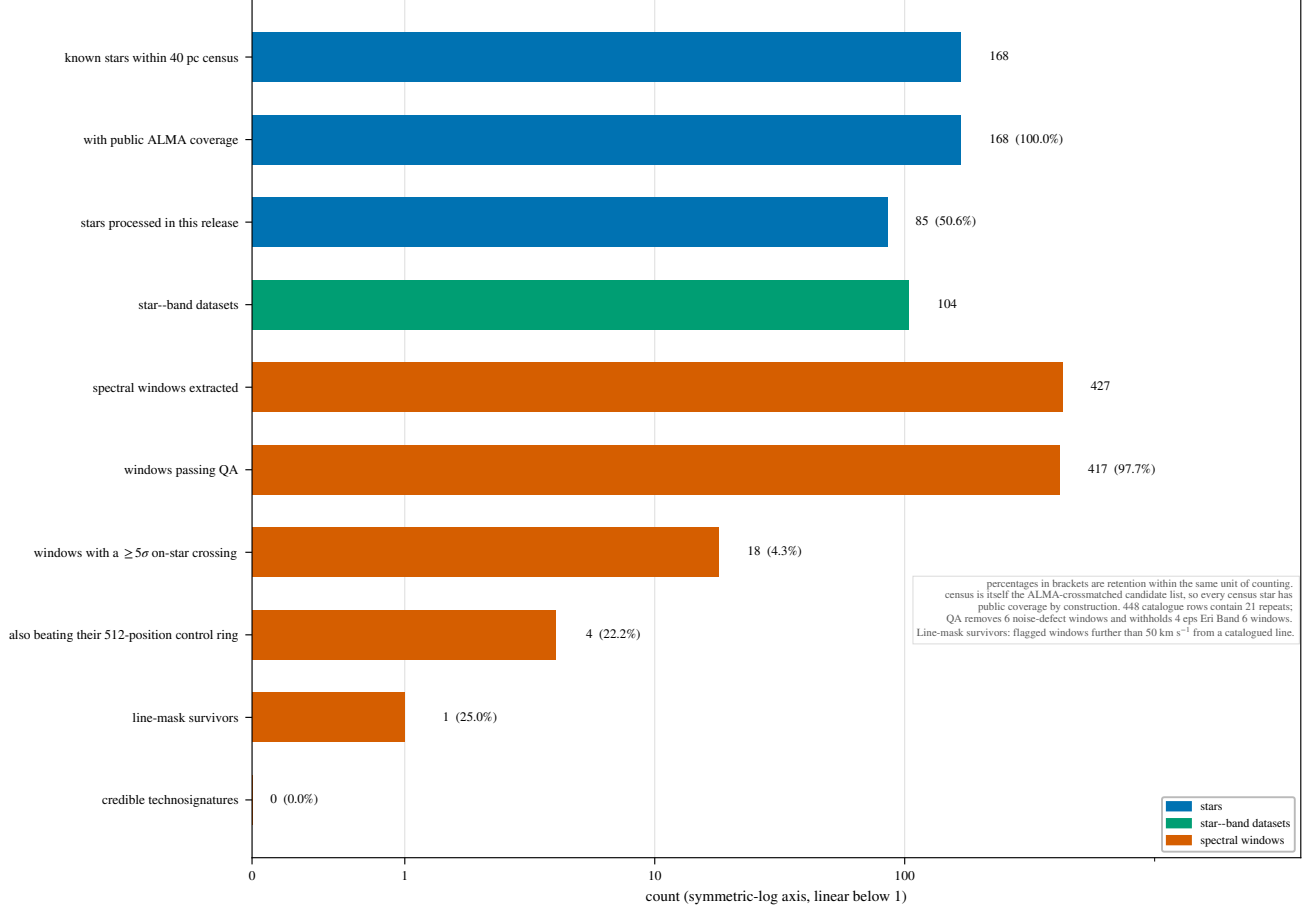


FIG. 10.— The survey’s selection funnel, from archive availability to the zero-candidate outcome. Top: star selection. Bottom: the archive-availability accounting, on the  $\leq 20$  pc subset and quoting its counts – public on-star execution blocks from the proper-motion-matched archive census, the star-execution-block pairs in scope, the window-level records with their correct unit ( $\sim 4$  spectral-window rows per pair), and the search outcome. The searched release spans 100 execution blocks; the difference between availability and scope is the recurrence pool drawn on by the AU Mic second-epoch test (§5.4).

TABLE 16  
CONSOLIDATED FAMILY-WISE TRIALS ACCOUNTING FOR THIS RELEASE; THE BOOKKEEPING LADDER IS IN APPENDIX P.

| Trials factor                                       | Unit           | Count                                 |
|-----------------------------------------------------|----------------|---------------------------------------|
| Channels per window (median; maximum)               | window         | 118; 3770                             |
| Drift trials per window (median; maximum)           | window         | 4; 1944                               |
| Channel $\times$ drift trials                       | release        | $3.97 \times 10^7$                    |
| Effectively independent channel cells <sup>b</sup>  | release        | $8.5 \times 10^4 - 1.1 \times 10^5$   |
| Control positions / $N_{\text{eff}}$                | window         | 512 / $\sim 30-43$                    |
| Windows (star-level tests)                          | release        | 417                                   |
| Windows containing an on-star hit                   | release        | 18                                    |
| Gaussian-expected chance crossings                  | release        | 11.4                                  |
| Cross-target frequency pairs tested                 | release        | 946                                   |
| Star-beats-ring rate $\hat{p}_{\text{sbr}}$         | window/release | 64/417 (15.3%)                        |
| Execution blocks spanned                            | release        | 100                                   |
| Principal false alarm: EB-block permutation         | release        | $P(F \geq 3) = 6.1\%$ ( $E[F]=1.04$ ) |
| Expected flags under the symmetric null ( $N/513$ ) | snapshot       | 0.81                                  |
| Wilson-rate bracket on $P(F \geq 3)$                | release        | 4–19%                                 |

<sup>b</sup>From the measured lag-one channel autocorrelation of the drift-tracked statistic,  $\rho_{\text{lag1}} = 0.989-0.991$ .

The first result is structural. The per-channel time-median subtraction that removes each position’s continuum also absorbs any phase-steady carrier dwelling in one channel for more than roughly 40 per cent of the track. On coarse windows this is total – every in-grid drift is sub-channel over the track, so *every* in-grid carrier is static in this sense – and the pipeline-identical variant recovers 0 of 500 injections at every amplitude from  $4\sigma$  to  $10\sigma$ . The coarse-window EIRP<sub>5 $\sigma$</sub>  limits therefore bound transmitters time-variable on track timescales, amplitude-, phase-, or duty-cycle-modulated below the  $\sim 50$  per cent dwell threshold, and say nothing about phase-steady carriers. The fine window measures the same boundary in drift space: carriers with  $f_{\text{drift}} \leq 0.25$  dwell  $\geq 40$  per cent of the track per

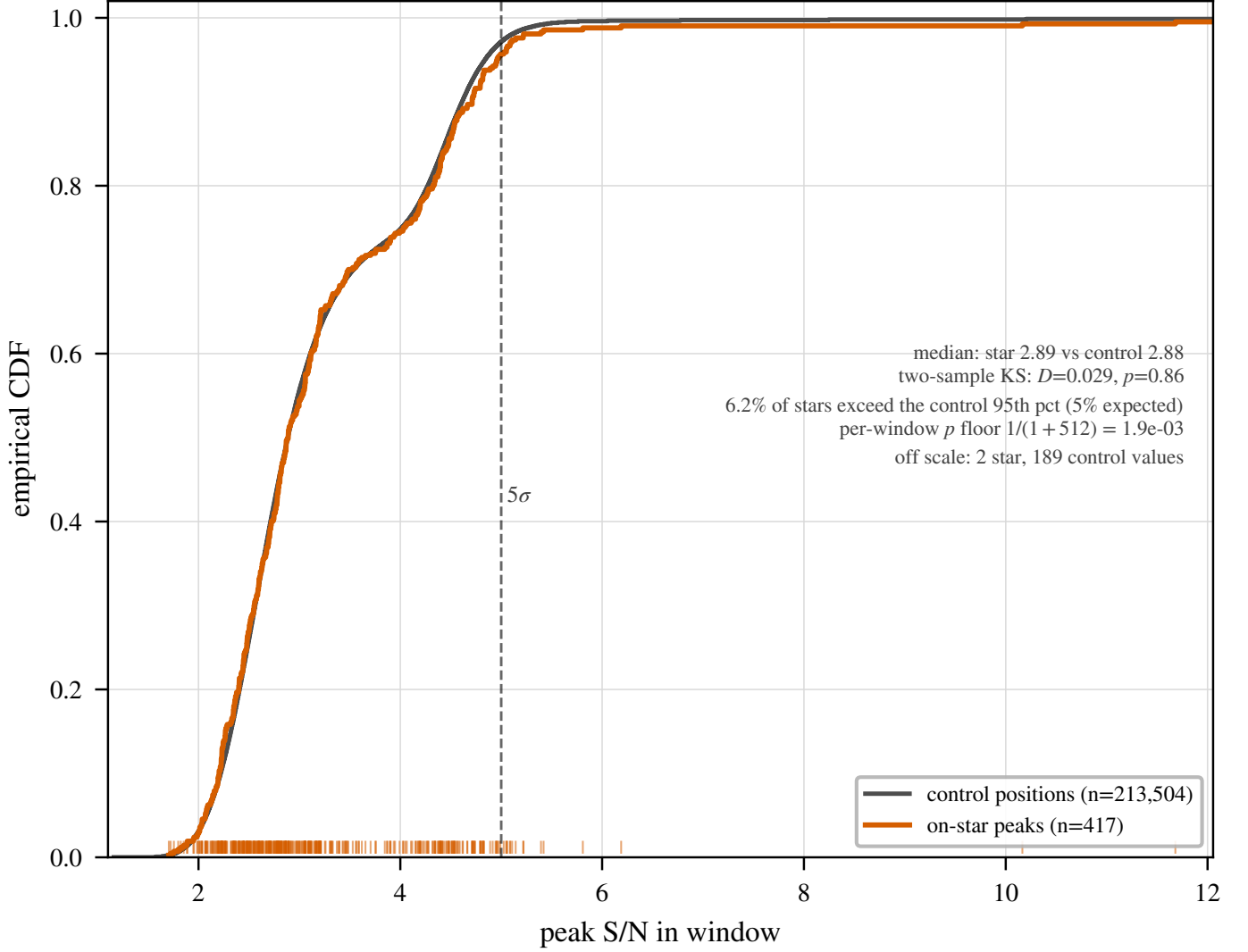


FIG. 11.— The symmetric star-versus-control null on the seven locally reprocessed windows. Each window’s  $p$  is the add-one rank of the star statistic  $T_*$  against the 512 control statistics under identical treatment, Uniform(0,1) under the null. All seven lie at  $p \geq 0.81$ ; with  $n = 7$  the 95 per cent Kolmogorov band is wide, so the content is the absence of star-side excess, not a precise distribution.

channel and are absorbed; those with  $f_{\text{drift}} \geq 0.5$  survive and constitute the searched class.

The second result is the completeness curve. On the fine window, drifting carriers ( $f_{\text{drift}} \geq 0.5$ , pooled over grid phase and sub-channel placement) are recovered 17, 42, 58, 75 and 83 per cent of the time at 4, 5, 6, 8 and  $10\sigma$ , saturating by twice threshold power (Fig. 4). The machinery alone (tone injected after the subtraction) is amplitude-faithful to within  $-9/+14$  per cent for channel-centred tones at every drift fraction; boundary-straddling placement costs a factor 0.6–0.8 in the statistic and recovers 0 per cent at  $5\sigma$  against 83 per cent for centred ones (67 and 100 per cent at  $10\sigma$ ), with the 50-per-cent crossing near  $1.4\times$  threshold amplitude and full recovery at  $\sim 1.6\times$  threshold power. Survey completeness uses the placement-pooled curve, conservative for the boundary-straddling case (§6). For that class  $\text{EIRP}_{50} = 1.10 \text{EIRP}_{5\sigma}$  (the interpolated half-recovery point,  $5.5\sigma$ ), while  $\text{EIRP}_{90} > 2.0 \text{EIRP}_{5\sigma}$  is a bound only: the curve reaches 83 per cent at the largest amplitude injected, so the 90-per-cent point lies beyond the measured range. Coarse windows carry no searched-class  $\text{EIRP}_{50}$  by construction; Table 17 collects the per-class reading. The scope is configuration-level: six configurations on three stars, directly covering Bands 3, 6, and 7. Band 8 windows share the coarse-channel construction, so the static-carrier result holds there by the same mechanism, but their drifting-class completeness is transferred rather than measured (§6). Not spanned: other target environments, primary-beam-offset positions, integration durations, and edge-phased drift grids; campaigns on Band 8 data and on the  $\beta$  Pictoris windows are infeasible with the measurement sets retained locally, and are declared rather than approximated.

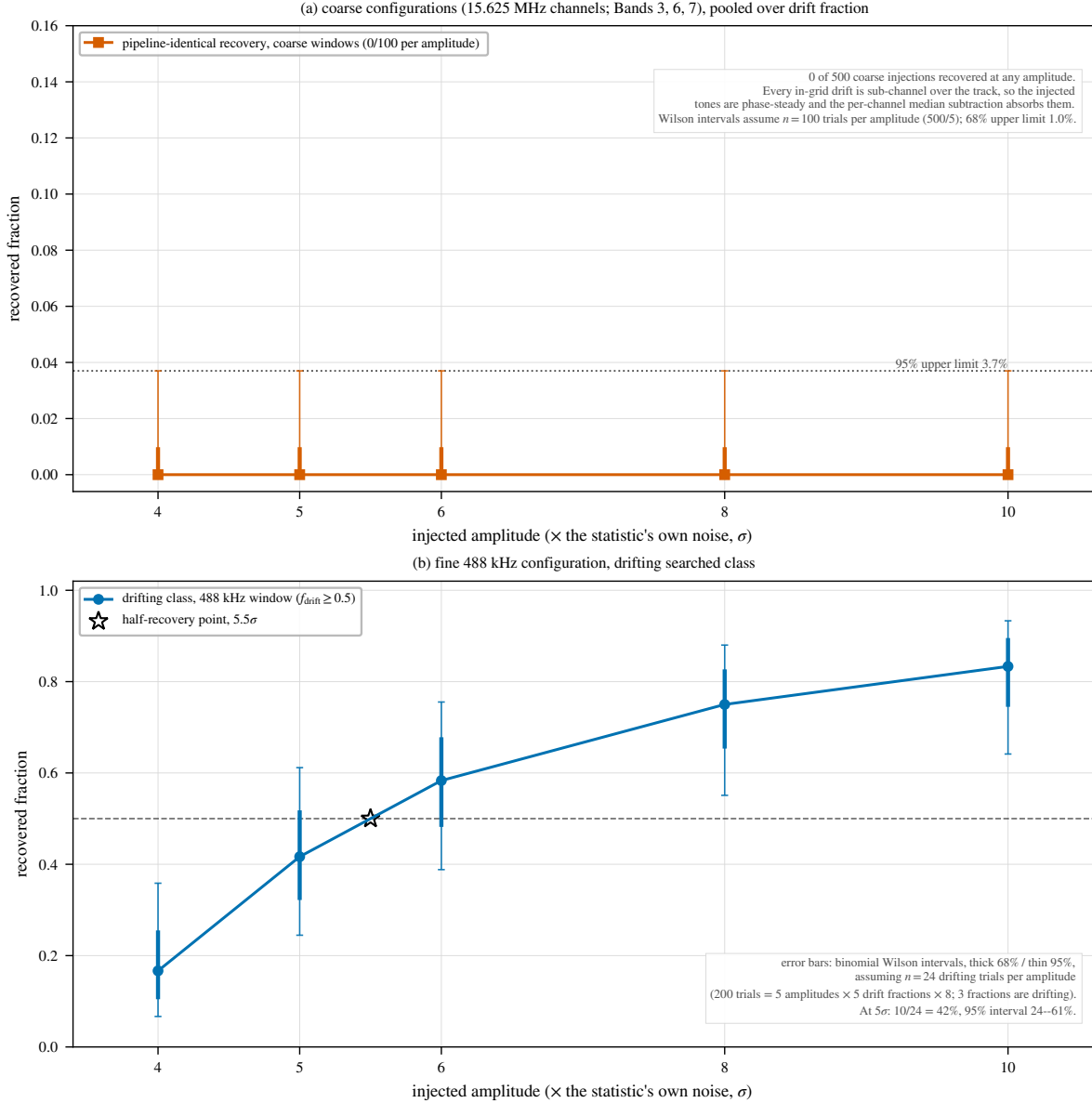
#### CLOSURE-PHASE POINT-SOURCE VETTING: SUMMARY AND FULL SPECIFICATION

Threshold-crossing candidates face a further automated test, one single-dish and beamformed surveys lack because it needs multiple independent baselines. Its sensitivity limit (§4.4) is quantified below: at these flux densities it has no discriminating power, so every flagged window’s formal consistency with a point source carries no evidential weight (§4.4).

TABLE 17  
PER-CLASS READING OF THE QUOTED LIMITS.  $C_{5\sigma}$  IS THE MEASURED RECOVERY OF THE CLASS AT THE NOMINAL THRESHOLD.

| Window class         | Searched class | $C_{5\sigma}$               | EIRP <sub>50</sub>      | Static-carrier recovery |
|----------------------|----------------|-----------------------------|-------------------------|-------------------------|
| Fine ( $\leq 1$ MHz) | drifting       | 0.42                        | 1.10 EIRP <sub>50</sub> | 0/20 (absorbed)         |
| Coarse (15.625 MHz)  | time-variable  | not calibrated <sup>a</sup> | ...                     | 0/500 (by construction) |

<sup>a</sup>The injected tones were all phase-steady on a coarse window (see the appendix text), so recovery of the time-variable modulation defining the coarse searched class is unmeasured. Static carriers are suppressed on both classes.



Measured injection recovery. Denominators are reconstructed from the campaign design, not recorded in the retained products: the fine-window value  $n = 24$  is the unique design-consistent choice that reproduces all five quoted percentages exactly.

FIG. 12.— Measured injection recovery on real ALMA data (1200 trials, this appendix). Top: the five coarse-channel configurations, pooled over drift fraction and grid phase: the pipeline-identical variant (solid) recovers nothing by construction, leaving only the machinery's detection fraction (dashed) meaningful. Bottom: the fine-channel configuration, where the drift dimension is real and pipeline-identical recovery  $C(A, f_{\text{drift}})$  rises with amplitude for drifting carriers, static and near-static ones being absorbed. Boundary-straddling tones can fail the frozen criterion's peak-drift clause at high drift while still detected – hence the non-monotonic drift dependence near the ceiling.

The closure phase, the sum  $\arg(V_{ij}) + \arg(V_{jk}) - \arg(V_{ik})$  around a closed triangle of baselines  $i-j-k$ , is exactly zero for an unresolved point source at any sky position, whose visibility phase is linear in  $(u, v)$  and cancels around any closed loop, and is immune to per-station calibration errors. Being position-independent, it asks only whether the signal is a coherent point source at all: a coherent far-sidelobe interferer (satellite, aircraft) would itself produce near-zero closure phase and pass.

A z-score assesses consistency with zero: the mean drift-tracked closure phase over its standard error across the closed triangles, with three properties handled explicitly. Triangles of one observation are not independent: any two sharing a baseline share its thermal noise, so the independent closure quantities are fewer than the triangles (of order  $10^4$  for a 40-antenna configuration) and the aggregate score must not treat them as independent trials. Closure phases wrap at  $\pm\pi$ , making the statistic circular (unit-phaser averaging). And its null distribution is calibrated empirically against the observation’s own integrations, not assumed Gaussian, absorbing the shared-baseline covariance too.

The calculation was validated on synthetic visibilities: zero closure phase for a noiseless point source, consistency with zero for a noisy one, inconsistent scatter for an RFI-like non-point source. As positive control we drove continuum detections through the machinery at their own frequencies on four fields of the local reprocessing set: AT Mic Band 7 (the known  $\sim 2$ -arcsec binary, as resolving control), AU Mic Band 6 and Band 3 continuum, and the AU Mic Band 6 frequency 230.8252 GHz. Triangle closure phases came from time- and channel-averaged unflagged cross visibilities, with  $\sigma_{CP}$  propagated from per-baseline scatter between integrations. Per-baseline S/N is 0.5–0.7 at the median and never exceeds 5.1 (0.02 per cent of triangles have all three baselines above S/N=5), so every field’s closure phases are indistinguishable from noise, including the resolved binary, whose  $\sim 0.4$  mJy total flux sits below the per-baseline noise of these short integrations. The test triggers only on threshold crossing, at negligible cost.

#### CHIRPED-DRIFT AND PERIODICITY SCREEN: SUMMARY AND FULL SPECIFICATION

Stacking at constant linear drift rate makes the primary search blind by design to a transmitter whose drift rate changes during the observation, and to pulsed or periodically-modulated carriers. We implemented a chirped (quadratic-drift) extension of the same stacking, on a coarser grid of 21 linear-drift  $\times$  5 chirp-rate trials, and a per-channel time-domain periodicity search – a per-position FFT along each channel’s time axis, its peak power against a robust per-channel noise estimate. Both vet the star’s dynamic spectrum against 8 retained control positions, as the primary search does its control ring; those 8 are a strict subset of that window’s 512-position ring, not an independent ensemble (§4), and exist for every processed target, so such analyses re-extract nothing.

On the 122 retained spectra, with a 9-member ensemble and coarser grids, this is a demonstration rather than survey-grade: it establishes only that the extension runs and behaves correctly under the null, which it does. The exceedance rates (10.7% chirp, 13.1% periodicity, against the  $\sim 11\%$  chance rate of a star topping such an ensemble) are as expected, no instance is treated as a candidate, and no limit derives from it. Extending both to the full 512-position ring, at negligible cost since the spectra are retained, would promote the screen to survey grade; that is deferred to the completed survey.

#### CROSS-TARGET FREQUENCY-OCCUPANCY CHECK: SUMMARY, KERNEL, COORDINATES, AND SCOPE

This single-pointing archival survey has no ON/OFF cadence against radio-frequency interference; its analogue is the many unrelated stars sharing similar correlator setups, where a narrowband feature recurring at the same observed *sky frequency* is almost certainly RFI or instrumental. We cross-match threshold crossings towards different targets falling within  $k = 3$  native channels of the coarser width (kernel below), so the null rejects sky-frequency recurrence only. Across the 417 windows and 107 tabulated crossings, no pair falls within the kernel, or within 0.5 MHz. The kernel ( $\sim 46$  kHz) is narrower than the  $\beta$  Pictoris line offsets of 10.6 and 21.9 MHz, so those candidates were dispositioned by line-catalogue inspection, not this check.

*Matching kernel.* Across every target/spw result in this release, the check cross-matches both the best-fit-channel frequency (whether or not it formally crossed the threshold) and the list of formal  $\geq 5\sigma$  crossings, pairing two crossings at frequencies  $\nu_1, \nu_2$  towards different targets when  $|\nu_1 - \nu_2| < \max(k \Delta\nu_1, k \Delta\nu_2)$ : the kernel scales with the coarser native channel width, not a fixed tolerance. The 5 MHz tolerance of the original formulation corresponds to  $k$  between  $\sim 0.2$  and  $\sim 330$  across this release’s native channel widths.

*The three frequency coordinates.* The *sky frequency* of a hit, reconstructed from archive spectral-window metadata in the observation’s recorded frame; the *topocentric frequency*, differing from a barycentric convention by up to the  $\pm 30 \text{ km s}^{-1}$  barycentric term ( $\sim 23$  MHz at 230 GHz, far wider than the matching kernel); and the *correlator channel index* of the native spectral window. Our cross-match probes the first; a frame-convention inconsistency between two observations, or an artefact fixed in the second, would defeat it, so the frame-convention disclosure queued with the symmetric-null recomputation (§5.4) feeds this check too. Artefacts fixed in the correlator frame – birdies, local-oscillator and quantization spurs, bandpass-edge ripple – recur at the same intermediate frequency or channel number, not the same sky frequency. A cross-match on the third coordinate (same channel number, identical correlator setup) is possible with the retained products and also returns zero pairs. A full intermediate-frequency reconstruction, pairing artefacts across tunings, is not recoverable from the retained metadata and stays with the completed survey.

*Exposure and exclusions.* The null’s power is the count of target pairs sharing frequency coverage. On the  $\leq 20$  pc subset this check has run over, 366 of the 946 star pairs (39%) overlap in searched frequency by at least 0.01 GHz, 359 (38%) by 0.5 GHz and 345 (36%) by 1 GHz. Of those, 109 involve one of the five stars carrying formal crossings, so every crossing frequency was tested against genuinely overlapping independent targets (Table 18). Two execution

TABLE 18  
TARGET PAIRS WITH THE DEEPEST SEARCHED-FREQUENCY OVERLAP; A SHARED TUNING IS WHERE A RECURRING SKY-FREQUENCY FEATURE WOULD BE CAUGHT.

| Target pair                 | Overlap (GHz) |
|-----------------------------|---------------|
| $\gamma$ Lep – $\gamma$ Vir | 14.25         |
| $\gamma$ Vir – TRAPPIST-1   | 10.78         |
| $\gamma$ Lep – TRAPPIST-1   | 10.77         |
| HD 10647 – HR 1010          | 10.13         |
| AU Mic – $\beta$ Pic        | 8.18          |
| GL 3379 – HD 285968         | 7.32          |
| GJ 849 – LHS 1140           | 7.32          |
| TRAPPIST-1 – Wolf 359       | 7.31          |
| Proxima Cen – Wolf 359      | 7.31          |
| Proxima Cen – TRAPPIST-1    | 7.31          |

TABLE 19  
PER-BAND SUMMARY OF THE SEARCHED WINDOWS AT THE SNAPSHOT EPOCH.  $n$  IS THE WINDOW COUNT, “STARS” THE DISTINCT STARS CONTRIBUTING, AND THE EIRP COLUMNS THE MEDIAN AND BEST NOMINAL  $5\sigma$  THRESHOLDS. “FINE” COUNTS WINDOWS CHANNELISED BELOW 1 MHz.

| Band | $n$ | Stars | Median EIRP (W)      | Best EIRP (W)        | Fine |
|------|-----|-------|----------------------|----------------------|------|
| 3    | 44  | 9     | $9.8 \times 10^{13}$ | $1.6 \times 10^{13}$ | 4    |
| 4    | 5   | 1     | $2.5 \times 10^{13}$ | $2.3 \times 10^{13}$ | 0    |
| 5    | 5   | 1     | $4.0 \times 10^{13}$ | $3.4 \times 10^{13}$ | 0    |
| 6    | 217 | 53    | $1.3 \times 10^{15}$ | $2.8 \times 10^{12}$ | 58   |
| 7    | 156 | 38    | $2.8 \times 10^{15}$ | $2.0 \times 10^{12}$ | 50   |
| 8    | 12  | 3     | $6.1 \times 10^{16}$ | $3.2 \times 10^{15}$ | 3    |

blocks are shared between catalogue entries – the UV Ceti binary’s (one field, the G 272-61A/B entries) and the AT Mic pair’s (AT Mic B and AT Mic AB; §3) – whose intra-pair “coincidences” are identities by construction, excluded from the cross-target test.

A bookkeeping caveat bounds the null on that subset: the per-window hit lists of the interim retained product are capped at 50 frequencies, so the true count over the subset is 171 crossings against 107 tabulated. The difference is one window,  $\beta$  Pic Band 6 ultra-fine, with 114 formal crossings of which the 50 strongest are tabulated; those 50 span 0.748 MHz, and the four crossings towards HD 285968, the only other star with crossings there, lie 13.7 MHz from that extent but inside the window’s full 54-MHz searched span, so the 64 untabulated crossings, whose frequencies the retained products do not carry, could not be tested: the null is demonstrated for the tabulated set, unresolved for the remainder – a limitation of the retained product, not of the criterion, which the public release removes by shipping uncapped hit lists (Data Availability).

#### PER-TARGET RESULTS

At this scope a typeset table would run to many pages, so these results – data rather than argument – ship with the manuscript source as `per_target_results_v3.29.csv`, summarised here.

One row per extracted spectral window (448 at the snapshot epoch, each with its quality-control status) carries star name and heliocentric distance, ALMA band and execution block, searched frequency range and native channel width, nominal  $\text{EIRP}_{5\sigma}$  threshold, line-excluded continuum rms, stellar and control-ring peak signal-to-noise ratios, the count of controls reaching or exceeding the star, the empirical  $p$ -value of §5.4, drift ceiling, on-source time, and nearest catalogued transition with velocity offset. Every quantity in this paper is reproducible from it.

$\text{EIRP}_{5\sigma}$ , the nominal threshold, is the minimum EIRP that would have produced a  $5\sigma$  detection in that window,  $\text{EIRP}_{5\sigma} = 4\pi d^2(5\sigma_{\text{comb}})\Delta\nu$ , on the same threshold and primary-beam correction as the continuum column, so the row’s two limits are comparable. The per-channel flux-density threshold follows as  $S_{\text{min}} = \text{EIRP}_{5\sigma}/(4\pi d^2 \Delta\nu_{\text{ch}})$ , the narrowband per-channel noise and *not* the tabulated rms, that target’s line-excluded full-bandwidth continuum noise.

Bands 6 and 7 supply 83 per cent of the windows, an imprint of the archive rather than a design choice. Band 8 windows are an order of magnitude less sensitive in EIRP at fixed distance, and Bands 4 and 5 have one star each, so neither carries weight in the aggregate limits.

#### THE VELOCITY-SPACE LINE MASK: JUSTIFICATION AND RETROSPECTIVE APPLICATION

The one-channel line check failed the  $\beta$  Pictoris candidates (§5.4): a native channel spans a fraction of a  $\text{km s}^{-1}$ , while circumstellar kinematics shift real emission by many channels. The replacement mask takes its tolerance from the kinematic budget of circumstellar gas – Keplerian motion where these molecules survive ( $\gtrsim 10$  au)  $\lesssim 13 \text{ km s}^{-1}$ ; cold-gas linewidths  $\lesssim 1 \text{ km s}^{-1}$ ; the eight masked species cold dense-gas tracers, no bulk-wind term for the dwarfs dominating the sample; the uncorrected barycentric term bounded by  $\pm 30 \text{ km s}^{-1}$  – and is empirically conservative: the largest

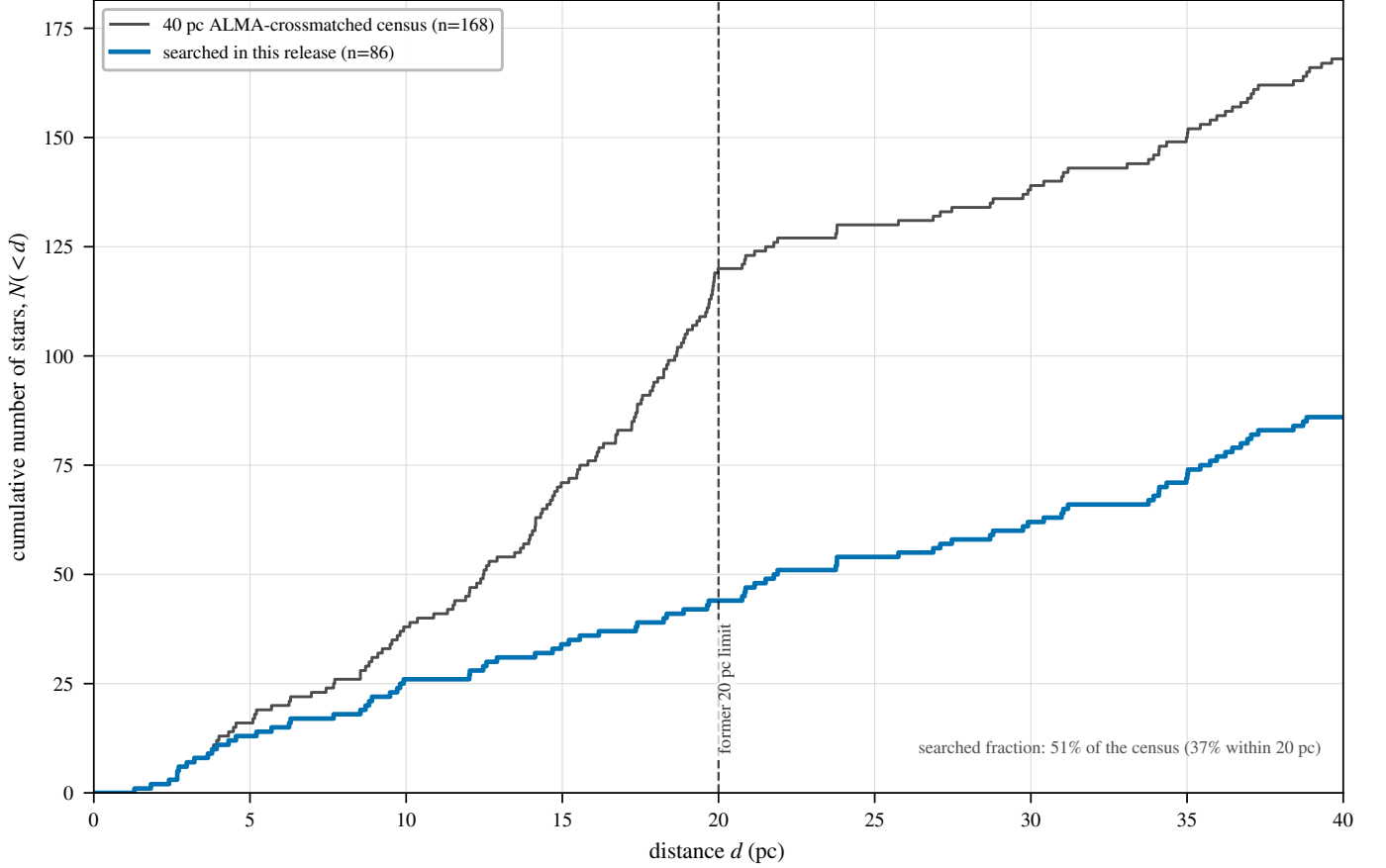


FIG. 13.— The ALMA-covered subset versus the stellar population within 20 pc: cumulative  $N(<d)$ . Grey: the reference Gaia DR3 catalogue (2357 stars); blue: this paper’s census sample, only  $\sim 5$  per cent of the total. The curves are not expected to track one another (§3).

TABLE 20

PER-BAND CHARACTERISATION OF THE 417 SEARCHED WINDOWS. “Sys.” COUNTS UNIQUE STELLAR SYSTEMS IN THE BAND, SO ONE OBSERVED IN TWO BANDS ENTERS BOTH ROWS (79 SYSTEMS, 85 STARS IN TOTAL); “EBs” THE CONTRIBUTING EXECUTION BLOCKS, BY DE-DUPLICATED-WINDOW PROVENANCE; “Win.” THE SEARCHED WINDOWS;  $\Delta\nu_{\text{ch}}$  THE NATIVE CHANNEL WIDTH IN MHz (MEDIAN, RANGE IN BRACKETS); “Union” THE UNIQUE SKY FREQUENCY AFTER MERGING OVERLAPPING WINDOWS;  $\text{EIRP}_{5\sigma}$  THE MEDIAN NOMINAL THRESHOLD OVER ITS WINDOWS;  $\dot{\nu}_{\text{ceil}}$  THE DRIFT-RATE CEILING RANGE IN  $\text{Hz s}^{-1} \text{GHz}^{-1}$ . THE “ALL” ROW IS OVER THE WHOLE SAMPLE, NOT A COLUMN SUM: EBs AND SYSTEMS RECUR ACROSS BANDS, AND THE FREQUENCY UNIONS ARE DISJOINT BY CONSTRUCTION.

| Band | Sys. | EBs | Win. | $\Delta\nu_{\text{ch}}$ (MHz) | Union (GHz) | Med. $\text{EIRP}_{5\sigma}$ (W) | $\dot{\nu}_{\text{ceil}}$ |
|------|------|-----|------|-------------------------------|-------------|----------------------------------|---------------------------|
| 3    | 8    | 9   | 40   | 15.63 (0.244–15.63)           | 20.6        | $1.1 \times 10^{14}$             | 12.0–13.3                 |
| 4    | 1    | 1   | 4    | 15.63                         | 6.9         | $2.6 \times 10^{13}$             | 12.0                      |
| 5    | 1    | 1   | 4    | 15.63                         | 6.9         | $4.4 \times 10^{13}$             | 12.0                      |
| 6    | 52   | 52  | 206  | 15.63 (0.015–31.25)           | 28.7        | $1.4 \times 10^{15}$             | 12.0–13.3                 |
| 7    | 33   | 34  | 151  | 15.63 (0.061–15.63)           | 22.6        | $2.9 \times 10^{15}$             | 12.0–13.3                 |
| 8    | 3    | 3   | 12   | 15.63 (0.061–15.63)           | 6.6         | $6.1 \times 10^{16}$             | 12.0                      |
| All  | 79   | 100 | 417  | 15.63 (0.015–31.25)           | 92.4        | $1.3 \times 10^{15}$             | 12.0–13.3                 |

attributed circumstellar component ( $\beta$  Pic,  $-2.7$  to  $-3.6 \text{ km s}^{-1}$ ) sits at 7 per cent of it, and *zero* candidate frequencies lie  $13\text{--}50 \text{ km s}^{-1}$  from their nearest transition, so a  $\pm 13 \text{ km s}^{-1}$  Keplerian-only mask would have suppressed the same candidates and released the same five: the wider tolerance costs no yield and buys robustness against radial-velocity catalogue error.

Two of the six remaining crossing windows fall inside the mask (the  $\beta$  Pic coarse window without a candidate at  $-3.6 \text{ km s}^{-1}$ ; HD 285968 at  $+8.7 \text{ km s}^{-1}$ , already rejected by its control ring, maximum 10.9 against a star peak of 6.5); the rest lie far outside any circumstellar tolerance (AU Mic  $+379$ , LHS 1140  $+115 \text{ km s}^{-1}$ ). Both products are reported: of the 18 windows with an on-star hit, 4 also beat their control ring, three accounted for by the mask, leaving one unattributed crossing dispositioned by its spatial behaviour. But the transitions that make astrophysical false positives cheap are also natural “Schelling points” for a deliberate communicator (Cocconi & Morrison 1959), so the released products keep a known-line-coincident list – every on-source  $5\sigma$  crossing inside the masked intervals, with its offset from each nearby transition – for a future search with independent discrimination to revisit.

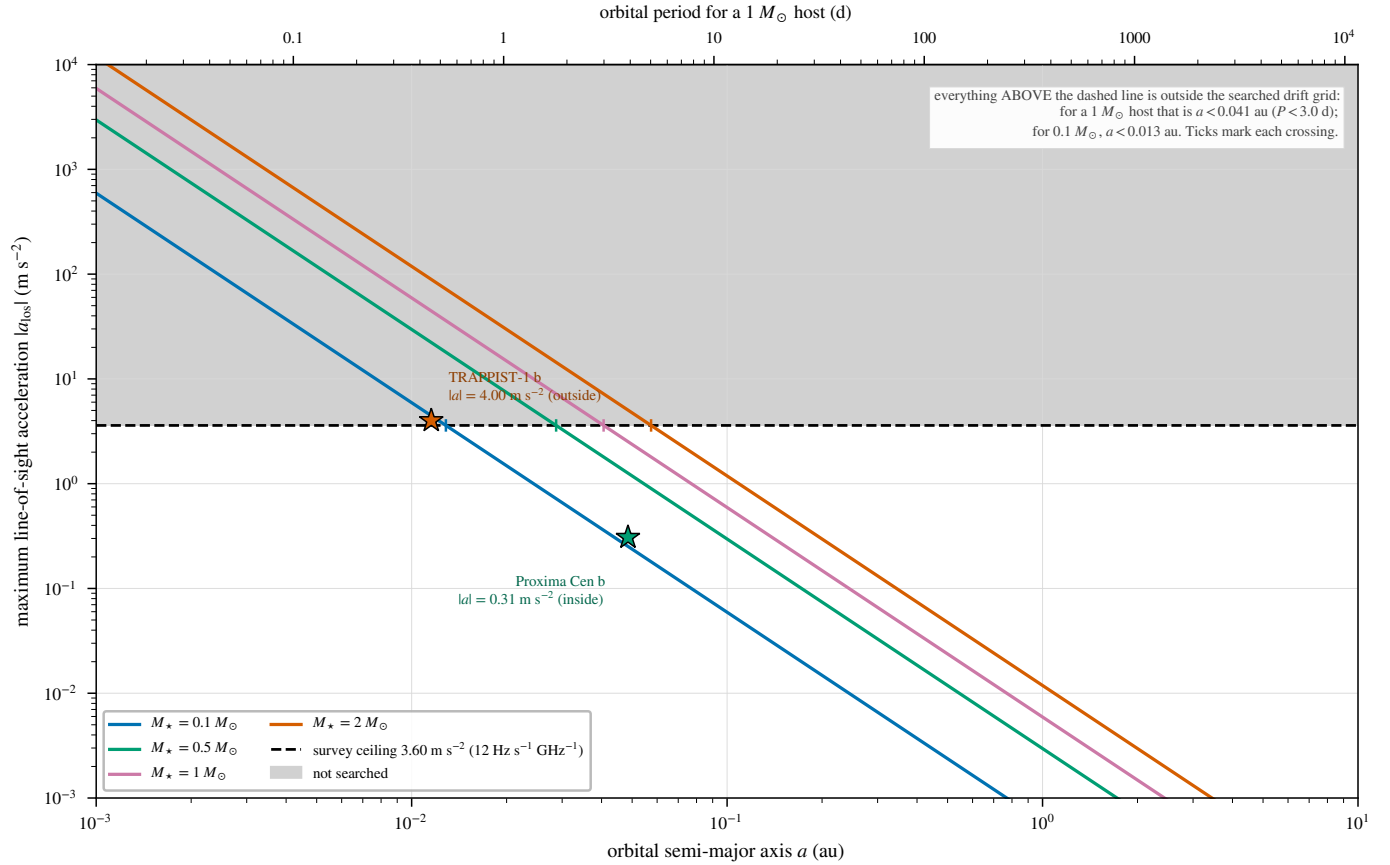


FIG. 14.— The drift-rate ceiling as a physical selection function. Maximum star–planet relative acceleration, edge-on and maximised over orbital phase, as the carrier drift rate it induces ( $\text{Hz s}^{-1}$  per  $\text{GHz}$  of carrier frequency), against orbital period, for host-star masses  $0.09$ – $1.8 M_\odot$  (lines) and the known planets of the planet-hosting targets here (points). The shaded band is the survey’s drift-search ceiling (§5.4); above it a transmitter is outside the searched grid. Only TRAPPIST-1 b reaches it (§6).

#### AU MIC: THE THREE TESTS IN FULL

*Localisation.* The first-epoch block was reprocessed from the raw science data model with the observatory pipeline’s own calibration recipe – retaining 310 integrations against the release pipeline’s 249, on a finer drift grid – and the symmetric star-versus-control statistic of §5.4 run on the candidate window: the star does not beat its control ring. The frozen products agree only asymmetrically: the source-region peak exceeds the 512-control maximum by 0.12, below that statistic’s window-to-window scatter, and the candidate fails the single-position variant too. It joins the six validation windows in showing no localised excess at the stellar position.

*Recurrence.* The one further public Band 6 block covering 230.83 GHz, the same programme executed 2014 June 5, was searched identically: no channel within  $\pm 10$  of the first-epoch flagged channel exceeds 3.1, and nothing is localised to the stellar position. Its band rms (1.48 mJy per 488-kHz channel) matches the first epoch’s 1.9 mJy, so a comparable excess would have been recovered, and the frequency recurs at no other target here. Intermittent or precisely-scheduled transmissions outside the sampled windows remain unconstrained, a caveat quantified by the duty-cycle analysis of §6.

*Statistical scale.* The on-source peak (5.97) sits at the 92.8th percentile of the control-maxima distribution over all 417 windows (median 4.55, 90th percentile 5.84; Fig. 15): thirty (7.2 per cent) match or exceed it, add-one empirical  $p = 0.074$ , before any correction across 417 windows. That is the kind of marginal excess the false-alarm calculation predicts. It matches no catalogued molecular transition (287 MHz from CO  $J=2 \rightarrow 1$ ); closure-phase vetting finds it consistent with a point source, one-sided evidence only (Appendix F).

Two astrophysical contexts are recorded for future re-examination; neither grounds the rejection. AU Mic is an exceptionally active flare star, but the block’s own data test that: its broadband continuum sits at a quiescent 0.15 mJy, the two flares of MacGregor et al. (2020) (16.8 and 6.1 mJy, optically thin, GHz-wide) belong to a later epoch, and an 11 mJy narrowband event over it would be a factor  $\sim 70$  the continuum does not show. A flare origin needs a spectrally narrow coherent process leaving the continuum untouched, which the data neither require nor establish, and the same-tuning second-epoch twin shows no burst domination. Second, the controls share the primary-beam footprint, so smooth structure such as AU Mic’s debris belt raises star and controls alike and cannot alone produce a candidate.



TABLE 21

PER-STAR OBSERVING DUTY CYCLE, ORDERED BY DISTANCE, FOR THE SUBSET WITH COMPILED EPOCH ACCOUNTING. THE TWO SPATIALLY UNRESOLVED BINARIES SHARE EVERY OBSERVATION: SEPARATE CATALOGUE ROWS, SINGLE TRIALS, GIVING 79 INDEPENDENT SYSTEMS.

$N_{\text{win}}$ : SEARCHED SPECTRAL-WINDOW MEASUREMENTS IN THIS RELEASE, AFTER DEDUPLICATION;  $N_{\text{EB}}$ : DISTINCT EXECUTION BLOCKS SEARCHED;  $t_{\text{on}}$ : SUMMED ON-SOURCE INTEGRATION IN MINUTES, TAKING EACH EXECUTION BLOCK’S MEDIAN ACROSS ITS WINDOWS;  $T_{\text{span}}$ : SUMMED ELAPSED SPAN OF THOSE BLOCKS IN HOURS, INCLUDING CALIBRATION AND OVERHEADS, SO EXCEEDING  $t_{\text{on}}$ . RECONSTRUCTED FROM THE ALMA SCIENCE ARCHIVE (QUERIED 2026 SEPTEMBER 8) RATHER THAN RETAINED BY THE FROZEN EXTRACTION PRODUCTS, THESE WERE AUDITED AGAINST THE ARCHIVE’S EXPOSURE ACCOUNTING (ONLINE-ONLY APPENDIX Q), WHOSE TWO EXCEPTION GROUPS ARE CARRIED AT ARCHIVE-CONSISTENT EXPOSURES. CALENDAR DATES AND MJDs ARE IN APPENDIX A; PER-WINDOW VALUES ACCOMPANY THE MACHINE-READABLE TABLE.

| Target         | $N_{\text{win}}$ | $N_{\text{EB}}$ | $t_{\text{on}}$ (min) | $T_{\text{span}}$ (h) |
|----------------|------------------|-----------------|-----------------------|-----------------------|
| Proxima Cen    | 4                | 1               | 50.0                  | 1.05                  |
| Barnard’s Star | 4                | 1               | 50.0                  | 1.01                  |
| Wolf 359       | 4                | 1               | 50.0                  | 1.10                  |
| Sirius B       | 12               | 3               | 17.2                  | 0.27                  |
| G 272-61B      | 4                | 1               | 31.0                  | 0.62                  |
| G 272-61A      | 4                | 1               | 31.0                  | 0.62                  |
| CD-23 14742    | 4                | 1               | 50.0                  | 1.10                  |
| eps Eri        | 4                | 1               | 42.8                  | 0.89                  |
| tau Cet        | 1                | 1               | 23.2                  | 0.47                  |
| BD+05 1668     | 4                | 1               | 23.7                  | 0.46                  |
| HD 33793       | 1                | 1               | 14.1                  | 0.26                  |
| Wolf 28        | 1                | 1               | 48.4                  | 0.98                  |
| GJ 674         | 4                | 1               | 50.0                  | 1.04                  |
| GL 3379        | 4                | 1               | 45.9                  | 0.91                  |
| LSR J1835+3259 | 4                | 1               | 43.1                  | 0.80                  |
| HN Lib         | 4                | 1               | 24.2                  | 0.49                  |
| GJ 581         | 4                | 1               | 50.0                  | 1.03                  |
| lp 876-10      | 4                | 1               | 46.5                  | 0.74                  |
| 61 Vir         | 4                | 1               | 37.8                  | 0.95                  |
| chi01 Ori      | 4                | 1               | 46.4                  | 0.92                  |
| GJ 849         | 4                | 1               | 44.4                  | 0.88                  |
| gam Lep        | 8                | 2               | 63.7                  | 1.23                  |
| HD 285968      | 4                | 1               | 46.9                  | 0.93                  |
| AU Mic         | 12               | 3               | 136.1                 | 2.93                  |
| AT Mic B       | 4                | 1               | 14.6                  | 0.32                  |
| AT Mic AB      | 4                | 1               | 14.6                  | 0.32                  |
| gam Vir        | 8                | 2               | 67.7                  | 1.28                  |
| HR 1010        | 8                | 2               | 70.1                  | 2.02                  |
| TRAPPIST-1     | 12               | 3               | 168.4                 | 2.91                  |
| HD 69830       | 8                | 2               | 92.4                  | 2.04                  |
| CD-38 10980    | 4                | 1               | 9.6                   | 0.18                  |
| LP 349-25      | 4                | 1               | 44.2                  | 0.82                  |
| GJ14           | 4                | 1               | 51.1                  | 0.90                  |
| LHS 1140       | 4                | 1               | 43.3                  | 0.95                  |
| HD 38858       | 4                | 1               | 0.3                   | 0.03                  |
| HD 207129      | 4                | 1               | 47.9                  | 1.04                  |
| HD 23484       | 4                | 1               | 45.9                  | 0.91                  |
| HD 10647       | 8                | 3               | 107.2                 | 1.98                  |
| g Lup          | 3                | 1               | 8.7                   | 0.12                  |
| eta Crv        | 8                | 2               | 98.6                  | 2.04                  |
| HD 53143       | 4                | 1               | 41.8                  | 0.84                  |
| Wolf 219       | 4                | 1               | 1.0                   | 0.02                  |
| bet Pic        | 15               | 3               | 93.7                  | 1.81                  |
| eta Cru        | 4                | 1               | 30.2                  | 0.59                  |

#### THE POPULATION-LEVEL CONTINUUM ANOMALY SCREEN: SUMMARY AND FULL SPECIFICATION

Being a census rather than an interest-selected list, the sample supports an ancillary screen: whether a star’s continuum flux is anomalous against its own photospheric expectation (Planck function with Gaia DR3 parameters, Pecaut & Mamajek fallback radius where GSP-Phot is unavailable) and against same- $T_{\text{eff}}$  comparisons in coarse bins of at least five. It is exploratory, not a headline technosignature constraint: the samples are small (57 measurements: 17 detections, 40 limits; bins of 4–7 stars) and a millimetre excess is at best an indirect, model-dependent observable of waste-heat technology. This release yields one outlier,  $\chi^1$  Ori at Band 3 (excess ratio 2.52, the largest of the 17 detections; resampled tail probability 0.53 in its six-star bin, a ranked outlier of a small sample, not a significant departure), consistent with a spectroscopic G0V binary’s unresolved companion or chromosphere (Appendix M).

*Radius sourcing.* Where a Gaia GSP-Phot radius is unavailable – close to half the sample by construction – the pipeline substitutes a coarse Pecaut & Mamajek main-sequence  $T_{\text{eff}}$ –radius relation, whose scatter propagates into the excess-ratio distribution; stars can also depart from a mean relation through age, metallicity or unresolved multiplicity.

TABLE 22

MASKED BANDWIDTH (GHz) BY SPECIES AND ALMA BAND ON THE OPERATIVE MASK (DEFINED IN THE TEXT). COLUMNS B3–B8 AND ‘TOTAL’ ARE WINDOW-SUMMED (DENOMINATOR: THE GROSS SEARCHED BANDWIDTH); ‘UNION’ MERGES THE SAME INTERVALS ACROSS WINDOWS AND STARS (DENOMINATOR: THE 92.4 GHz UNIQUE-FREQUENCY UNION). SPECIES ROWS OVERLAP WHERE TWO MASKS SHARE A CHANNEL, SO THEY EXCEED THE ALL-SPECIES TOTALS. PER-STAR INTERVALS ARE IN THE RELEASED VETO-MASK EXPORT.

| Species              | B3   | B4   | B5   | B6    | B7   | B8   | Total | Union |
|----------------------|------|------|------|-------|------|------|-------|-------|
| CO                   | 0.12 | 0.00 | 0.00 | 1.51  | 0.86 | 0.00 | 2.49  | 0.33  |
| <sup>13</sup> CO     | 0.35 | 0.05 | 0.07 | 1.71  | 0.51 | 0.00 | 2.69  | 0.74  |
| C <sup>18</sup> O    | 0.04 | 0.00 | 0.00 | 0.00  | 0.00 | 0.00 | 0.04  | 0.04  |
| HCN                  | 0.00 | 0.00 | 0.00 | 0.00  | 0.12 | 0.00 | 0.12  | 0.12  |
| HCO <sup>+</sup>     | 0.10 | 0.00 | 0.07 | 0.00  | 0.36 | 0.00 | 0.53  | 0.50  |
| CS                   | 0.00 | 0.00 | 0.07 | 0.57  | 0.11 | 0.00 | 0.75  | 0.29  |
| CN                   | 0.87 | 0.00 | 0.00 | 11.04 | 2.59 | 0.00 | 14.50 | 2.69  |
| SiO                  | 0.00 | 0.00 | 0.00 | 0.27  | 0.81 | 0.00 | 1.08  | 0.26  |
| All species (merged) | 1.48 | 0.05 | 0.21 | 15.10 | 5.36 | 0.00 | 22.20 | 4.96  |

TABLE 23

ONE ROW PER SEARCHED STAR, ORDERED BY DISTANCE, IN TWO SIDE-BY-SIDE BLOCKS READ DOWN THE LEFT, THEN THE RIGHT.  $d$  IS THE DISTANCE USED BY THE SEARCH; “BANDS” THE ALMA BANDS WITH AT LEAST ONE SEARCHED WINDOW FOR THAT STAR;  $N_w$  THE NUMBER OF SEARCHED WINDOWS; EIRP<sub>5 $\sigma$</sub>  THE DEEPEST (SMALLEST) NOMINAL 5 $\sigma$  THRESHOLD AMONG THEM, IN W. STAR NAMES ARE THOSE OF THE RELEASED MACHINE-READABLE TABLE, CROSS-REFERENCING IT DIRECTLY. 85 STARS, 79 INDEPENDENT SYSTEMS, 417 SEARCHED WINDOWS; DE-DUPLICATED, NOISE-DEFECT AND WITHHELD WINDOWS ARE EXCLUDED HERE BUT PRESENT, AND LABELLED, IN THE RELEASED FILE.

| Star                             | $d$ (pc) | Bands | $N_w$ | EIRP <sub>5<math>\sigma</math></sub> | Star                           | $d$ (pc) | Bands | $N_w$ | EIRP <sub>5<math>\sigma</math></sub> |
|----------------------------------|----------|-------|-------|--------------------------------------|--------------------------------|----------|-------|-------|--------------------------------------|
| Proxima Cen                      | 1.30     | 6     | 4     | 3.30e+13                             | PM J034331958                  | 20.76    | 6     | 4     | 1.01e+15                             |
| NAME Barnards star               | 1.83     | 6     | 4     | 6.15e+13                             | V star TX PsA                  | 20.83    | 7     | 4     | 2.75e+14                             |
| Wolf 359                         | 2.41     | 6     | 4     | 7.84e+13                             | V star WW PsA                  | 20.87    | 7     | 4     | 2.76e+14                             |
| alf CMa B                        | 2.67     | 3,4,5 | 12    | 2.34e+13                             | j1256-1257                     | 21.15    | 7     | 4     | 8.32e+14                             |
| G 272-61B                        | 2.67     | 3     | 4     | 1.62e+13                             | hd92945                        | 21.51    | 7     | 4     | 2.65e+14                             |
| G 272-61A                        | 2.72     | 3     | 4     | 1.67e+13                             | WD2039-202                     | 21.77    | 6     | 4     | 1.29e+15                             |
| CD-23 14742                      | 2.98     | 6     | 4     | 1.12e+14                             | HD 45184                       | 21.89    | 6     | 4     | 3.03e+15                             |
| tau Cet                          | 3.65     | 6     | 1     | 3.79e+13                             | LP 476-207 783296              | 23.75    | 7     | 4     | 4.85e+14                             |
| BD05 1668                        | 3.79     | 6     | 4     | 2.71e+14                             | LP 476-207 384128              | 23.79    | 7     | 4     | 4.73e+14                             |
| HD 33793                         | 3.93     | 6     | 1     | 5.62e+14                             | HD202628                       | 23.79    | 6     | 4     | 1.77e+14                             |
| Wolf 28                          | 4.31     | 6     | 1     | 6.67e+13                             | HD 31392                       | 25.76    | 6     | 4     | 4.85e+15                             |
| GJ 674                           | 4.55     | 6     | 4     | 2.54e+14                             | CD-57 1054                     | 26.87    | 7     | 4     | 7.63e+14                             |
| GL 3379 Gaia DR3                 | 5.21     | 6     | 4     | 1.12e+14                             | LP 353-51                      | 27.10    | 7     | 4     | 1.23e+15                             |
| 3316364602541746048              |          |       |       |                                      |                                |          |       |       |                                      |
| LSR J18353259                    | 5.69     | 3     | 4     | 5.12e+13                             | hd107146                       | 27.47    | 7     | 4     | 7.26e+15                             |
| HN Lib                           | 6.25     | 6     | 4     | 1.49e+14                             | L 836-122                      | 28.70    | 7     | 4     | 7.11e+14                             |
| GJ 581                           | 6.30     | 6     | 4     | 6.01e+14                             | HD172555                       | 28.79    | 6     | 4     | 1.21e+14                             |
| lp 876-10 Gaia DR3               | 7.68     | 7     | 4     | 2.70e+13                             | HD161868                       | 29.75    | 6,7   | 8     | 1.88e+15                             |
| 6623351805412369024              |          |       |       |                                      |                                |          |       |       |                                      |
| 61 Vir                           | 8.53     | 7     | 4     | 4.52e+14                             | 51 Eri                         | 29.91    | 6     | 4     | 1.09e+15                             |
| chi01 Ori                        | 8.70     | 3     | 4     | 1.08e+14                             | HD 155555C                     | 30.41    | 7     | 4     | 6.91e+14                             |
| GJ 849                           | 8.81     | 6     | 4     | 8.30e+14                             | 2MASS J05241914-1601153 551040 | 30.99    | 7     | 3     | 3.62e+15                             |
| gam Lep                          | 8.91     | 6,7   | 8     | 9.27e+13                             | WD0346-011                     | 31.04    | 6     | 4     | 2.54e+15                             |
| HD 285968                        | 9.49     | 6     | 4     | 3.45e+14                             | 2MASS J05241914-1601153 717696 | 31.19    | 7     | 3     | 3.66e+15                             |
| AU Mic                           | 9.71     | 3,6   | 12    | 5.26e+13                             | UCAC2 20312880                 | 33.77    | 7     | 4     | 1.06e+15                             |
| V AT Mic B                       | 9.81     | 7     | 4     | 5.60e+13                             | HR 2562                        | 33.93    | 7     | 4     | 3.39e+15                             |
| NAME AT Mic AB Gaia DR3          | 9.92     | 7     | 4     | 5.82e+13                             | TWA 7                          | 34.10    | 7     | 4     | 7.39e+14                             |
| 6792436799475128960              |          |       |       |                                      |                                |          |       |       |                                      |
| gam Vir                          | 12.02    | 6,7   | 8     | 2.24e+14                             | WD 0407179                     | 34.11    | 6     | 4     | 2.76e+15                             |
| HR 1010 Gaia DR3                 | 12.04    | 6,7   | 8     | 6.50e+14                             | UCAC3 176-23654                | 34.35    | 7     | 4     | 9.42e+14                             |
| 4722135642226902656              |          |       |       |                                      |                                |          |       |       |                                      |
| TRAPPIST-1                       | 12.47    | 3,6,7 | 12    | 5.76e+13                             | GJ 2006B                       | 34.98    | 7     | 4     | 8.04e+14                             |
| HD 69830                         | 12.57    | 6,7   | 8     | 3.91e+13                             | ALMA J153702653-33192492       | 35.01    | 6,7   | 9     | 3.20e+15                             |
| CD-38 10980                      | 12.91    | 6     | 4     | 4.14e+14                             | GJ 2006A                       | 35.03    | 7     | 4     | 7.95e+14                             |
| LP 349-25                        | 14.13    | 3     | 4     | 2.57e+14                             | TYC 2211-1309-1                | 35.42    | 7     | 4     | 9.12e+14                             |
| GJ14 Gaia DR3 383426372059603712 | 14.69    | 6     | 4     | 1.21e+14                             | HD14055                        | 35.74    | 6,7   | 8     | 7.85e+14                             |
| LHS 1140                         | 14.96    | 6     | 4     | 6.24e+14                             | HD 48370                       | 35.95    | 6,8   | 8     | 2.86e+15                             |
| HD 38858 Gaia DR3                | 15.21    | 6     | 5     | 6.96e+14                             | HD 89452                       | 36.23    | 6     | 4     | 8.54e+15                             |
| 3023711269067191296              |          |       |       |                                      |                                |          |       |       |                                      |
| HD 207129 Gaia DR3               | 15.56    | 6     | 4     | 7.62e+14                             | HD61005                        | 36.45    | 6,8   | 8     | 7.60e+14                             |
| 6564091190988411520              |          |       |       |                                      |                                |          |       |       |                                      |
| HD 23484 Gaia DR3                | 16.17    | 6     | 4     | 1.70e+15                             | CP-72 2713                     | 36.72    | 6,7   | 8     | 8.26e+14                             |
| 4856592239127350400              |          |       |       |                                      |                                |          |       |       |                                      |
| HD 10647                         | 17.35    | 6,7   | 8     | 1.49e+14                             | HD170773                       | 36.93    | 6,7   | 8     | 5.47e+14                             |
| g Lup                            | 17.40    | 6     | 3     | 2.55e+14                             | TWA 3A 696000                  | 37.05    | 6     | 4     | 3.13e+14                             |
| eta Crv                          | 18.24    | 7,8   | 8     | 4.36e+13                             | Barta 161 12                   | 37.28    | 7     | 4     | 1.13e+15                             |
| HD53143 Gaia DR3                 | 18.34    | 6     | 4     | 2.84e+15                             | HD377                          | 38.40    | 6     | 4     | 1.48e+16                             |
| 5479222240596469632              |          |       |       |                                      |                                |          |       |       |                                      |
| Wolf 219                         | 18.88    | 6     | 4     | 8.43e+14                             | HD 139084B 921024              | 38.72    | 7     | 4     | 1.71e+15                             |
| bet Pic                          | 19.63    | 3,6   | 15    | 2.34e+13                             | HIP 490                        | 38.83    | 6     | 4     | 4.36e+14                             |
| eta Cru                          | 19.69    | 6     | 4     | 1.11e+15                             |                                |          |       |       |                                      |

Rerunning the screen with those parameters imposed on every star judges that scatter against the outlier threshold: the sole flag survives, the M and G/F bin scales are unchanged (already fallback-dominated), and the A/F-hot median moves from 4.3 to 3.9.

*Error model.* Evaluated before any flag, the error model is  $\sigma_{\text{tot}}^2 = \sigma_{\text{map}}^2 + (f_{\text{cal}} S_{\nu})^2 + \sigma_{\text{model}}^2$ :  $\sigma_{\text{map}}$  the image rms,  $f_{\text{cal}}$  the band-dependent absolute-flux systematic (5 per cent in Bands 3–6, 10 per cent in Bands 7–8),  $\sigma_{\text{model}}$  the photospheric-model uncertainty propagated from the Gaia temperature and radius errors and the main-sequence-relation scatter for fallback stars. The per-target file’s  $\sigma_{\text{tot}}$  column carries the map and calibration terms only; the model term enters the population-level excess-ratio screen, not the tabulated per-star errors. Read excess ratios near unity with both terms in mind: the model term dominates for fallback stars, the calibration term for the brightest detections.

*Binning and the outlier statistic.* Excess ratios are compared against similar- $T_{\text{eff}}$  stars in coarse, equal-width bins through a robust median/MAD outlier statistic, the bin width a fixed pipeline constant published with the release and chosen so that populated bins meet the five-comparison-star minimum. At that minimum the bin median rests on two or three values, the MAD scale is small-sample, and repeating the comparison for every star and bin brings a small expected number of chance flags, so the screen stays exploratory at this depth: it flags stars for inspection, and is not a per-star hypothesis test at a fixed false-alarm rate. Raising the minimum to ten comparisons is deferred to the completed survey, which will reach it in the occupied bins.

*Exact status in this release.* The screen is live: stellar parameters were ingested for every continuum-measured target (Gaia GSP-Phot for 16), and the detections are binned M/G/F/A-hot (7/6/4 members of the three populated bins, so every bin statistic is small-sample), with per-bin medians and log-scatters in the machine-readable release. The one measurement crossing the outlier threshold ( $z = 4.1$ ) is dispositioned above, with the two committed robustness reruns (fallback-only parameters; leave-one-out bin scatter).

#### CONTINUUM SEARCH (EXPERIMENTAL ANCILLARY ANALYSIS)

The continuum lane is ancillary and experimental, not a co-equal search (Appendix L).

The seventeen detections and forty non-detections of the lane are plotted in Figure 8 (right panel); an “upper limit” is  $5\times$  the map rms at the star’s position, primary-beam corrected like the EIRP thresholds; the per-target rms of Appendix I rescales it to any confidence level. The deepest non-detection is towards TRAPPIST-1 in Band 3. Under the combined error model of Appendix L sixteen of the seventeen stay above  $5\sigma$ ; the exception ( $\eta$  Corvi Band 7,  $4.7\sigma$  combined) is marginal. Bright peaks far from the target ( $\epsilon$  Eri Band 6 at  $10.2\sigma$ , 18 arcsec off the phase centre; GJ 674 and CD–23 14742 at  $10.7$  and  $9.9\sigma$ ) fall outside the 3-arcsec tolerance and are correctly recorded as non-detections; whether the  $\epsilon$  Eri peak is real off-target emission or a primary-beam artefact at  $\sim 0.7$  FWHM offset is undecidable from the retained products (holography-beam recomputation committed, §4).

Every detection is astrophysically unsurprising. The G 272-61A/B pair is the unresolved UV Ceti binary (0.85 mJy, attributable to neither component). Six others are young, active systems with debris discs or chromospheric activity ( $\tau$  Cet,  $\beta$  Pic B3/B6, AU Mic, AT Mic A/B). Six show the photospheric flux expected of their spectral types ( $\gamma$  Lep and  $\gamma$  Vir in both bands,  $\chi^1$  Ori,  $\eta$  Cru;  $\eta$  Corvi’s Band 8 limit is likewise photosphere-consistent). Two are late-M dwarfs of documented strong magnetic activity: LP 349–25 ( $10.5\sigma$ , 0.071 mJy) and the auroral emitter LSR J1835+3259 ( $6.7\sigma$ , 0.078 mJy; Hallinan et al. 2007), which merits deeper follow-up. Closure phase cannot discriminate structure for any of the seventeen at these flux densities (Appendix F), and the anomaly screen’s single outlier is astrophysically dispositioned (Appendix L).

#### A SERENDIPITOUS MOLECULAR-LINE CATALOGUE

The continuum search’s line-exclusion step (§4) flags every  $\geq 5\sigma$  outlier channel as a byproduct. No existing mm/submm line survey of the solar neighbourhood rests on a distance-limited census of archival coverage, so even a modest catalogue has value beyond its SETI motivation. Twelve of the outlier groups scanned to date are single- or few-channel features with no known transition within 300 MHz, most plausibly statistical. Six are plausibly real: CO( $J=1\rightarrow 0$ ) and CO( $J=2\rightarrow 1$ ) towards  $\beta$  Pictoris (the stellar-frame component of Table 6 and §5.4); a marginal CO( $J=2\rightarrow 1$ )-consistent feature recurring across two epochs towards HD 285968, tentative only (1–2 channels of significance per epoch); and a single-channel, 195-MHz-offset feature near SiO( $J=5\rightarrow 4$ ) towards HD 53143, more likely statistical than real. All are reported openly; the machine-readable catalogue (per-group target, band, channel coordinates, significance, nearest transition) ships with the tagged public release, which the completed survey’s full-sample re-scan will extend.

#### LESSONS FOR ARCHIVAL PIPELINES

Three of the four audit items of §5.3 are failure modes generic to automated archive mining. First, every lane needs its own geometric guard: the narrowband step aborts on item (ii)’s 52-arcmin pointing mismatch while the continuum-imaging step had none, an asymmetry the corrected release closes. Second, window-level bookkeeping anomalies yield to cross-target pattern recognition, not per-target sanity checks: item (iii) recurred with matching correlator parameters across two unrelated targets, and the recurrence, not either instance, marked the artifact. Third, name resolution is not position resolution: the  $\gamma$  Lupi entry was misidentified through the SIMBAD identifier list, not by positional error, since HD 139664 carries the historical main identifier “\* g Lup”, so a name lookup of “ $\gamma$  Lupi” returns

it, not the B2IV star seven times farther off (parallax 7.75 mas,  $3.7^\circ$  away). The cheap check that caught it should be standard: the star’s Gaia DR3 parallax, 57.476 mas, recovered digit-for-digit by inverting the pipeline distance, with  $T_{\text{eff}} \approx 6600$  K and cross-identifiers HR 5825, HIP 76829 and GJ 594 consistent with F4V. Archival surveys inherit the catalogues’ historical artefacts; position-anchored verification is the inexpensive antidote.

#### FALSE-ALARM ACCOUNTING UNDER THE SYMMETRIC CRITERION

Versions up to 3.24 carried a four-model bookkeeping ladder (execution-block permutation, hypergeometric, block bootstrap and a Poisson rate heuristic, spanning  $P(F \geq 3) = 6.1\text{--}9.2$  per cent) to calibrate a false-alarm rate for an *asymmetric* statistic with no closed-form per-window null rate. The symmetric statistic of §5.4 removes that need, giving a per-window rate of 1/513 by construction and a survey-level expectation of  $417/513 = 0.81$  flags, so the ladder is retired rather than updated.

The 1/513 rank floor of §5.4 can be lowered only by enlarging the control ring, which the primary beam bounds; and windows are not fully independent, grouped into 100 execution blocks sharing weather, calibration and correlator setup, so  $N/513$  is an expectation over a mildly correlated sample rather than  $N$  independent trials. Both limits are conservative for the conclusion drawn, that the flagged windows are consistent with chance: they widen rather than narrow the plausible null range.

#### THE SYMMETRIC STAR-VERSUS-CONTROL NULL IN FULL

The per-window  $p$ -value of the test of §5.4 is  $p = (1 + \#\{T_i \geq T_\star\})/(N + 1)$ , and its seven measured windows are the six validation windows of the local reprocessing set plus the AU Mic candidate’s reprocessed first-epoch window (§5.5). It is the designated replacement for the empirical star-versus-ring calibration as the remaining sets are re-run. That candidate also fails the single-position variant of the release statistic (star 5.22 against its ring maximum 5.85, §5.5).

Table 16 consolidates the family-wise trials factors: every per-window factor applies identically at the 512 control positions, making the control ring and the empirical star-beats-ring rate built on it the operative calibration, not any analytic correction.

#### ROBUSTNESS OF THE EMPIRICAL CALIBRATION

The release evaluates  $3.97 \times 10^7$  channel $\times$ drift cells; a Gaussian one-sided  $5\sigma$  tail ( $2.87 \times 10^{-7}$ ) over that count predicts 11.4 chance crossings, against the 18 windows with an on-star crossing. Raw counts run higher because the drift-tracked statistic is strongly correlated between adjacent channels ( $\rho_{\text{lag1}} = 0.989\text{--}0.991$ , i.e.  $8.5 \times 10^4\text{--}1.1 \times 10^5$  effectively independent cells, not  $3.97 \times 10^7$ ), so crossings arrive in contiguous runs. Collapsed to clusters (kernel of 3 native channels) they leave far fewer independent events – the count, not the raw count, that the trials expectation must be compared with, and the two agree. The per-crossing census for the extended sample is in the released products.

No single stratum drives the star-beats-ring rate. The stratification was run on the  $\leq 20$  pc subset, whose counts it quotes. By band: 20 per cent (B3,  $n = 40$ ), 10.9 per cent (B6,  $n = 128$ ), 17.8 per cent (B7,  $n = 45$ ), zero in the sparsely sampled B4/B5/B8 ( $n = 4$  each). By channelisation: 17.0 per cent fine ( $n = 47$ ) against 12.4 per cent coarse ( $n = 178$ ). By primary-beam offset: 15.7 per cent above the median offset (0.10 arcsec) against 11.0 per cent below. The disc-hosting caveat inverts empirically: disc- or planet-hosting stars contribute every crossing window and every spatially significant hit but a *lower* star-beats-ring rate (9.8 per cent) than non-hosts (18.3 per cent), so circumstellar emission shared by star and ring does not drive the statistic. Excluding the fifteen  $\beta$  Pictoris windows leaves one unattributed crossing among 402 windows, against 0.78 expected ( $P(\geq 1) = 54$  per cent).

Beyond the shared primary-beam gain of §5.4, neighbouring control positions are correlated below the synthesised-beam scale, so the independent spatial trials fall from 512 to  $N_{\text{eff}} \simeq 30\text{--}43$ , as the symmetric reprocessing runs measure directly – a floor to which the empirical 15.3 per cent rate, measured on the correlated ensembles themselves, is immune by construction. A continuum source in the ring raises the narrowband statistic only if it also emits a narrow line in-window, and the cross-target occupancy screen (Appendix H) covers recurring frequencies.

#### INTEGRATION-TIME AUDIT: METHOD AND EXCEPTIONS

The archive cross-check of §5.6 uses the ALMA Science Archive’s `ivoa.obscore` service (queried 2026 September 8), whose rows give the science-intent exposure `t_exptime` per observing unit set and pointed science source, summed over its executions. For each of the 57 searched execution blocks the pointed science source came from the archive’s field list, aliases resolved by hand (Sirius B enters through the Sirius A pointings, the UV Ceti and G 272-61 pairs share one field each, AT Mic’s block points at the pair), and its on-source time compared with the archive exposure for that source and set. `obscore` counts science-intent scans only, so target time in calibrator-intent scans is invisible, making the archive figure a lower bound and any pipeline over-count conservative; per-execution durations are not exposed, so a multi-execution set is compared through its aggregate.

It covers every window-level record of the subset, by star and execution block: the 59 pairs of the level-1 bookkeeping, with the G 272-61 and AT Mic binaries each collapsed to their shared pointing and AT Mic’s narrow flagband window kept separate from its wide ones. It is conservative for both candidate hosts’ windows (§5.4):  $\beta$  Pictoris’s windows (1824–1845 s) sit inside an archive science exposure of 47444 s and AU Mic’s inside 109334 s, with no disposition change; the archive values lower the exposure-weighted coverage total of §5.4 by less than 1 per cent.

The AT Mic block shows the signature of a dump-counting defect: three windows report 208 integrations where 145

TABLE 24

EXCEPTIONS OF THE INTEGRATION-TIME AUDIT (§5.6). THE 53 REMAINING STAR-EXECUTION-BLOCK GROUPS ALL HAVE PIPELINE ON-SOURCE TIME AT OR BELOW THE ARCHIVE SCIENCE EXPOSURE, SO THE COMPARISON IS CONSERVATIVE FOR THEM; THREE MORE (WOLF 28, THE NARROW AT MIC WINDOW, G 272-61) AGREE TO WITHIN 1%.

| Star / block                              | Pipeline (s) | Archive (s) | Ratio |
|-------------------------------------------|--------------|-------------|-------|
| AT Mic (A002_Xcd97a1_X3f1e, wide windows) | 1258         | 877         | 1.43  |
| CD-38 10980 (A002_Xe5808e_X80a1)          | 617          | 575         | 1.07  |

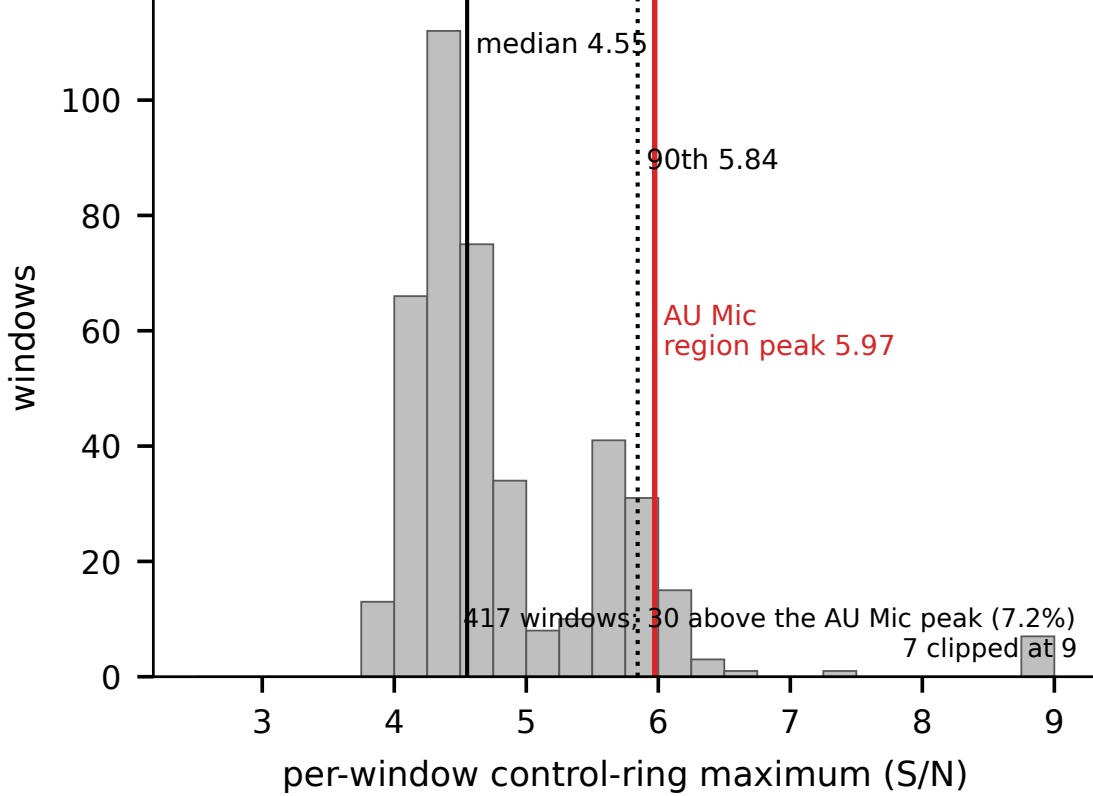


FIG. 15.— Diagnostic distribution for the marginal AU Mic candidate: control-ring peak signal-to-noise ratios of all 417 windows of this release (histogram) – the maxima expected from noise plus background structure alone. Dashed: the formal  $5\sigma$  crossing threshold. Solid: the on-source peak of the AU Mic candidate window; dotted: that window’s own control maximum, both at the values quoted in the text. Three windows lie beyond the plotted range, marked at upper right: HD 285968 Band 6 at 10.9, and  $\beta$  Pictoris Band 6 at 9.1 and 15.2, the latter the CO-filled control ring of the text.

exist, and more on-source time than the block’s own 1140 s elapsed span. The duty table uses the archive-consistent value for both AT Mic entries; the over-counted values survive only in per-window machine-readable fields, to be corrected next release. The CD-38 10980 excess is smaller than the intent-filter’s conservative margin and its limit is unaffected: thresholds derive from measured rms, not integration time. A Wolf 359 discrepancy in the raw comparison is an audit-side artefact – an over-aggressive calibrator-source exclusion in the query – and disappears once the star’s science-source exposure (9463 s, ample against the 2999 s searched) is included: exactly the error the audit is designed to catch.

#### EXCLUSION AND DEFECT FORENSICS

##### THE FOUR CLASSES OF EXCLUDED ENTRY, ITEMISED

The two process-level failures of §5.3 are item (ii) below. The six windows failing the noise-defect test lie across five stars (51 Eri, HD 10647, HD 139664, TWA 7, HD 92945 twice). The supporting audit is clean across all 417 windows bar the four  $\chi^1$  Ori Band 3 windows (elevated system temperature; limits weakened, not invalidated), and all 168 census identifiers re-resolve through SIMBAD and *Gaia* to 168 distinct sources inside 40 pc. The four classes of entry absent from the per-target file are itemised below, so their absence is auditable.

(i) **Narrowband process failures.** Four target/bands (Barnard’s Star [B7], Wolf 359 [B7], SCR J1845–6357, Wolf 358) have no narrowband result after process-level search failures; each keeps a valid continuum non-detection and carries no EIRP threshold.

(ii) **Giclas 9–38 pair.** Giclas 9–38A and Giclas 9–38B [B3] are excluded entirely: the pair’s only matched member observation unit set points  $\sim 52$  arcmin from either star,  $35\times$  its own primary-beam FWHM. The narrowband step

correctly aborts; the continuum-imaging step lacks that guard (Appendix O), so the target/band waits on one.

**(iii) Window-level noise defect.** One spectral window in each of two Band 6 targets with similar correlator setups (HD 10647, HD 139664) reports 12.1 s of on-source integration against 393–520 s for its siblings in the same block, yet a combined noise estimate  $\sim 1000\times$  deeper than those siblings and than the survey’s best continuum limit – the opposite of radiometer-noise scaling for a  $\sim 40\times$  shorter integration. The 12.1 s and the noise estimate come from the same machine-written per-window product carrying every other window’s values, and are defective together. That window’s narrowband result is excluded and the combined continuum measurement withheld for both target/bands, being built from all configured windows; their other narrowband windows are retained.

**(iv) The “ $\gamma$  Lupi” resolution.** The object observed under the catalogue entry “ $\gamma$  Lupi” is the F4V debris-disc host HD 139664 at 17.40 pc (Gaia DR3 5989102478619519616), not the naked-eye  $\gamma$  Lupi (Appendix O). All such entries are relabelled; no measured value changes and item (iii)’s defect status is unaffected.

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